TSG Administrator's Guide

Version 21.08

Publish Date: *September 10, 2021*

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Getting Started

The following topics provide system overview and detailed steps to help you logging into Tiangou Secure Gateway (TSG). Then it goes on to elaborate how to set up a basic security policy and a basic proxy policy. Administrators can configure, manage, and monitor Tiangou Secure Gateway using the web interface, CLI, and TSG Integration API.

> System Overview
> Logging into the System
> Set Up a Basic Security Policy
> Set Up a Basic Proxy Policy
> Command Line Interface
> TSG Integration API
> TSG Administration

System Overview

Purpose

The Tiangou Secure Gateway (TSG) can be used for any purpose where keeping track of the traffic flowing in a network is useful.

The following are examples of such purposes:
- Protecting the network from malicious traffic.
- Enforcing network policies.
- Lawful interception.
- Network performance optimization.
- Network visualization

System Introduction

Tiangou Secure Gateway (TSG) is a scalable traffic management product for all types of network environments. TSG performs deep packet and flow inspection on Internet Protocol (IP) packets, and classifies their content using stream-based analysis engine. TSG Firewalls are devices or programs that control the flow of network traffic between networks or hosts that employ differing security postures. TSG firewall uses a network stack to process the packet, like the OSI model. When a network packet passes through, it will be parsed and resembled to a network session. And the reassembled network session is decoded to identify the embedded content.

Tiangou Secure Gateway's Proxy module enables authorities to perform layer 4-7 advanced manipulation of application and user traffic for interception. The Proxy is deployed in transparent mode; thus, no proxy settings on browser side.

TSG enables service providers and organizations to gain insight into their network and control traffic in high-performance environments, such as large data centers and high-bandwidth network perimeters. TSG allows content visibility of HTTP, DNS, MAIL, FTP, SSL and SIP. TSG identifies and controls applications as well as evasive tools blocking. The TSG is able to modify HTTP sessions, as well as override redirect request, modify headers, inject scripts, replace texts and respond with an uploaded file. The TSG has an SSL Proxy allows all decrypted traffic to be mirrored to a third-party system for additional analysis.

Who is this Guide for?

This manual is for TSG operators, system administrators and implementation personnel.

Logging into the System

Logging into the Web Interface

You can use the Web Interface to perform configuration and monitoring tasks with relative ease. This graphical interface allows you to access TSG using HTTP and it is the best way to perform administrative tasks.

It is recommended to use the following web browsers:

- Chrome 83+

However, the best practice is to install the latest version.

STEP 1. Using a browser, open the home page of the system, for example, (http://<IP address>). You can use the IPv4 or IPv6 address.

STEP 2. Enter your username and password defined for the TSG, select your own **Language** and set the **Authentication Mode** to **LOCAL**, then click **Login**.

STEP 3. You can see your User Name at the top right of the web interface. Click it and you can change the language settings to English, Chinese or Russian. You can also Sign Out from here.

✎*Lightweight Directory Access Protocol (LDAP) is a standard protocol for accessing information directories. You can connect to an LDAP server when you set the Authentication Mode to LDAP.*

Changing Your Password

You should periodically change your password. The following procedure explains how to change the password while logged into TSG.

STEP 1. Select **Administration** > **Admins** and find your account item in the list.

STEP 2. Click the item in the list, then click **Edit**.

STEP 3. Check your account information and enter your new password.

STEP 4. Click **OK**.

Logged In Admins

TSG administrator can view users' login status, time, IP address, and disconnect logged in users. When two system users login with the same account, the latter will disconnect the former automatically.

Admins and Two-Factor Authentication

To configure current account preference, you can click **My Account** in the upper right corner. You can configure default Language and/or Line per page here.

To prevent unauthorized users from gaining access to an account with nothing more than a stolen password. TSG users can enable Two-Factor authentication strengthen the security of an account. Two-factor authentication is a combination of two of the following: your password and a text with a code sent to your smartphone or other device. It is recommended to use cloud-based mobile authenticator apps such as GOOGLE Authenticator, Microsoft Authenticator.

Login Restrictions

TSG restricts administrator logins to improve system security. An IP address will be Lockout after maximum login attempts. And you can specify limited IP addresses to be able to login the system. Configure Login Restrictions by the following procedure:

STEP 1. Select **Administration** > **Login Restrictions**.

STEP 2. If you wish to set the IP addresses that are allowed to log in, enable **Set to allow login IP**. By default, it is off, and all IP addresses are allowed to log in. Once enabled, up to 256 IPv4 CIDRs can be configured. For example, 192.168.0.1/32, 192.168.1.1/24.

STEP 3. (Optional)Add **IP** addresses if you enabled Set to allow login IP.

STEP 4. Configure **Maximum Login Attempts**, if the number of attempts reaches the limit, the client IP will be locked.

STEP 5. Specify **Lockout Time**. Within the lockout time period, this client IP cannot log into the system even with the correct user name and password.

STEP 6. Click **OK**.

Set Up a Basic Security Policy

Use the following workflow set up a very basic Security policy. This gives you a brief idea of policies so that you can verify that you have successfully configured TSG.

STEP 1. Launch the Web Interface.

1. Select **Policies** > **Security** and click **Create**.
2. Enter a descriptive **Name** for the rule.
3. Set the **Action** to **Allow**.
4. Specify a **Source** IP Address or leave the value set to **any**.
5. Specify a **Destination** IP Address or leave the value set to **any**.
6. Select the **Application**. You can select multiple applications. For more details, see Applications.
7. (Optional)**Add** a **Filter** or leave the value empty.
8. (Optional)Specify a **Tag** or leave the value empty. You can select one or more Policy Tags which are created from **Policies** > **Tags** previously. Optionally, you can click the plus icon to create new Tags. After you click the icon, a page will slide in on the right. Enter the **Tag Category**. Pick a **Color**. And add one or multiple **Tags**. Then you can select the tag you just created from the list.
9. (Optional)Specify **Effective Devices** by choosing Device Tags or leave the value empty, which means the policy is effective on all devices by default.
10. (Optional)Select a **Schedule** or leave the value set to always.
11. Verify that **Log Session** is enabled. Only traffic that matches the Security policy rule will be logged in Security Events.
12. (Optional)Enter a **Description** for the rule.
13. Verify that Enabled is enabled.
14. Click **OK**.

STEP 2. (Optional)To verify that you have set up your basic policies effectively, test whether your Security policy rules are being evaluated and determine which

Security policy rule applies to a traffic flow. For example, to verify the policy rule that will be applied for a client with the IP address 192.168.0.1 when it sends a HTTP request to the 172.16.0.1 server:

1. Select **System** > **Trouble Shooting** > **Policy Verify**.
2. Select **Security Policy Match** from the **Select Test** drop-down.
3. Enter the **Client IP** and **Server IP** addresses.
4. Specify the **Client Port** and **Server Port**.
5. Select the **Protocol** and **APP ID** from the drop-down.
6. Click **Verify** to execute the **Security policy match** test.

STEP 3. After the policy has been hit, view Logs to monitor the policy rule status and determine the effectiveness of the policy rule. Select **Logs** > **Security Events** and view relative information about the policy.

✎*When you are creating or editing policies and objects, click the sidebar menu will not navigate you to the corresponding page. A prompt window will appear to remind you that changes you made are not saved.*

Set Up a Basic Proxy Policy

Security Policies with Intercept actions intercept HTTP/HTTPS traffic for proxy, which is a prerequisite for proxy policy.

You can perform the following to set up a basic proxy policy.

STEP 1. Add a rule.

1. Select **Policies** > **Proxy** and click **Create** to add a new rule.
2. Enter a descriptive **Name** for the rule.
3. Define what action you want TSG to take for traffic that matches the rule. Select an **Action**. See Proxy Policy Actions for a description of each action.
4. Define the matching criteria for the source fields in the packet. Specify a **Source** IP Address/ User or leave the value set to any.
5. Define the matching criteria for the destination fields in the packet. Specify a **Destination** IP Address or leave the value set to any.

6. (Optional)Specify a **Filters** as match criteria for the rule. For example, select a **Category** for **Host**. If you select a category, only web traffic will match the rule and only if the traffic is destined for that specified category.
7. (Optional)If the Action is Redirect, specify the **Application**.
8. (Optional)Specify a **Tag** or leave the value empty. You can select one or more Policy Tags which are created from **Policies** > **Tags** previously. Optionally, you can click the plus icon to create new Tags. After you click the icon, a page will slide in on the right. Enter the **Tag Category**. Pick a **Color**. And add one or multiple **Tags**. Then you can select the tag you just created from the list.
9. Specify **Effective Devices** by choosing Device Tags or leave the value empty, which means the policy is effective on all devices by default.
10. (Optional)Enter a **Description** for the rule.
11. Select a **Schedule** or leave the value set to always.
12. Verify that **Log Session** is enabled if you wish to have proxy event logs. When Log Session is enabled, select Metadata only for **Log Options** or set it to All by default.
13. Verify that **Enabled**is on.
14. Click **OK** to save the policy rule.

STEP 2. (Optional)To verify that you have set up your basic policies effectively, test whether your Proxy Policy rules are being evaluated and determine which Proxy Policy rule applies to a traffic flow.

1. Select **System** > **Trouble Shooting** > **Policy Verify**.
2. Select **Proxy Policy Match** from the **Select Test** drop-down.
3. Enter the **Client IP** and **Server IP** addresses or leave it set to default.
4. Specify the **Client Port** and **Server Port** or leave it set to default.
5. Select the **Protocol** and **APP ID** from the drop-down.
6. Click **Verify** to execute the Proxy Policy Match test.

STEP 3. Go to **Logs** > **Proxy Events** and view **Logs** to monitor the policy rule status, verify if the proxy rule has been hit and determine the effectiveness of the policy rule.

Command Line Interface

You can use the TSG Command Line Interface (CLI) to perform a series of tasks by entering commands in rapid succession over SSH. The TSG CLI is a TSG specific command shell. By leveraging industry-standard tools and utilities, the CLI provides a set of commands that you can use to monitor and configure TSG devices. TSG CLI supports two types of command, TSG specified commands and Linux-like system operational commands. TSG specified commands help control policy and object, configure and check devices status. The commands related with policy and object control work on all TSG devices cluster. The other CLI commands only work on local device. When you become familiar with the nesting structure and syntax of the commands, the CLI provides quick response times and administrative efficiency.

For more details, please view **TSG CLI User Guide**.

TSG Integration API

TSG Integration API is a web service implemented using HTTP requests and responses following architecture style REST the RESTful API. You can use this RESTful API to streamline your operations and integrate with existing, internally developed applications and repositories.

For more details, please refer to **TSG Integration API Specification**.

TSG Administration

Configure TSG Users

The following table list three authentication mode for TSG:
To restrict system access to authorized users, TSG provides role-based access control (RBAC). The basic concept of RBAC is that permissions are associated with roles, and users are made members of appropriate roles, thereby acquiring the roles' permissions. This leads to "user-role-permission" authorization model. In TSG system, the relationship between users and roles is one-to-one, and the relationship between roles and functional permissions is one-to-many. That is, a user can only have one role, and a role can be assigned multi-level permissions to different features.

Authentication Mode	Privileges
LOCAL	The administrative account credentials mechanisms are local to TSG and provide full access to the web interface.
SSH Keys	The CLI specified administrative accounts are local to the TSG, but authentication to the CLI is based on SSH keys.
External service	The administrative accounts you define locally on TSG serve as references to the accounts defined on an external LDAP server. The external server performs authentication.

For each feature, there are three permission levels: Enable, Read Only and Disable. When the user has Read Only permission to a feature which normally is related with certain menu, the user cannot click Create, Edit, Delete, Enable and Disable button and the detail pages of Policies and Objects are locked.

- Enable—Read/write access to the selected feature.
- Read Only—Read-only access to the selected feature.
- Disable—No access to the selected feature.

Users and Roles

Perform the following steps to add a LOCAL administrative account on TSG.

STEP 1. Select **Administration** > **Admins**, select tab Users and click **Create**.

STEP 2. Enter a **Name** to identify the account.

STEP 3. Enter your **User Name** which is the login name and **Password**.

STEP 4. Please **Confirm Password**.

STEP 5. If you enable **Required Password Change**, fill in the **Required Password Change Period**. A message will show up when you login after the period expires to enforce you to change to a new password.

STEP 6. Verify that the account is **Enabled**.

STEP 7. Select **Role** from dropdown list. Each account must and can only have one role, which defines different user permissions. For details, see Roles and Permissions.

STEP 8. Click **OK**.

STEP 9. (Optional)To verify that you have add a TSG account effectively, you can **Sign Out** and log into the system with the new account. Select **System Logs** > **Login Log** and you can view your login information.

If compliance, audit, or security requirements stipulate that the default administrative account must be removed from your devices, you can remove it after you create at least one other superuser administrative account. You cannot remove the default administrative account until you configure at least one other superuser administrative account on the device. Perform the following steps to delete an account on TSG.

STEP 1. Select tab Users of **Administration** > **Admins** and find the item you want to delete in the list.
STEP 2. Click **Delete** at the top left. Click **Delete** to confirm.

If you wish to temporarily disable an account, you can search it by User Name or Name at the top right search box. Then click the switch under **Enable**. Please make sure your account is assigned the proper role with proper permission before you do **Delete** or **Enable**. See Roles and Permissions to find more details.

✎*And TSG supports 100 concurrent users at present.*

Roles and Permissions

There are two predefined roles in TSG, including:
• superuser: full permissions to all features and can create new accounts.
• superreader: read only permissions to all features.

✎*If a user's role permissions enable TSG feature "Administrator-Users-Users" and at least include "Administrator-Users-Roles" Read Only access, the user can create new users, and modify other users' role permissions. This ability can affect the access rights of all users, please authorize with caution.*

Perform the following to create a new role:

STEP 1. Select **Administration** > **Admins**, select tab Roles and click **Create**.
STEP 2. Enter a **Name** to identify the role.
STEP 3. (Optional)Enter your **Description**.
STEP 4. Click the icon before each **Menu** to switch the permission levels, which defines different permissions.
STEP 5. Click **OK**.

✎*It is recommended to configure the same access permission for Policies, Objects and System menu, because their data are related. Make sure Devices are enabled before*

you enable Dashboard, because Devices affects the reading of data for device module in Dashboard.

Enroll LDAP Servers

Configuring TSG to connect to a LDAP server enables you to login in LDAP Authentication Mode with LDAP account. Perform the following steps to add a LDAP server on TSG.

STEP 1. Select **System** > **Server Profiles** > **LDAP Server** and click **Create**.
STEP 2. Define a **Name** to specify the LDAP server.
STEP 3. Enter your **Host** and **Port** of the LDAP server.
STEP 4. Enter your **User Name**, which is the administrative user of LDAP server, and **User Mapper** which specifies the hierarchy of LDAP user.
STEP 5. Enter the **Password** of user in STEP 4. Verify that **Enabled** is on.
STEP 6. **Test Connection**. After success, click **OK**.
STEP 7. (Optional)To verify that you have add a LDAP server effectively, you can view related information on the **LDAP Server** list page.

After setting LDAP server, you can login using the LDAP accounts of enrolled LDAP server. After a LDAP account login TSG first time, the LDAP accounts will show on the tab **Admins** of **Administration** > **Admins**. The column Source indicates the type of account which is shown as LDAP for LDAP account. The column User name includes the full path for LDAP user, and the value of "uid" is the actual login username on TSG. When logging into TSG system for the first time with LDAP user, TSG system will assign the user the role supperreader by default. If the LDAP user requires other role permissions, you need to login by other users who have permission to modify a user's role to modify it.

In **Server Profiles** > **LDAP Server** page, you can view the LDAP Server list. Operator displays who has modified the item, and it can be a LOCAL or LDAP account. Select the item you wish to change in the list and click **Edit** to modify LDAP server information. You can delete or disable the LDAP server and after that you will not be able to log into the system with the LDAP account.

Audit Log

If you perform an operation which influence the running of TSG, TSG will generate a log about this action. For example, Audit Log will record the operations of adding

or deleting or updating an object or policy, or clearing traffic logs, etc. You can view **System Logs** > **Audit Log** to see details. You can query audit logs within certain time range by ID, Source IP or Target Type. Audit logs can be exported as trace evidence. And when you are editing a policy or an object, you will find a link to audit log about this policy or object.

Mail Server

Configure Mail Server to send mail alerts, which is currently used to send reports. Perform the following to create a Mail server profile:

STEP 1. If you perform an operation which influence the running of TSG, TSG will generate a log about this action. For example, Audit Log will record the operations of adding or deleting or updating an object or policy, or clearing traffic logs. You can view **Administration** > **Audit Logs** to see details.

STEP 2. Select **Server Profiles** > **EMail Servers**.

STEP 3. For Simple Mail Transport Protocol (SMTP) server (email server), Add a **Server** and **Port**.

STEP 4. Enable **Need Authentication**.

STEP 5. Define a **Name** to identify the SMTP server (1-32characters). This field is just a label and doesn't have to be the hostname of an existing email server. Define **E-mail**, the name to show in the **From** field of the email.

STEP 6. Enable **SSL**.

STEP 7. Click **OK** to save the Email server profile.

To verify if the mail server works well, you can generate a Report and **Enable Notification** to send emails. For details, see View and Manage Reports.

Policies

The policy is the axis around which most features of TSG revolve. Many TSG settings end up relating to or being associated with the policies and the traffic they govern. Any traffic going through TSG has to be associated with a policy. These policies are essentially discrete compartmentalized sets of instructions that control the traffic flow going through TSG. These instructions control where the traffic goes, how it is processed, if it is processed, and whether or not it is allowed to pass through the gate.

When TSG receives a connection packet, it analyzes the source, destination, application and corresponding filters. Using this information, TSG attempts to locate a security policy that matches the packet. If a policy matches the parameters, then TSG takes the required action for that policy. If it is Allow, the traffic is allowed to proceed to the next step. If the action is Deny, the traffic is not allowed to proceed. If the action is Monitor, the traffic is scanned and a log is generated. If it is Intercept, the traffic is intercepted and relative proxy policy will be applied.

> Policy Concepts
> Security Policy
> Proxy Policy
> Schedules
> Verify Policy Rules

Policy Concepts

Policies allow you to enforce rules and take action. From TSG's perspective, network packets are first resembled to a session, and identified as many manageable attributes. For example, a TLS session could be parsed to attributes as IP address, SNI, certificate and so on. For effective management, user could define policy rules with TSG user interface.

A policy rule combines with several conditions and one action. The action determines how to control the traffic, and action parameters are managed in policy profiles. Profiles are elaborated in chapter Certificate Managements and chapter Proxy Profiles. The conditions describe attributes of applicable traffic, and they are organized by policy objects. While policy objects enable you to identify traffic to enforce policies, policy profiles help you define further action. Traffic must match all conditions of a policy before taking its action. Traffic matching one item of an object means it satisfies the object. In short, objects in one policy are "AND" relation, items and subordinate object in one object are "OR" relation.

You can identify a policy rule by its name or ID number. The policy ID for a rule never changes even if you modify the rule, such as when you change the rule name. The policy ID allows you to track the rule across rules even after you disable the rule.

✎*Note that activation of policy rules on all devices is no more than 1 minute. For details about capacities of policies and URLs/URIs, and other system parameters, see product datasheet.*

Rule Types

TSG supports a variety of policy types that work together to safely enable applications on your network. The different types of policy rules that you can create on TSG are:

Policy Type	Description
Security	Determine whether to allow, deny, monitor or intercept a session based on traffic attributes such as the IP addresses, subscriber ID, APP ID and protocol fields. For more details, see Security Policy.

Proxy	Instruct the proxy how to manipulate (including allow, deny, monitor, redirect, replace, hijack and insert), a session based on traffic attributes such as the source and destination IP address, the subscriber ID, and filter, including host, URL, request header, response header, request content and response content. Encrypted traffic Identified will be decrypted for further control. Manipulation requires targeted sessions are intercepted in security policies. For more details, see Proxy Policy.

Policy Conditions

Each rule has a set of conditions. All criteria need to match if the rule action is to be used. The order of each condition in the list is irrelevant. A condition consists of an object type from the list of object types used for traffic identification, an operator (only support equals for now) and a specific object of that type. If a connection matches all the objects set to "equals" in the rule, the rule will apply to the connection.

Policy objects server as policy conditions. There are four types of conditions at most in each policy, namely: Source, Destination, Application and corresponding Filter. You can view which objects are referenced as conditions by selecting **Policies** > **Security** or **Policies** > **Proxy**, then double click the policy in the list.

A policy object consists of collective items that groups discrete identities such as IP addresses, URLs, applications, or user accounts. You can define an object with a set of items or a set of subordinate objects and reuse it in multiple policies.

✎*Consider a Source, Destination, Application and each member of a Filter as a condition. A policy contains 8 conditions at most. The relation between each condition is "and". The relation within each condition is "or". Take HTTP as an example, possible values of its Filter can be Host, URL, Request Header etc. If you add two Hosts for Filter, the relation between each Host is also "and".*

For more details about policy objects, see Objects.

Policy Hierarchy and Evaluation Order

TSG firewall uses a network stack to process the packet, like the OSI model. When a network packet passes through, it will be parsed and resembled to a network session. And the reassembled network session is decoded to identify the embedded content. At each step of the decoding process, all applicable fields of the application protocol, the application, and the file are extracted and stored. These extracted attributes provide context to the content that was extracted from the data. This extracted attributes and content provide comprehensive visibility into network traffic. With this deep visibility, security policies can be fine grained crafted to detect and manage network threats.

Based aforementioned design, security policy rules are enforced as following **inspection layers**:

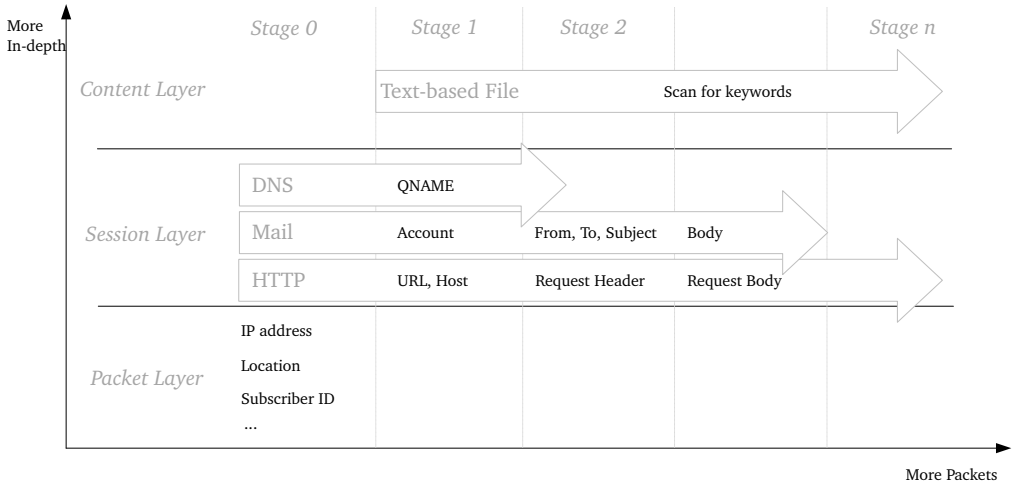
1. Packet layer: The firewall parses IP fragments, reassembles using the defragmentation process, and then parses IP header and TCP/UDP protocol headers if applicable. As application identification is on all layers, the identified application as well as IP address, ASN and Subscriber ID is used as key to find rule match. Because the application is not necessarily known in the first packets, it can take several packets to determine what the underlying application is, and rules without specifying application will be enforced in advance.
2. Session layer: Packets are reassembled to network session and decoded to traffic attributes, e.g. TLS SNI (TLS handshake option), HTTP Host, Mail addresses and DNS QNAME.
3. Content layer: Content that extracted from one session or more sessions are applicable for keywords scan, e.g. HTTP request body and Mail attachment.

Within the layer, traffic attributes are not acquired simultaneously, which is known as **parsing stages**. This design offers stateful security functions at the application layer, and the effectiveness of policy enforcement. The parsing stages of common protocols are listed in table below.

Protocols	Attributes Parsing Stages
HTTP	1. Host, URL 2. Request Header 3. Request Body 4. Response Header 5. Response Body

TLS/SSL	1. SNI 2. CN, SAN
DNS	QNAME
MAIL	1. Account 2. From, To, Subject 3. Content 4. Attachment File Name 5. Attachment File
FTP	1. Account 2. URI 3. Content

As the firewall receives more packets and do more in-depth inspection, more traffic attributes and content are ready for policy enforcement, which is shown in figure below.



It is highly likely that even after only a relatively small number of policies have been created that there will be some that overlap or are subsets of the parameters that the policies use to determine which policy should be matched against the incoming traffic. When this happens there has to be a method to determine which policy should be applied to the packet. The method which is used by TSG is the policy hierarchy. In another words, policies with lower inspection layers and earlier parsing stages are precedent. For example, a HTTP session, its URL appears before response body. That's why policy rules that inspect HTTP response body, are always evaluated after ones that inspect URL.

✎*A policy rule that combines with condition fields of different layers and stages are evaluated at the last layer and stage. Rule stage is decided by policy condition fields; and rule modification may change the rule stages.*

As firewall maybe unable to determine the exact application from the first packet, a session may be allowed through the firewall as undecided until the application is identified, at which time the security rules are re-evaluated and an appropriate action is taken.

For more details about how TSG process packet flow, please see Appendix E TSG Packet Flow.

Security Policy

Security policies determine whether to deny, allow, monitor or intercept a session based on initial session attributes, such as the subscriber ID, IP addresses, ports, protocols and applicable protocol fields (e.g. TLS SNI). On TSG, security policy rules are enforced by firewall module.

All traffic passing through the firewall is matched against a session and each session is matched against a security policy rule. When a session match occurs, the firewall applies the matching security policy rule to bidirectional traffic in that session (client to server and server to client). For traffic that doesn't match any defined rules, the default rule applies. The default rule is predefined to allow all traffic for firewall policy check. Although the default rule is read-only by default, you can override them and change a limited number of settings, including the tags, action (allow or deny) and log settings.

A packet is matched against the first rule that meets the defined criteria and, after a match is triggered, subsequent rules are not evaluated. Session that matches a rule generates a log entry at the end of the session in the security event log if you enable logging for that rule. The logging options are configurable for each rule.

Components of a Security Policy Rule

The Security policy rule construct permits a combination of the required and optional fields as detailed in the following table:

Required/ Optional	Field	Description
Required	Name	A label (up to 128 characters) that identifies the rule.
	Action	Specifies an Allow, Deny, Monitor or Intercept action for the traffic based on the criteria you define in the rule. For more details, see Security Policy Actions.
	Application	The application that you wish to control. It provides application control and visibility in creating security policies that block unknown applications, while enabling, inspecting, and shaping those that are allowed. For more information, see **Objects** > **Applications**.
Optional	Source	Define host IP addresses, address groups, Subscriber ID, IP Learning, or Geographic enforcement.
	Destination	The location or destination for the packet. Define host IP addresses, address groups, IP Learning, or Geographic enforcement.
	Filter	All web traffics are compared against the filtering, giving you a way to safely control how your users interact with online content. You will have Filter available when you select only one of the following protocols: HTTP, SSL, DNS, MAIL, FTP, QUIC and SIP in Applicationfield.
	Tag	A keyword or phrase that help you to identify the policy.
	Effective Devices	Select the devices that the security rule will be applied.
	Schedule	Schedule when (day and time) the security rule should be in effect.
	Log Session	When enabled, traffic that match security policies will be logged in Security Events.
	Description	A text field, up to 1024 characters, used to describe the rule.
	Enabled	Only enabled policies will be enforced.

Actions

For traffic that matches the attributes defined in a security policy, you can apply the following actions:

Action	Description
Allow	The traffic is allowed to proceed to the next step.
Deny	The traffic is not allowed to proceed. There are 5 Sub Action to select, they are Drop, RST (TCP reset attack), Block, Alert and Redirect. Sub Action of Deny: **Drop**, silently drops the traffic. **RST**, session was terminated and a TCP reset is sent to both the sides of the connection. **Block**, sends HTTP 403/404 response code and customized content to the client. **Alert**, sends HTTP 200/204 response code and customized content to the client. **Redirect**, redirects to the URL set in the policy configuration. **Rate Limit**, limit the bandwidth of the session that matches the policy to the set value. Rate supports float number. The unit is Kbps, Mbps and Gbps.
Intercept	Intercept HTTP/HTTPS traffic for proxy. If the traffic use SSL/TLS, it will be decrypted.
Monitor	Scan all allowed traffic and generate a detailed log. Support Action Parameters: **Mirror Traffic** which is not enabled by default. If you enable mirror traffic, the current and subsequent packets that hit the policy will be mirror to external device. Enable Add VLAN ID and specify VLAN ID to designate mirror destination address. Note that the mirror traffic function is only effective on ATCA (TSG9140). Its corresponding Security Event logs will record mirrored_pkts and mirrored_bytes of packet mirroring and support retrieval by these fields.

Applications and Filters

AppSketch is a traffic classification system available in TSG firewalls, determines what an application is irrespective of port, protocol, encryption or any other evasive tactic used by the application. It applies multiple classification mechanisms, (including application signatures, application protocol decoding, heuristics, and machine learning) to your network traffic stream to accurately identify applications.

Applications enables visibility into the applications on the network, so you can learn how they work and understand their behavioral characteristics and their relative risk. When you define policy rules to control traffic, applications begin to classify traffic without any additional configuration.

Applications Type		Description
Application	Basic Protocols	Network protocols ensure that computer network devices can transmit and receive data using a common language regardless of their different designs, hardware or infrastructures. List all 50 basic protocols: Unknown_TCP, Unknown_UDP, Unknown_Other, DNS, FTP, FTPS, HTTP, HTTPS, ICMP, IKE, IMAP, MAIL, IMAPS, IPSEC, XMPP, L2TP, NTP, POP3, POP3S, PPTP, QUIC, SIP, SMB, SMTP, SMTPS, SPDY, SSH, SSL, SOCKS, TELNET, DHCP, RADIUS, OPENVPN, STUN, TEREDO, DTLS, DoH, ISAKMP, MDNS, NETBIOS, NETFLOW, RDP, RTCP, RTP, SLP, SNMP, SSDP, TFTP, BJNP, LDAP, RTMP, RTSP.
	Predefined Applications	TSG has various predefined applications.
	Customized Applications	TSG allows you create custom applications and reference the customized application objects in policy.
Selector	Selector	TSG allows you dynamically group applications based on application properties that you define into one object Selector. It is available for Action Allow and Deny. For more details, see Application Selector.

Group	Group	An application group is an object that contains applications that you want to treat similarly in a policy. It is available for Action Allow and Deny. For more details, see Application Group.

You will have Filter available when you select only one of the following protocols: HTTP, SSL, DNS, MAIL, FTP, QUIC and SIP. The following table lists different Filter fields for each protocol and the available object type you can select for each filter field, it also shows whether each filter field support allow, deny, monitor and intercept action or not.

Protocol	Filter Field	Available Object	Applicable Actions			
			Allow	Deny	Monitor	Intercept
HTTP	Host	FQDNs Category	✓	✓	✓	✓
	URL	URLs	✓	✓	✓	✗
	Request Header	HTTP Signatures	✗	✓	✓	✗
	Response Header		✗	✓	✓	✗
	Request Content	Keywords	✗	✓	✓	✗
	Response Content		✗	✓	✓	✗
SSL	SNI	FQDNs Category	✓	✓	✓	✓
	CN		✗	✓	✓	✗
	SAN		✗	✓	✓	✗
QUIC	SNI	FQDNs Category	✓	✓	✓	✗
DNS	QNAME	FQDNs Category	✗	✓	✓	✗
MAIL	Account	Accounts	✗	✓	✓	✗
	From		✗	✓	✓	✗
	To		✗	✓	✓	✗
	Subject	Keywords	✗	✓	✓	✗
	Content		✗	✓	✓	✗
	Attachment Name		✗	✓	✓	✗

	Attachment Content		✗	✓	✓	✗
FTP	URI	URLs	✗	✓	✓	✗
	Content	Keywords	✗	✓	✓	✗
	Account	Accounts	✗	✓	✓	✗
SIP(used in VoIP)	Originator Description	Accounts	✓	✓	✓	✗
	Responder Description		✓	✓	✓	✗

The Evaluation Order for security policy is described as following.

Security policy is evaluated before Proxy policy. Proxy manipulation requires targeted sessions are intercepted in security policies. That is to say an intercept security policy rule is the prerequisite for proxy policy.

Traffic can match one or multiple security policy rules with monitor action. This means traffic can hit multiple security policy rules only with monitor actions. For security policies with allow, deny and intercept actions, traffic can only hit one policy that meets the defined criteria with top priority. After a match is triggered, subsequent rules are not evaluated. It is important to know the policy evaluation order to create policies accordingly.

There are three factors in evaluation of policies. They are condition, action and policy ID in sequence.

1. The ranked evaluation order for condition is:
   IP > FQDN/URL/Request Header > Request Content > Response Header >Response Content
2. The priority queue of action is:
   Allow > Deny > Intercept > Monitor
   For policies that match SSL Decryption Exclusion are evaluated before intercept policies.
3. Prioritize policies in descending order by policy ID.

For more details about how TSG process packet flow, please see **Appendix E TSG Packet Flow**.

Voice over Internet Protocol

Voice over Internet Protocol (VoIP) has become more and more popular as an alternative to the traditional public switched telephone network (PSTN). VoIP mainly uses RTP as its media protocol to deliver multimedia sessions and SIP for signaling.

TSG supports allow, deny and monitor VoIP based on IP address and/or accounts and you can view its logs, including play the audio file. For now, TSG only supports the mentioned actions above with VoIP calls using SIP for signaling and RTP for delivering audio data.

To view detailed description about VoIP log fields, see **Appendix C Log Fields Description** > **Log Fields per Protocol** > **SIP** and **RTP**.

GPRS Tunneling Protocol(GTP)

GPRS Tunneling Protocol (GTP) allows mobile subscribers to use their phones to establish connections for network access on the move. GTP creates, modifies, and deletes tunnels for transporting IP payloads between the user equipment, the GPRS support nodes (GSNs) in the GPRS backbone network and the internet. GTP comprises three types of traffic—control plane (GTP-C), user plane (GTP-U), and charging (GTP' derived from GTP-C) traffic. TSG supports GTP, which allows you to inspect, validate, filter, and perform security checks on GTPv2-C, GTPv1-C.

To view detailed description about GTP log fields, see **Appendix C Log Fields Description** > **Log Fields per Protocol** > **GTP-C**.

Allow Rules

TSG allows the network traffic to pass through, without apply further policy checking. You can define traffic that you choose not to enforce policies because of business, regulatory, personal, or other reasons, such as financial, health, military, or government traffic.

Generally, allow policies need not generate logs. However, if you enable Log Session, traffic that matches the allow policy rule will also be logged in Event Logs. If you wish to have logs, it is recommended to create monitor rules with Log Session enabled, which will have the same effect.

Intercept Rules

Security policy with intercept action allows you to define traffic that you want the Proxy to terminate. Both HTTP and HTTPS sessions could be terminated. You can do further manipulation on intercepted traffic with Proxy Policy.

When an SSL session is set to intercept, the Proxy terminates the client-side session and initiates a new server-side session. Thus, the server certificate is replaced and content is decrypted.

You can specify different keyrings for individual intercept policy. If not, the Proxy uses the default keyring for trusted website. Keyrings are managed via **Certificate Managements** > **Decryption Keyrings**.

Intercept Options

While policy objects enable you to identify traffic to enforce policies, policy profiles help you define further action. When you create an Intercept policy and select the SSL application to enable access to encrypted sites, you will have three Action Parameters, namely Keyring, Mirror Decrypted Traffic and Decryption Profile. They serve as your intercept options.

• You can specify a Keyring or use TrustedDefault. To specify a different Keyring, select the item from drop-down or click the plus icon to create a new one. For more details, see Decryption Keyrings.

• If you enable Mirror Decrypted Traffic, you can select one item from drop-down or click the plus icon to create a new profile. For more details, see Traffic Mirror Profile.

• You can use decryption-default for your Decryption Profile, or you can select different one from drop-down and create new ones. For more details, see Decryption Profile.

Proxy Limitations

There are two types of traffic that exclude from decryption. They are the proxy limitations. One is that you choose not to decrypt because of business, regulatory, personal, or other reasons. The other is traffic that breaks decryption for technical reasons:

- Asymmetric Routing Traffic
- Certificate Pinning Apps:
    - Facebook, Instagram
    - YouTube, Gmail, Google Search
    - Twitter
    - Apple Store
- Mutual Authentication (Client Certificate Request)
- Some android System that unable to install root certificate
- Website using HPKP
- Nonstandard SSL/TLS implications
- SSLv1.0, SSLv2.0 are not supported
- Unsupported ciphers

For more details, see SSL Decryption Exclulsion.

If you wish to exclude FQDN from decryption, create an SSL Decryption Exclusion item and it will immediately become effective. For policies that match SSL Decryption Exclusion are evaluated before intercept policies.

Intercept Trouble Shooting

You can find out if the interception is successful by checking if the certificates are issued by your pre-configured Root CA. You should exercise with cautions because web applications may not cooperate with SSL interception.

- Your connection is not private/secure

Visit the website that should match the intercept rule. For Chrome, Microsoft Internet Explorer and Firefox, click the lock icon next to the address bar to view website certificates. If the browser warning that the connection is not secure, one possible reason is you haven't install/trust the root certificate yet.

Another possible reason is that the site uses an incomplete certificate chain, the Proxy doesn't automatically fix the chain as a browser would. Although Proxy is able to cache unseen intermediate certificates that issued by trusted root (**Certificate Managements** > **Cached Intermediate Certificates**), you still need to manually download the missing intermediate certificates occasionally. Missing certificates could be import via **Certificate Managements** > **Trusted Certificate Authorities**.

• Handling Web Sites Where Decrypt Re-sign Works for a Browser but not an App (SSL or Certificate Authority Pinning)

Some apps for smart phones and other devices use a technique called SSL (or Certificate Authority) pinning. The SSL pinning technique embeds the hash of the original server certificate inside the app itself. As a result, when the app receives the resigned certificate from TSG, the hash validation fails and the connection is aborted.

If internet users cannot connect to the web site using the site's app, but they can connect using the web browser, even when using the browser on the same device where the app fails. For example, users cannot use the Facebook iOS or android app, but they can point Safari or Chrome at https://www.facebook.com and make a successful connection.

Because SSL pinning is specifically used to avoid man-in-the-middle attacks, there is no workaround. You must choose between the following options:

- Support app users, by enabling Certificate Pinning dynamic bypass option, in which case you can only decrypt browser traffic to the site. For more details, see **Proxy Profiles** > **Decryption Profile**.

- Force users to use browsers only. If you must decrypt traffic to the site, you will need to inform users that they cannot use the site's app when connecting through your network, that they must use their browsers only.

Create a Security Policy Rule

STEP 1. Add a rule. Select **Policies** > **Security** and click **Create** to add a new rule. Enter a descriptive **Name** for the rule.

STEP 2. Define what action you want TSG to take for traffic that matches the rule. Select an **Action**. See Security Policy Actions for a description of each action.

STEP 3. Define the matching criteria for the source fields in the packet. Specify single or multiple **Source** IP Addresses/ Subscriber IDs or leave the value set to any. If you don't add entries to the list, the policy has no IP address restrictions.

STEP 4. Define the matching criteria for the destination fields in the packet. Specify one or multiple **Destination** IP Addresses or leave the value set to any.

✎*As a best practice, use address objects as the Destination Address to enable access to only specific servers or specific groups of servers especially for commonly exploited services, such as DNS and VPN.*

STEP 5. Select one or more **Applications**. There are built-in protocols and well-known Applications and customized Applications in the list. You can search the application you want to fill in. You can also use application **Selector** and **Group** as objects in policy. For more details, see **Security Policy** > **Applications and Filters** and **Objects** > **Applications**.

STEP 6. (Optional) Specify a **Filter** as match criteria for the rule. For example, select a **Category** for **Host**. If you select a category, only web traffic will match the rule and only if the traffic is destined for that specified category. For more details, see **Security Policy** > **Applications and Filters**.

✎*Filter is not available when multiple or no applications are selected.*

STEP 7. (Optional)Specify a **Tag** or leave the value empty. You can select one or more Policy Tags which are created from **Policies** > **Tags** previously. Optionally, you can click the plus icon to create new Tags. After you click the icon, a page will slide in on the right. Enter the **Tag Category**. Pick a **Color**. And add one or multiple **Tags**. Then you can select the tag you just created from the list.

STEP 8. Specify **Effective Devices** by choosing Device Tags or leave the value empty, which means the policy is effective on all devices by default. Device Tags are built-in labels to classify the devices according to user's requirement when TSG is deployed. Normally, the Device Tags will be classified by areas or ISPs.

STEP 9. (Optional)Enter a **Description** for the rule.

STEP 10. (Optional)Select a **Schedule** or leave the value set to always. For more details, see **Policies** > **Schedules**.

✎*If you select a schedule, the schedule will determine when the policy will be enabled and effective. This means you don't have to turn the Enabled switch on. Because the policy will automatically be effective according to your specified schedule. And even if you set the policy Enabled, the policy will not be effective, if it has not reached the schedule time frame.*

STEP 11. Verify that **Log Session** is enabled if you wish to have logs. Only traffic that matches a Security policy rule will be logged.

STEP 12. Verify that **Enabled** is enabled.

STEP 13. Click **OK** to save the policy rule.

STEP 14. (Optional)To verify that you have set up your policies effectively, test whether your Security policy rules are being evaluated and determine which Security policy rule applies to a traffic flow. For example, to verify the policy rule that will be applied for a client with the IP address 192.168.0.1 when it sends a HTTP request to the 172.16.0.1 server:

1. Select **Settings** > **Trouble Shooting** and select **Policy Verify** tab.

2. Select **Security Policy Match** from the **Select Test** drop-down.
3. Enter the **Client IP** and **Server IP** addresses or leave it set to default.
4. Specify the **Client Port** and **Server Port** or leave it set to default.
5. Select the **Protocol** and **APP ID**.
6. Click **Verify** to execute the **Security policy match** test.

STEP 15. After the policy has been hit. View **Logs** to monitor the policy rule status and determine the effectiveness of the policy rule. Select **Logs** > **Security Events** and view relative information about the policy.

You can view detailed information about the policy you just created. To edit and delete the policy, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left.

You can export the contents of policies to a json file. First, search policies according to ID, Name. Then, click the Export icon on the right to download and save the file to your local folder.

You can also import policies by clicking the import icon. Only json and txt formats can be uploaded. You can take the exported file as template for import.

Select the checkbox for policies in the list and Click **Watch** at the bottom to add to Watch List. And then you can click the star icon in the bottom right and select Policy tab to view the Watch List. You can search policies by ID and Name in the list. The Watch feature gives you the ability to observe policies and objects on which you wish to keep an eye any time. In policy list page, click Hit Count if there are any, to view corresponding log list. In policy detail page, click Log Count to jump to corresponding log list.

✎*Policy No. 0: For traffic that passes through TSG but does not hit any policies, the traffic is executed according to policy No. 0. The conditions of policy No. 0 are all any, and by default the action is Allow. Policy No. 0 is forbidden to edit.*

Proxy Policy

Proxy policy instructs the proxy how to manipulate a session. Manipulation requires targeted sessions are intercepted in security policies. An individual manipulation policy rules determine whether to allow, monitor, deny or manipulate a session based on traffic attributes. Valid objects depend on specific action.

Proxy policy works correctly on all platforms, including Windows, Linux, MacOS, Android, iOS, etc. and also supports console commands, such as wget and curl, mobile devices, tablets and workstations in all locations. But TSG does not support TV device that cannot install the certificate, and device with Android 7.0 and versions below.

For the site which breaks decryption for certificate pinning, the proxy only can bypass or deny the traffic based on your configuration. For more details, see Decryption.

Components of a Proxy Policy Rule

The Proxy Policy rule construct permits a combination of the required and optional fields as detailed in the following table:

Required/ Optional	Field	Description
Required	Name	A label (up to 128 characters) that identifies the rule.
	Action	Specifies an Allow, Deny, Monitor, Redirect, Replace, Hijack or Insert action for the traffic based on the criteria you define in the rule. For more details, see Proxy Policy Actions.
Optional	Source	Define host IP addresses, address groups, Subscriber ID, IP Learning, or Geographic enforcement.
	Destination	The location or destination for the packet. Define host IP addresses, address groups, IP Learning, or Geographic enforcement.
	Filter	All web traffics are compared against the filtering, giving you a way to safely control how your users interact with online content. You must fill one of the Source, Destination or Filter.
	Application	The application that you wish to control. It provides application control and visibility in creating proxy policies that block unknown applications, while enabling, inspecting, and shaping those that are allowed.

Tag	A keyword or phrase that help you to identify the policy.
Effective Devices	Select the devices that the security rule will be applied.
Schedule	Schedule when (day and time) the security rule should be in effect.
Log Session	When enabled, traffic that match security policies will be logged in Security Events.
Log Option	Metadata and All: Metadata only structured logs are recorded. All provides raw log files for some special log fields, such as content of HTTP request header or HTTP response content.
Description	A text field, up to 1024 characters, used to describe the rule.
Enabled	Only enabled policies will be enforced.

Actions

For traffic that matches the attributes defined in a proxy policy, you can apply the following actions:

Action	Description
Allow	The Proxy pass-through the matched sessions. The traffic is allowed to proceed to the next step.
Deny	The Proxy terminate matched HTTP session with an error page. More information of uploading a html file, see **Proxy Profile** > **Response Pages**.
Monitor	The Proxy produce a log to record matched HTTP session information.

Redirect	The Proxy redirect matched HTTP session to a predefined URL. Since redirection need to be performed before delivering response to client, condition of response body is not applicable in this action. You MUST configure the redirect response via Response Code and Redirect URL. The Destination URL should begin with a valid protocol (http:|https:). You SHOULD NOT select 301 as Response Code unless you exactly know what you are doing. This action produces a log. The Redirected URL can include Policy ID, Subscriber ID and/or Client IP. For example, http://www.example.com/query?pageid=12345&policy_id={{tsg_policy_id}}&user_id={{tsg_subscriber_id}}&source_ip={{tsg_client_ip}}
Replace	The Proxy Searches in a given HTTP part to Find a given string, and Replace any matches with another given string. If no match was found, the session remained untouched. For performance concerns, condition of request body and response body is not available in this action. For example, you can configure the Proxy to search in the response body of URL "www.example.com/index.html", find every "string1" and replace with "string2". This action produces a log. You can use regex to do replacement, e.g. case insensitive find "(?i)CaSEInSensitive(?-i)". You can add multiple replacement. ✎*Note that when searching and replacing content, there must be at least 512 characters of the replaced content. And the system support utf8 encoding and has the ability to replace content of different languages, including Kazakh, Russian and English.*
Hijack	The Proxy hijack a downloading file. Supported file type are img, exe, apk and html. More information of uploading a file for hijack, see **Proxy Profile** > **Hijack Files**.
Insert	The Proxy insert a "js" or "css" scripts to webpages. More details of uploading a script, see **Proxy Profile** > **Insert Scripts**.

Applications and Filters

Only two basic protocol HTTP and DoH are available for the Application in proxy policy. The following table lists different Filter fields for the two protocols and the available object type you can select for each filter field, it also shows whether each filter field support allow, deny, monitor, redirect, replace, hijack and insert action or not.

Only two basic protocol HTTP and DoH are available for the Application in proxy policy. The following table lists different Filter fields for the two protocols and the available object type you can select for each filter field, it also shows whether each filter field support allow, deny, monitor, redirect, replace, hijack and insert action or not.

Filter Type	Filter Fields Fit in Actions HTTP | DoH													
	Allow		Deny		Monitor		Redirect		Replace		Hijack		Insert	
Host	✓	✗	✓	✗	✓	✗	✓	✓	✓	✗	✓	✗	✓	✗
URL	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗
Request Header	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗
Response Header	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗
Request Content	✗	✗	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗
Response Content	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
QNAME	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗

Object type available for Filter:

Filter Type	Object Type				
	FQDN	Category	URL	HTTP Signature	Keyword
Host	✓	✓			
URL			✓		
Request Header				✓	
Response Header				✓	
Request Content					✓
Response Content					✓
QNAME	✓	✓			

The Evaluation Order for proxy policy is described as following.

Security policy is evaluated in preference to Proxy policy. Proxy manipulation requires targeted sessions are intercepted in security policies. That is to say an intercept security policy is the prerequisite for proxy policy.

For proxy policies with allow action, traffic can only hit one policy that meets the defined criteria. After a match is triggered, subsequent rules are not evaluated. Except allow action, proxy policies with other actions can be matched together only with monitor policies. This means traffic can hit a block proxy policy and a monitor proxy policy that meets the defined criteria at the same time. It is important to know the policy evaluation order to create policies accordingly.

There are three factors in evaluation of policies. They are condition, action and policy ID in sequence.

1. The ranked evaluation order for condition is:
   IP > FQDN/URL/Request Header > Request Content > Response Header >Response Content
2. The priority queue of action is:
   Allow > Block (Monitor) >Redirect (Monitor)> Replace (Monitor) > Hijack (Monitor) > Insert (Monitor)
3. Prioritize policies in descending order by policy ID.

For more details about how TSG process packet flow, please see **Appendix E TSG Packet Flow**.

Create a Proxy Policy Rule

STEP 1. Add a rule. Select **Policies** > **Proxy** and click **Create** to add a new rule. Enter a descriptive **Name** for the rule.

STEP 2. Define what action you want TSG to take for traffic that matches the rule. Select an **Action**. See Proxy Policy Actions for a description of each action.

STEP 3. Define the matching criteria for the source fields in the packet. Specify one or multiple **Source** IP Addresses/ Subscriber IDs or leave the value set to any. If you don't add entries to the list, the policy has no IP address restrictions.

STEP 4. Define the matching criteria for the destination fields in the packet. Specify single or multiple **Destination** IP Addresses or leave the value set to any.

✎*As a best practice, use address objects as the Destination Address to enable access to only specific servers or specific groups of servers especially for commonly exploited services, such as DNS and VPN.*

STEP 5. (Optional)If the Action is Redirect, specify the **Application**.

STEP 6. (Optional) Specify **Filters** as match criteria for the rule. For example, select a **Category** for **Host**. If you select a category, only web traffic will match the rule and only if the traffic is destined for that specified category.

For more details, see **Proxy Policy** > **Applications and Filters** and **Objects** > **Applications**.

STEP 7. (Optional)Specify a **Tag** or leave the value empty.

You can select one or more Policy Tags which are created from **Policies** > **Tags** previously. Optionally, you can click the plus icon to create new Tags. After you click the icon, a page will slide in on the right. Enter the **Tag Category**. Pick a **Color**. And add one or multiple **Tags**. Then you can select the tag you just created from the list.

STEP 8. Specify **Effective Devices** by choosing Device Tags or leave the value empty, which means the policy is effective on all devices by default. Device Tags are built-in labels to classify the devices according to user's requirement when TSG is deployed. Normally, the Device Tags will be classified by areas or ISPs.

STEP 9. (Optional)Enter a **Description** for the rule.

STEP 10. Select a **Schedule** or leave the value set to always. For more details, see **Policies** > **Schedules**.

✎*If you select a schedule, the schedule will determine when the policy will be enabled and effective. This means you don't have to turn the Enabled switch on. Because the policy will automatically be effective according to your specified schedule. And even if you set the policy Enabled, the policy will not be effective, if it has not reached the schedule time frame.*

STEP 11. Verify that **Log Session** is enabled if you wish to have proxy event logs. Only traffic that matches a Proxy policy rule will be logged. When **Log Session** is enabled, select one of **Log Options**, Metadata only or All. If option Metadata only is enabled, only structured logs are recorded for the proxy policy rule. While option All provides raw log files for some special log fields, such as content of HTTP request header or HTTP response content. For more details of logs, see **View and Manage Logs**.

STEP 12. Verify that **Enabled** is on if you don't have a schedule.

STEP 13. Click **OK** to save the policy rule.

STEP 14. (Optional)To verify that you have set up your policies effectively, test whether your Proxy Policy rules are being evaluated and determine which Proxy Policy rule applies to a traffic flow.

1. Select **Settings** > **Trouble Shooting** and select **Policy Verify** tab.
2. Select **Proxy Policy Match** from the **Select Test** drop-down.
3. Enter the **Client IP** and **Server IP** addresses or leave it set to default.
4. Specify the **Client Port** and **Server Port** or leave it set to default.
5. Select the **Protocol** and **APP ID** from the drop-down.

6. Click **Verify** to execute the **Proxy Policy Match** test.

STEP 15. (Optional)After the policy has been hit. View **Logs** to monitor the policy rule status and determine the effectiveness of the policy rule. Select **Logs** > **Proxy Events** and view relative information about the policy.

You can view detailed information about the policy you just created. To edit and delete the policy, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left.

You can export the contents of policies to a json file. First, search policies according to ID, Name. Then, click the Export icon on the right to download and save the file to your local folder.

You can also import policies by clicking the import icon. Only json and txt formats can be uploaded. You can take the exported file as template for import.

Select the checkbox for policies in the list and Click **Watch** at the bottom to add to Watch List. And then you can click the star icon in the bottom right and select Policy tab to view the Watch List. You can search policies by ID and Name in the list.

WAN NAT

This section describes Network Address Translation (NAT) and how to configure TSG for NAT.

The WAN NAT feature can hide and change the traffic's source and destination IP address. WAN NAT allows you to not disclose the real IP addresses of hosts that need access to public addresses.

When an internal real user visits an external service, the traffic will be steered to TSG. And the IP address will be included in TSG's IP pool, when the real user is disconnected, WAN NAT can disguise as the real user and send request to the external service for its own purposes. Thus, WAN-NAT provides larger address space. It offers a very large IP Pool to help you use Web crawler to gain insight and guide Public Opinion. And WAN NAT can reduce the blocking for web crawler. You can use WAN NAT to create Web Crawler, perform Network scan, effect verification and provide virtual service. TSG supports both source address translation (SNAT) and destination address translation (DNAT).

For SNAT, the source address is translated and thereby kept private. It is used for Web Crawler, Network scan, effect verification.

DNAT is commonly used for virtual service. TSG translates a destination address to a different destination address; for example, if you have a controlled DNS server, you can direct traffic visiting google DNS to the controlled DNS server.

Source NAT

Source NAT(SNAT) is typically used by internal users to access the Internet; the source address is translated and thereby kept private.

Perform the following tasks to configure SNAT:

STEP 1. Create an SNAT IP Pool object for the translated IP address you plan to use.

1. Select **Profiles** > **WAN NAT** > **SNAT IP Pool**, Click **Create**.
2. Add a **Name** for the IP Pool.
3. Specify a **Color**.
4. You can view **History Active IP** and select IP address from them.
5. Click **Reachability Test** to verify reachability of the IP address. It is recommended to use IP address which pass the test.

   ✎*There are 4 results for reachability test. They are reachable, unreachable, unknown, N/A. If all the data packets sent by TSG are received by server, the result is reachable. If all is not received by the server, the result is unreachable. If only parts of the data packets are received by server, the result is unknown. If the server did not respond, the result is N/A.*
6. Add a **Description**.
7. Click **OK**.

STEP 2. Create a VPN Account for the Original IP address.

1. Select **Profiles** > **WAN NAT** > **VPN Accounts**, Click **Create**.
2. Specify a **Color**.
3. Add a **VPN Name** and **Password**.
4. Click **OK**.

STEP 3. Create the NAT policy.

1. Add a **Name**.
2. Select **SNAT** Action.
3. Select the **VPN Account** you just created.

4. Select the IP Pools you just created.
5. For **Change Source IP**, if you wish to only translate IP address once, select **Never**; if you wish to translate IP address by time, select **By Time** and enter how many seconds change each time; if you wish to translate IP address by connection, select **By Connections** and enter how many connections change each time.
6. Specify **Connection Timeout**.
7. Add a **Tag** and **Description**.
8. Verify it is **Enabled**.
9. Click **OK**.

STEP 4. Verify the translation. Take packet capture from server which shows the Client IP changes.

Destination NAT

Destination NAT(DNAT) is performed on incoming packets when the firewall translates a destination address to a different destination address; for example, it translates a public destination address to a private destination address. Destination NAT also offers the option to perform port forwarding or port translation.

• Port Forwarding—Can translate a public destination address and port number to a private destination address but keeps the same port number.

• Port Translation—Can translate a public destination address and port number to a private destination address and a different port number, thus keeping the actual port number private. The port translation is configured by entering a Translate Port in the DNAT policy rule.

To translate the destination address for all packets on your TSG, as shown in the following procedure:

STEP 1. Create a VPN Account for the Translated Packet.

1. Select **Profiles** > **WAN NAT** > **VPN Accounts**, Click **Create**.
2. Specify a **Color**.
3. Add a **VPN Name** and **Password**.
4. Click **OK**.

STEP 2. Create the NAT policy.

1. Add a **Name**.
2. Select **DNAT** Action.
3. Select the **Source** and **Destination**.
4. Select the **VPN Account** you just created.
5. Specify the **Translate Port** or leave it set to default.
6. Add a **Tag** and **Description**.
7. Verify it is **Enabled**.
8. Click **OK**.

Schedules

Schedules is the time frame that is applied to the policy or report. Schedules allow you to control the time period for which security rules and proxy rules are in effect. This can be something as simple as a time range that the sessions are allowed to start such as between 8:00 am and 5:00 pm. Something more complex like business hours that include a break for lunch and time of the session's initiation may need assign multiple schedules to a policy because it will require multiple time ranges. You can then apply these schedules to the rules and reports.

Types of Schedules:

• Recurring: Recur daily, weekly or monthly during the specified time periods.

• One-time: Policy or report is effective once during the specified days and time period. You can apply one-time schedules to control policies or reports related to one-time events.

The schedule page displays the full list of predefined and custom schedules. To view the policies to which a schedule is attached, to apply a schedule to a policy or to change the applied schedule, go to the corresponding policy Edit page.

✎*You cannot delete a schedule that is currently applied to a policy. To do so, you must apply a different schedule to the policy or delete the policy itself.*

Perform the following to create a schedule.

STEP 1. Select **Policies** > **Schedules** and click create.
STEP 2. Add a **Name**.

STEP 3. Select **Repeat** mode from Daily, Weekly, Monthly and one-time. You can customize recurring mode.

STEP 4. Enter **Start Date**, **End Date**, **Start Time** and **End Time**.

STEP 5. Click **OK**.

Verify Policy Rules

Verify the policy rules in your running configuration to ensure that your policies appropriately allow and deny traffic and access to applications and websites in compliance with your business needs and requirements.

You can test and verify that your policy rules by executing policy match tests for your TSG directly from the web interface.

For example, a Proxy Policy has been defined with ID 4736, Name tsg-test-pxyverify. And the basic information about this policy is listed as following.

Source	Destination	Application	Filter	
192.168.50.71 0-65535	Any	HTTP	HTTP	www.vip.com www.json.cn index.com
			URL	microsoft.com
			Response Header	beer cute&nice&easy love&beauty

STEP 1. Logging into the Web Interface.

STEP 2. Select **System** > **Trouble Shooting** > **Policy Verify** to perform a policy match or connectivity test.

STEP 3. Enter the required information to perform the policy match test. In this example, just run a **Proxy Policy Match**.

1. Select **Proxy Policy Match** from the Select Test drop-down.
2. Enter the **Client IP** and **Server IP** addresses or leave it set to default. If you wish to match your specified policy, enter at least one Client IP that is in Source IP set and enter at least one Server IP that is in Destination IP set. If the Source or Destination is any in your policy, leave the corresponding Client IP or Server IP set to default. Here fill in Client IP 192.168.50.71.
3. Specify the **Client Port** and **Server Port** or leave it set to default.

4. Select the **Protocol** from Any, TCP and UDP.
5. Specify **Subscriber ID** if your specified policy references Subscriber ID object.
6. Select **APP ID** from the Applications side page, and you can search your application by name. Here select HTTP.
7. If you selected one of HTTP, SSL, DNS, MAIL, FTP and QUIC as **APP ID**, you need to add corresponding additional filter for your policy rule testing. And you need to fill in all the filter fields defined in the policy, but you can add only one item for each field. For example, click plus, and add "www.vip.com" for Host; click plus, and add "microsoft.com" for URL; click plus and add "beer" for Response Header.

STEP 4. Click **Verify** to execute the **Proxy policy match** test.

STEP 5. Review the **Policy Match** and **Object Match** result to see the policy rules.

Only when all conditions are matched, you can find the matched policy. The evaluation order for policy verify is the same as executing policies.

Objects

A policy object consists of one item or a set of collective items that groups discrete identities such as IP addresses, URLs, applications, or accounts. One policy object is allowed to reference same type objects as subordinate objects. Typically, when creating a policy object, you group objects that require similar permissions in policy. For example, you can group the set of server IP addresses as an address group policy object and reference the address group in the security policy. By grouping objects, you can significantly reduce the administrative overhead in creating policies. An object group is also considered as an object when referenced. You can reference the object group in policy instead of manually selecting multiple objects one at a time.

> Objects Type
> IP Addresses
> Subscriber IDs
> Categories
> Applications
> Configure Object Group

You can create an object or create an object group. A policy object consists of one or multiple items, while an object group is composed of one or multiple subordinate objects. One object is allowed to reference same type objects as subordinate objects.

To view object usages, please click column **Reference Count** of object list.

Then you will see
- Various policies for referencing the object;
- The group object that references the object.

Click the Graph button, you will see an object relationship graph. Policies, Proxy TCP Options, and parent objects that reference current object are displayed in the graph. As well as application signatures, which reference IP Address object when creating signatures with ip.src and ip.dst as Traffic Attribute. For more details, see **Advanced Setting** > **Proxy TCP Option** and **Objects** > **Applications** > **Signatures**.

Click the solid circle to unfold the referenced ancient object and click the hollow circle to fold the picture.

✎*Note that direct or indirect self-reference is prohibited, i.e. A->A or A->B->A. Within the TSG system, object references can have up to six levels (from root node to leaf node).*

You can reference objects and object groups in your policies. Thus, you reduce the administrative overhead in creating policies. You can identify an object by its name or ID number. The object ID never changes even if you modify the object, such as when you change the object name.

Objects Type

You can create the following policy objects on TSG. A policy object consists of one or multiple items, while an object group is composed of one or multiple subordinate objects.

Policy Object	Description

IP Addresses/ Address Group	IP Address contains three sub-types: IP, geography and IP Learning. The IP object can include an IPv4 or IPv6 address (single IP, range). Address Group Allow you to group specific source or destination addresses that require the same policy enforcement. You can then group a collection of address objects of the same type to create an address object group. IP learning type is not allowed to create group.
FQDNs/ FQDN Group	That is fully qualified domain name to identify traffic. Using an FQDN object or FQDN object group reduces issues in environments where the host is subject to dynamic IP address changes. Support exactly matching and suffix matching.
Subscriber IDs/ Subscriber ID Group	Allow you to create a list of Subscriber ID for RADIUS traffic. Support exactly matching only.
HTTP Signatures/ HTTP Signature Group	Allow you to add keyword in Request as User-Agent and Cookie, in Response as Set-Cookie and Content-Type. Support exactly matching, prefix matching, suffix matching and substring matching.
Keywords/ Keyword Group	A string you define that can be added as a filter in policy. You can enable Hex Mode. Support exactly matching, prefix matching, suffix matching and substring matching. ✎*Support maximum 8 substrings for AND expression. And when adding keyword for other objects, the same rule applies.*
URLs/ URL Group	A Uniform Resource Locator, colloquially termed a web address, is a reference to a web resource that specifies its location on a computer network and a mechanism for retrieving it. A typical URL could have the form http://www.example.com/index.html, which indicates a protocol (http), a hostname (www.example.com), and a file name (index.html). Here the protocol is not allowed when adding. Support exactly matching, prefix matching, suffix matching, and substring matching

Categories/ Category Group	Category classifies websites based on site content, features, and safety. Once created, the category can be selected as a filter of a policy. This means that a policy will only allow or block requests that match the category. For details, please refer to Categories.
Accounts/ Account Group	Stores the account information for your application. For example, you can add your email account as a filter when creating a policy using MAIL application. Support exactly matching, prefix matching, suffix matching and substring matching.
Mobile Identities/ Mobile Identity Group	Consists of IMSI and Phone Number. Both are string type, composed of decimal numbers with maximum 15 digits. IMSI only supports prefix matching. Phone Number supports exactly matching, prefix matching, suffix matching and substring matching.
APNs/ APN Group	Access Point Name of GTP users. String type with format like FQDN. Support exactly matching and suffix matching.
Applications	Applications, a patented traffic classification system available in TSG firewalls, determines what an application is irrespective of port, protocol, encryption or any other evasive tactic used by the application. It applies multiple classification mechanisms—application signatures, application protocol decoding, and heuristics—to your network traffic stream to accurately identify applications. An Application allows you to filter applications dynamically.

✎*Description of matching method:*

• *Substring matching, which has no wildcard character. E.g. "abc" is a substring of "abcde".*

• *Prefix matching, which is end with an asterisk character "*". E.g. "abc*" is a prefix of "abcde".*

• *Suffix matching, which is begin with an asterisk character "*". E.g. "*cde" is a suffix of "abcde".*

• *Exactly matching, which is begin with a dollar character "$". E.g. "$abcde" is exactly same as "abcde".*

• *AND expression, which contains maximum 4 substrings. E.g. "abc" & "cde" matches "abcde".*

You could input nonprintable keyword in Hex mode. A valid hexadecimal keyword only contains "0 9" and "a f", and has even character number, e.g. "1a2b3c4d".

You can perform the following to create an object.

STEP 1. Select **Objects**. Select the object menu you wish to create, and click **Create**.

STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.

STEP 3. Specify a **Color**.

STEP 4. Add an **Item**.

STEP 5. (Optional)Enter a **Description**. The description can have up to 1024 characters.

STEP 6. Click **OK**.

You can view detailed information about the object you just created. To edit and delete the object, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left.

You can export the contents of objects to a txt or csv file. First, search objects according to ID, Name, Keywords, Sub Object (ID), Description, Operator, Time and other conditions. Then, Click the Export icon on the right to download and save the file to your local folder.

You can also import objects by clicking the import icon. Only csv and txt formats can be uploaded. Duplicated items are automatically omitted when you import objects. You can take the exported file as template for import. In addition, the exported file of object can also be used to back up the object of the current system. After export, it can be directly imported into the same version (or the official version of TSG compatible with the exported version).

✎*The TSG system only provides the export of objects with items, but object group with subordinate object are not allowed to be exported.*

TSG allows searching objects based on ID, Name, Description, Operator, Time etc.

Select the checkbox for objects in the list and Click **Watch** at the bottom to add to Watch List. And then you can click the star icon in the bottom right and select Object tab to view the Watch List. You can search objects by ID and Name in the list.

IP Addresses

An address object is a set of IP addresses that you can manage in one place and then use in multiple policy rules. You can reference the same address object in multiple policy rules without needing to specify the same individual addresses in each use. Furthermore, create an address object on TSG to group IP addresses, and then reference the address object in a policy rule to avoid having to individually specify multiple IP addresses in the rule. For example, you can create an address object that specifies an IPv4 address range and then reference the address object in a Security policy rule.

There are three Sub Types of address object: IP, Geography and IP Learning. IP Sub Type include an IPv4 or IPv6 address (single IP, range). Geography are IP addresses organized by geographical scope. You can select a country or a city as an item. For more details, please see **IP Libraries**. IP Learning can learn from FQDNs whose host IP addresses change frequently.

✎*At present, the system supports geography selection at both national and urban levels.*

Initially, the IP Learning object is empty and contains no addresses. When the client tries to resolve a FQDN address, TSG will analyze the DNS response. The IP address(es) contained in the answer section of the DNS response will be added to the corresponding IP Learning object. When the IP Learning gets the resolved IP addresses, TSG loads the addresses into policy for traffic matching. At any given time, a single IP Learning object may have up to 10000 IP addresses.

You can perform the following to create an IP object:

STEP 1. Select **Objects** > **IP Addresses** and click **Create** and select Address.

STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.

STEP 3. Specify a **Color**.

STEP 4. Select a sub **Type**. There are three sub Types, namely: IP, Geography, and IP Learning.

STEP 5. (Optional)If you select **IP** as your sub Type, you can add one or more items, you can add IPv4, IPv6 and IP range. Also, add port, port range or leave the value set to empty.

STEP 6. (Optional)If you select **Geography** as your sub Type, select one or more countries and cities as your Item.

STEP 7. (Optional)If you select **IP Learning** as your sub Type, you need to fill in the following fields.

1. Add one or more FQDNs for **Learn from FQDNs**.
2. Select HTTP, SSL from drop-down for **Learn from Protocols**.
3. Select 1 Degree or 2 Degrees for **Learning Depth**.
4. Specify **Aging Times**. IP domain name learning after a certain period of time, the value density will be reduced. That is to say, the IP addresses obtain between the first discovery and the last discovery of the service IP returned by the domain name. Aging Times Cannot be 0, exceed maximum 2147483647, and be empty.
5. Specify **Vote Clients Number**. Total number of independent client IP supporting IP learning. Cannot be 0, exceed maximum 10000, and be empty.
6. Specify **Learned IP limits**. Maximum number of IP addresses to learn. Cannot be 0, exceed maximum 10000, and be empty.

STEP 8. (Optional)Enter a **Description** or leave the value set to empty. Description can have up to 1024 characters.

STEP 9. Click **OK**.

IP Libraries

IP Libraries map geographic locations to IP addresses. TSG provides built-in IP libraries, so you can select Geography Sub Type when creating IP Address object. You can also import your own geography and IP addresses mappings.

The following steps guide you to Create Geography:

STEP 1. Select **System** > **IP Libraries**, and click **Create**.

STEP 2. Create Geography.

1. Select geography **Type** between Country and City, if you select Country, you need to select **Continent** field. Here, select City as an example.
2. Select **Country** from slide page Geographic Locations.
3. Add **Geo Name ID**.
4. Specify **City**. The City name is case-sensitive and can have up to 128 characters.
5. Add **IP Range**. Take Format IPv4 Range x.x.x.x-y.y.y.y, IPv6 Range x:x:x::x-y:y:y::y as an example.
6. (Optional)Specify **Longitude** and **Latitude**.

7. Click **OK**.

You can **Edit** or **Delete** imported Geography. When editing built-in geography, please operate with caution.

Subscriber IDs

You can create Subscriber ID to keep track of Radius traffic user. After you create Subscriber ID object, you can use it in your policy rule and Active Subscriber ID will be shown in your dashboard.

You can perform the following to create a Subscriber ID:

STEP 1. Select **Objects** > **Subscriber IDs** and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.
STEP 3. Specify a **Color**.
STEP 4. Add an **Item**. You can view **History Subscriber** for your reference. Subscriber ID must be exactly matching. E.g. $test62.
STEP 5. (Optional)Enter a **Description**. The description can have up to 1024 characters.
STEP 6. Click **OK**.

Categories

Category classifies websites based on site content, features, safety and so on. TSG firewall has built-in categories.TSG allows users to create user-defined categories. One FQDN may belong to multiple categories.

Please refer to Appendix A Built-in Category for more details.

Applications

An application is any program, or group of programs, that is designed for the end user to perform an activity. Applications enables visibility into the applications on the network, so you can category them and understand their characteristics and their relative risk. This application knowledge allows you to create and enforce security

policy rules to allow and inspect applications and deny unwanted applications. When you use policy rules to control traffic, applications can classify traffic without any additional configuration.

AppSketch is a traffic classification system available in TSG firewalls, determines what an application is irrespective of port, protocol, encryption or any other evasive tactic. It applies multiple classification mechanisms to your network traffic stream to accurately identify applications. These classification mechanisms include application signatures, application protocol decoding, and heuristics.

The firewall identifies application with predefined and customized signature. The TSG firewall uses protocol decoding in the content inspection stage to determine one application from the other. After the firewall identifies the session application, security policy can be enforced as configured. The identified application as well as IP, port, protocol, Subscriber ID, FQDN and URL in the session is used as key to find rule match.

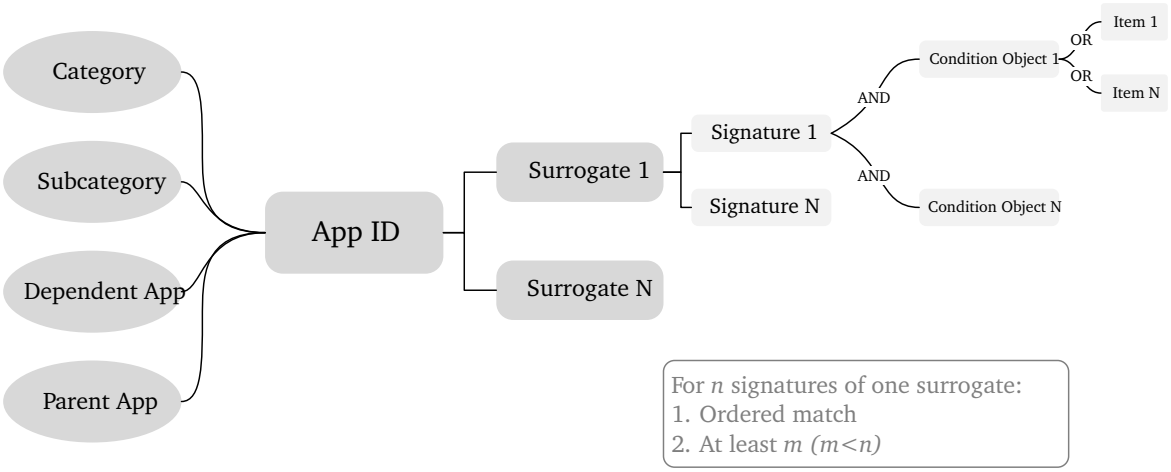
When creating a security policy, there are built-in protocols and well-known Applications and customized Applications in the list. You can search the application you want to fill in. You can also use application selector and group as objects in policy. TSG reports enable you to show statistics about bytes sent and received based on Application Label and IP address. See **Monitoring** > **View and Manage Reports** for details.

Signatures

In TSG, application is composed of App ID, Properties and Signature Sequence. App ID is the unique identification of Application. Application Properties include Category, Subcategory, Technology, Risk and Characteristics. You can create Application Selector based on application Properties. Signature refers to the expression of network traffic attributes in a specific scope. Traffic Attribute is a piece of information which obtained from network transfer unit. Signature Sequence is the signatures of the application that appear in a certain order. There is a sequential "and" relationship between signatures.

The following demonstrates how to create a customized signature.

STEP 1. Select **Objects** > **Applications**, select tab **Signatures** and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.
STEP 3. Specify a **Color**.



STEP 4. (Optional)Enter a **Comment**.

STEP 5. Add **Conditions**. You can add one or multiple conditions. The relation between New Conditions is "and", and the relation within existing condition is "or". Select **Attribute Name** and this affects the rest of the available selections. Fill in the corresponding content. Note that valid keywords length is from 4 to 1024 bytes.

STEP 6. Click **OK**.

✎*Within the same signature, attributes from different protocols are not allowed to serve as Conditions, except for TCP/IP/General Attributes.*

You can **Edit** or **Delete** your signature and reference one or multiple signatures when creating application object. You can also import or export user-defined signatures in json format.

Customized Attributes

The traffic attribute is the information obtained after the analysis of the network transmission unit. The attributes used by the App recognition can be found in **Appendix F Best Practices** > **Custom Application**.

You can also upload a Lua script to create your own traffic attributes.

The following is a basic example of how to create a customized attribute.

STEP 1. Select **Objects** > **Applications**, select tab **Customized Attributes** and click **Create**.

STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.

STEP 3. Select **Parent Attribute** from traffic attribute list.

STEP 4. Please upload **Uploaded File**.
STEP 5. Select **Attribute Type** from: Bool, Numeric Value, and String.
STEP 6. Specify **Maximum Execution Time**.
STEP 7. Click **OK**.

✎*Lua is a lightweight, high-level, multi-paradigm programming language designed primarily for embedded use in applications. TSG is able to interact with Lua scripts when process network traffic.*

✎*It is best practice to construct a Pre-Signature to improve performance when a customized attribute is referenced by Signature as a Condition.*

You can **Edit** or **Delete** your customized attributes and download the Uploaded File.

Predefined Applications

TSG supports a variety of built-in protocols and Applications. When the application is identified, the policy check determines how to treat the application. You can view all the predefined applications in the web interface.

The following table list some examples of system built-in applications:

App Name	App ID	Category	Subcategory	Description
twitter	503	general-internet	internet-utility	Online microblogging service that enables its users to read and send text-based short messages.
teamviewer	545	networking	remote-access	TeamViewer is an application that enables a connection to a remote computer in order to perform maintenance operations. It is also possible to show the current display to a remote computer, to transfer files, and to create a VPN tunnel.
archive	564	general-internet	internet-utility	Archive.org is the site of the Internet Archive, a non-profit digital library offering free universal access to books, movies & music, as well as more than 300 billion archived web pages.

qqlive	585	media	photo-video	QQLive is an application intended to watch TV in Peer-to-Peer mode. QQlive also classifies QQ live streaming on web browser
tango	598	collaboration	voip-video	Tango is an embedded smartphone application dedicated to audio/video-conference.

Application Customization

Applications allow you classify all traffic, across all ports, all the time. To ensure that your internal custom applications do not show up as unknown traffic, you can create a custom application. Then practice granular policy control over these applications to minimize the range of unidentified traffic on your network.

To create a custom application, you must define the application attributes: its characteristics, category and sub-category, risk, port, timeout. In addition, you must define patterns or values that TSG can use to match to the traffic flows (the signature). Finally, you can attach the custom application to a policy that allows or denies the application (or add it to an application group or match it to an application selector).

✎*In order to collect the right data to create a custom application signature, you'll need a good understanding of packet captures and how to analyze data pattern. If the signature is created too broadly, you might inadvertently include other similar traffic; if it is defined too narrowly, the traffic will evade detection if it does not strictly match the pattern.*

The following is a basic example of how to create a custom application.

STEP 1. Gather information about the application to create custom signatures.

To do this, you need have an understanding of the application and how you want to control access to it. For example, you may want to limit what operations users can perform within the application, such as uploading, downloading, or live streaming.

• Capture application packets so that you can find unique characteristics about the application on which to base your custom application signature. One way to do this is to run a protocol analyzer, such as Wireshark, on the client system to capture the packets between the client and the server. Perform different

actions in the application, such as uploading and downloading, so that you will be able to locate each type of session in the resulting packet captures (PCAPs).

• Because TSG supports packet captures for all traffic, you can take packet captures using TSG. See Take Packet Captures.

• Use the packet captures to find patterns or values in the packet contexts that you can use to create signatures that will uniquely match the application. For example, look for string patterns in HTTP request or response headers, URI, or hostnames.

STEP 2. Add the custom application.

1. Select **Objects** > **Applications** and click **Create**.
2. Enter a **Name** and a **Description** for the custom application that will help other administrators understand why you created the application.
3. Verify that **Enabled** is enabled.

   Policy rules referencing applications only match to and enforce traffic based on enabled applications.

   Predefined applications cannot be disabled and only allow a status of enabled. Disabling a base application could cause applications which depend on the base application to also be disabled. For example, disabling Facebook-base will disable all other Facebook applications.
4. Define the application Properties and Characteristics.

   Select the **Category**, **Subcategory**, **Technology** and **Risk** from drop-down. Add **Parent App** if there is any. Enable **Continue Scanning** if you need to. Select the checkbox for characteristics, including Evasive, Excessive Bandwidth, Prone to Misuse, SaaS, Transfer Files, Tunnels Other Apps, Used by Malware, Vulnerability and Widely Used.
5. Define the timeout values or leave the value set to default.

STEP 3. Define the Surrogates which is the criteria that the firewall will use to match the traffic to the new application.

You will use the information you gathered from the packet captures to specify unique string context values that the firewall can use to match patterns in the application traffic.

Select Signatures or click plus icon to create a signature.

1. On create a Signature page, define a Signature **Name** and optionally a **Comment** to provide information about how you intend to use this signature.
2. Pick a **Color** or use the default color.

3. Specify **Conditions** to define signatures.

   If the order in which the firewall attempts to match the signature definitions is important, make sure to enable the Ordered Match and then order the conditions so that they are evaluated in the appropriate order. Select a condition and click Move Up or Move Down.

STEP 4. Click **OK**.

STEP 5. Validate that traffic matches the custom application as expected.

1. Select **Policies** > **Security** and **Create** a security policy rule to allow the new application.
2. Run the application from a client system that is between the firewall and the application and then check the logs to make sure that you see traffic matching the new application (and that it is being handled per your policy rule).

✎*TSG enables you to import or export custom applications in batch with json format.*

Application Selector

An application selector is an object that dynamically groups applications based on application attributes that you define, including category, subcategory, technology, risk and characteristics. This is useful when you want to enable access to applications that you do not explicitly sanction, but that you want users to be able to access. For example, you may want to enable employees to choose their own office programs, such as Evernote, Google Docs, or Microsoft Office, for business use. To enable these types of applications, you could create an application selector that matches on the Category business-systems and the Subcategory office-programs. As new applications office programs emerge, these new applications will automatically match the selector you defined; you don't have to make any additional changes to your policy rules to enable any application that matches the attributes you defined for the selector.

STEP 1. Select **Objects** > **Applications** > **Selectors**.

STEP 2. Create a selector and give it a descriptive **Name**.

STEP 3. Define the selector by selecting attribute values from the Category, Subcategory, Technology, Risk, and Characteristics sections. As you select values, notice that the list of matching applications at the bottom of the dialog narrows. When you have adjusted the filter attributes to match the types of applications you want to safely enable, click **OK**.

Application Group

An application group is an object that contains applications that you want to treat similarly in a policy. Application groups are useful for allow or deny access to applications that you explicitly sanction or forbid. Grouping sanctioned or forbidden applications simplifies administration of your rules. Instead of having to update individual policy rules when there is a change in the applications you sanction or deny, you can update only the affected application groups.

When deciding how to group applications, consider how you plan to enforce access to your applications and create an application group that aligns with each of your policy goals. For example, you might have some applications that you will allow, and other applications that you want to deny. In this case, you would create separate application groups for each of these policy goals.

STEP 1. Select **Objects** > **Applications** > **Groups**.
STEP 2. Create a group and give it a descriptive **Name**.
STEP 3. Add the applications you want in the group.
STEP 4. (Optional) Enter a **Description** or leave the value set to empty. Description can have up to 1024 characters.
STEP 5. Click **OK**.

Configure Object Group

A policy object consists of one or multiple items, while an object group is composed of one or multiple subordinate objects. An object group is also considered as an object. Typically, when creating a policy object, you organize objects that require similar permissions in policy. One object is allowed to reference same type objects as subordinate objects, but not allowed to add items in object. For example, An IP object defines a set of single address, whereas an IP object group can define more than one address object. By grouping objects, you can significantly reduce the administrative overhead in creating policies. You can create object group for all types of objects.

The following procedure explains how you can create objects group directly through the Objects page.

STEP 1. Select **Objects**, then select the menu you want to create from the following: IP addresses, FQDNs, Subscriber IDs, HTTP Signatures, Keywords, URLs, Categories, Accounts.

STEP 2. To create a group, let's take FQDN as an example, click **Create** and select FQDN group.

STEP 3. Add a **Name**. The name is case-sensitive and can have up to 128 characters.

STEP 4. Specify a **Color**.

STEP 5. Add one or more **Sub Objects**. Note that you cannot add sub objects and items at the same time.

STEP 6. (Optional)Enter a **Description**. The description can have up to 1024 characters.

STEP 7. Click **OK**.

Decryption

Except firewall, TSG has a proxy which utilizes MITM (Man-in-the-middle) technologies and enables you to perform layer 4-7 advanced manipulation of network traffic. The Proxy is deployed in transparent mode; thus, no proxy settings on browser side. The proxy can decrypt and inspect traffic to control protocols and certificate verification. The proxy handles encrypted traffic according to your configured security settings. Traffic will be reconstructed according to the TCP/IP protocol stack with the original headers (source IP, source Port, destination IP, destination Port, Protocol, etc.) and decrypted payload. Decryption prevents malicious encrypted content from entering your network and sensitive content from leaving your network concealed as encrypted traffic. Enabling decryption need preparing the keys and certificates required, creating decryption profiles and configuring traffic mirror profile.

> Decryption Concepts
> Keys and Certificates
> Certificate Managements
> Proxy Profiles

Decryption Concepts

The Secure Sockets Layer (SSL) and Transport Layer Security (TLS) encryption protocols secure traffic between two entities, such as a web server and a client. Without special instructions, SSL in this document refers to SSL/TLS. SSL encapsulate traffic, encrypting data so that it is meaningless to entities other than the client and server with the certificates to affirm trust between the devices and the keys to decode the data.

The proxy uses certificates and keys to decrypt traffic to plaintext, and then enforces security settings on the plaintext traffic. After decrypting and inspecting traffic, the proxy re-encrypts the plaintext traffic as it exits the proxy to ensure privacy and security.

SSL decryption requires certificates to establish the proxy as a trusted third party, and to establish trust between a client and a server to secure an SSL/TLS connection. You can also use certificates when excluding servers from SSL decryption for technical reasons (the site breaks decryption for reasons such as certificate pinning, unsupported ciphers, or mutual authentication).

You can integrate a hardware security module (HSM) with TSG to enable enhanced security for the private keys. To learn more about integrating an HSM, see Manage Keys with a Hardware Security Module.

Keys and Certificates

Keys are strings of numbers typically generated using a mathematical operation involving random numbers and large primes. Keys transform strings—such as passwords and shared secrets—from unencrypted plaintext to encrypted ciphertext and from encrypted ciphertext to unencrypted plaintext. Keys can be symmetric (the same key is used to encrypt and decrypt) or asymmetric (one key is used for encryption and a mathematically related key is used for decryption). Any system can generate a key.

X.509 certificates establish trust between a client and a server to establish an SSL connection. A client attempting to authenticate a server (or a server authenticating a client) knows the structure of the X.509 certificate and therefore knows how to extract identifying information about the server from fields within the certificate, such as the FQDN or IP address (called a common name or CN within the certificate)

or the name of the organization, or user to which the certificate was issued. A certificate authority (CA) must issue all certificates. After the CA verifies a client or server, the CA issues the certificate and signs it with a private key.

When you decrypt traffic, a session between the client and the server is established only if the firewall trusts the CA that signed the server certificate. In order to establish trust, the firewall must have the server root CA certificate in its Trusted Certificate Authorities list and use the public key contained in that root CA certificate to verify the signature. The firewall then presents a copy of the server certificate for the client to authenticate. You can also configure the firewall to use an enterprise CA for SSL Proxy. If the firewall does not have the server root CA certificate in its Trusted Certificate Authorities list, the firewall will present a copy of the server certificate signed by the Forward Untrust certificate to the client. The Forward Untrust certificate ensures that clients are prompted with a certificate warning when attempting to access sites hosted by a server with untrusted certificates.

The change of the static expiration time for new generated certificates will not cause a service shutdown or other problems that negatively affect the decryption.

TSG allows you to delete installed certificates, including default certificate, change global and default certificate; but if the certificate is already referenced by a policy, TSG does not allow deleting it. You can modify the referenced certificate instead. For detailed information on certificates, see Certificate Management.

Certificate Managements

The digital certificates are used to ensure trust between parties in a secure communication session. Each certificate contains a cryptographic key to encrypt plaintext or decrypt ciphertext. Each certificate also includes a digital signature to authenticate the identity of the issuer. The issuer must be in the list of trusted certificate authorities (CAs) of the authenticating party.

Trusted Certificate Authorities

TSG trusts the most common and trusted authorities (CAs) by default. These trusted certificate providers are responsible for issuing the certificates TSG requires to secure connections to the internet. The additional CAs you might want to add are trusted

enterprise CAs that your organization requires. You can perform the following to import a certificate:

STEP 1. Select **Profiles** > **Decryption** and select **Trusted Certificate Authorities** tab, Click **Import**.

STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters. It can use only letters, numbers, hyphens, and underscores.

STEP 3. Click **Please upload** and upload a PEM (base64-encoded) format file.

STEP 4. Click **OK**.

Go back to Trusted Certificate Authorities tab, you can view detailed information about the CA you just created. To edit and delete CAs, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. The system will periodically check whether the CA certificate has expired. If the CA certificate expires, the system will automatically set the status of the CA certificate to disable. To download it, you can click the cloud icon under **File**, and wait a few seconds for the file to be downloaded to your local folder. You can search CAs based on ID, Name, Issuer, Common Name and Certificate Fingerprint, or the combination. Enter search conditions in search bar and click search icon.

Decryption Keyrings

If your enterprise has its own public key infrastructure (PKI), you can import a certificate and private key into TSG from your enterprise certificate authority (CA). Enterprise CA certificates (unlike most certificates purchased from a trusted, third-party CA) can automatically issue CA certificates for applications such as SSL/TLS decryption.

Note that the built-in certificate with ID 1(#1) means trusted certificate, and built-in certificate with ID 0(#0) means untrusted certificate. You can add trusted certificate to TSG with two methods. One is local management with TSG interface through the following procedure; the other is integrate an external HSM device, the certificate will be saved to the HSM for specified website. For more details about HSM, see Manage Keys with a Hardware Security Module.

✎*If the HSM is down, the firewall can process decryption for sites of HSM mode for which it has cached the response from the HSM, meanwhile the firewall will deploy default certificates (#0 or #1) for those un-cached sites of HSM mode.*

You can perform the following to Import a Certificate and Private Key:

STEP 1. Select **Profiles** > **Decryption** > **SSL Decryption Keyrings**, Click **Create**

STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.

STEP 3. **Please Upload** a **Certificate**. For Intermediate CA, certificate must be a complete chain.

STEP 4. **Please Upload** a **Private Key File** separately. It supports PEM (base64-encoded) format only. If you have your digital keys stored in HSM, please select **HSM**, and fill in **Slot ID**.

STEP 5. Enter customized **Reissue Expiry Hours** or select Mirror Server Certificate.

STEP 6. Select a **Type** from Root Certificate, Intermediate Certificate and End-entity.

STEP 7. Select **Public Key Algorithm** from RSA 1024, RSA 2048, SECP 256r1 and SECP 384r1.

STEP 8. Enter **Certificate Revocation List** address or leave the value set to empty.

STEP 9. Enable **Include root in client-side certificate chain** if you wish to.

Go back to Decryption Keyrings tab, you can view detailed information about the Keyrings you just created. To edit and delete Keyrings, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. To download it, you can click the cloud icon under **Private Key** and **Certificate**, and wait a few seconds for the file to be downloaded to your local folder. You can search Keyrings based on ID and Name. Enter search conditions in search bar and click search icon.

Manage Keys with a Hardware Security Module

A hardware security module (HSM) is a physical device that manages digital keys. An HSM provides secure storage and generation of digital keys. It provides both logical and physical protection of these materials from non-authorized use and potential adversaries. HSM clients integrated with TSG enable enhanced security for the private keys used in SSL/TLS decryption.

You can integrate Hardware Security Module (HSM) device on TSG and reference it in Decryption Keyrings.

You can integrate an HSM device by the following procedure.

STEP 1. Select **Devices** > **HSM** and click **Create**.

STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.

STEP 3. Select **HSM Server Type**. For now, only support CERTEX HSM.

STEP 4. Specify **Server IP** and **Partition Password**. Password only allow English letters, numbers, underscore _, minus sign , English dot (.) and its minimum 6 bits, maximum 16 bits.

STEP 5. Click **Reachability Test** to know the status of HSM.
STEP 6. Select **Data Center** of the HSM.
STEP 7. Click **OK**.

SSL Decryption Exclusion

SSL Decryption Exclusion can exclude two types of traffic from decryption:

• Traffic that breaks decryption for technical reasons, such as using a pinned certificate, unsupported ciphers, or mutual authentication (decrypting blocks the traffic). If you encounter sites that break decryption technically and are not on the SSL Decryption Exclusion list, you can add them to list manually by server hostname.

• Traffic that you choose not to decrypt because of business, regulatory, personal, or other reasons, such as financial-services, health-and-medicine, or government traffic. You can choose to exclude traffic based on FQDN.

✎*To increase visibility into traffic and reduce the attack surface as much as possible, don't make decryption exceptions unless you must.*

Perform the following to exclude a Server from Decryption:

STEP 1. Select **Profiles** > **Decryption** > **SSL Decryption Exclusion**, Click **Create**.
STEP 2. Enter an **FQDN**, it supports suffix matching and exactly matching only. E.g. *.example.com, $www.example.com.
STEP 3. Enter a **Description**. The description can have up to 255 characters.
STEP 4. Click **OK**.

If you create an SSL Decryption Exclusion entry, actually TSG will create an FQDN object which contains only one item. But this FQDN object can only be seen in SSL Decryption Exclusion and will be referenced in TSG built-in Policy ID 1.

Go back to SSL Decryption Exclusion tab, you can view detailed information about the SSL Decryption Exclusion list you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. You can search exclusion list based on ID and Name. Enter search conditions in search bar and click search icon.

Cached Intermediate Certificates

TSG will automatically cache intermediate certificates. You can select **Profiles** > **Decryption** and select **Cached Intermediate Certificates** to view detailed information about these Intermediate Certificates. These Intermediate Certificates are issued by Trusted Certificate Authorities, which is an effort to amend the incomplete certificate chain. TSG will collect the following information: source website, issuer by, issuer to, Cn, and expiry date.

To download it, you can click the icon under **File**, and wait a few seconds for the file to be downloaded to your local folder. You can also enable and disable it by clicking the switch under **Enabled**. The system will periodically check whether the intermediate certificate has expired. If the intermediate certificate expires, the system will automatically set the status of the intermediate certificate to disable. You can search intermediate certificates based on ID, Source Website, Issuer, Common Name and Certificate Fingerprint, or the combination. Enter search conditions in search bar and click search icon.

SSL Fingerprint

With the improvement of people's security awareness, more and more apps support Pinning. And JA3 fingerprinting is no longer a luxury and is a hard requirement. You can use shared JA3 hash across the network to help accurately identify Pinning applications and then configure the app to Dynamic Bypass or not accordingly in TSG. It can mean the difference between a rapid response and a missed detection.

It is difficult to collect JA3 hash for Pinning Apps, but as a traffic inspection device, TSG can determine exactly which apps are Not Pinning. And It is relatively easy to collect JA3 hash for Not Pinning Apps. Over time, more and more JA3 hashes of Not Pinning have been collected. If an SSL connection exhibits Pinning characteristics and is not included in the collected JA3 hash of Not Pinning, it is more accurate than ever to tell the APP is Pinning. Under the circumstances, the Dynamic Bypass is recommended in profile. If an SSL connection exhibits Pinning characteristics and is included in the collected JA3 hash of Not Pinning, it is more accurate than ever to tell the APP is a browser without installed root certificate. Under the circumstances, the Intercept is recommended in profile. To configure Dynamic Bypass or Intercept for Pinning Apps, see Decryption Profile.

The overall process is as follows:

1. The firewall collects JA3 hash through security event logs or session records. And you can view SSL.JA3 hash field and query by it in the specific logs page.
2. Analyze JA3 hash through TSG report. For example, you can create the following JA3 hash related reports.
   a) In session records, perform top N statistics on JA3 hash according to different SNI numbers.
   b) In security event logs, perform top N statistics on SNI according to the number of unique JA3 hash.
   c) In security event logs, perform top N on the combination of JA3 hash and SNI according to the bytes transmitted.
   d) In security event logs or session records, according to the number of unique JA3 hash, perform top N statistics on Client IP or Internal IP.
3. TSG administrator imports JA3 hashes that meet the requirements in the report analysis result into the DB through TSG interface.
4. JA3 hash is synchronized from DB to Redis. Redis delivers JA3 hash to Proxy. And Proxy uses JA3 hash for Pinning identification.

Perform the following steps to create an SSL fingerprint:

STEP 1. Select **Profiles** > **Decryption** > **SSL Fingerprint**, and click **Create**.
STEP 2. Add a **JA3 Hash**.
STEP 3. Select Yes or No for **Pinning**.
STEP 4. (Optional)Enter a **Description**. The description can have up to 1024 characters.
STEP 5. Click **OK**.

Go back to SSL Fingerprint tab, you can view detailed information about the SSL Fingerprint list you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. You can search fingerprint list based on ID and JA3 Hash. Click the Import or Export icon on the right to import or export csv file for SSL fingerprint. You can also upload User-Agent using json formats. The User-Agent string is often used for content negotiation, where the origin server selects suitable content or operating parameters for the response. The concept of content tailoring is built into the HTTP standard in RFC1945 “for the sake of tailoring responses to avoid particular user agent limitations.” The information in the User-Agent string contributes to the information that the client sends to the server, since the string can vary considerably from user to user.

Proxy Profiles

A policy rule combines with several conditions and one action. The action determines how to control the traffic, and action parameters are managed in policy profiles. While policy objects enable you to identify traffic to enforce policies, policy profiles help you define further action.

Response Pages

When the Proxy Policy or Security Policy terminates matched HTTP session with a response page in Deny action, you can specify a Response Code and a Response Content to generate an error page or you could upload a html file via **Proxy Profile** > **Response Pages**.

STEP 1. Select **Profiles** > **Response Pages** tab, and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.
STEP 3. Please Upload a **File**. Allow html/htm format only.

Go back to Response Pages tab, you can view detailed information about the page you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. To download it, you can click the cloud icon under **File**, and wait a few seconds for the file to be downloaded to your local folder. You can search page list based on ID and Name. Enter search conditions in search bar and click search icon.

Insert Scripts

The Proxy Policy can insert a “js” or “css” scripts to webpages. You can upload a script via **Proxy** > **Insert Scripts**.

STEP 1. Select **Profiles** > **Proxy** > **Insert scripts**, and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.
STEP 3. Please Upload a **Script**. Allow js” and “css” only.
STEP 4. Select a **Script Type** from drop-down.

Go back to Insert Scripts tab, you can view detailed information about the scripts you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. To download it, you can click the cloud icon under **File**, and wait a few seconds for the file to be downloaded to your local folder.

You can search scripts list based on ID and Name. Enter search conditions in search bar and click search icon.

Hijack Files

The Proxy Policy can hijack a downloading file or page. You can upload a file, img or html for hijack via **Proxy** > **Hijack Files**.

STEP 1. Select **Profiles** > **Proxy** > **Hijack Files**, and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 64 characters.
STEP 3. Please Upload a **File**. Allow img, exe, apk, and html type only.
STEP 4. Enable Mirror Server Response or enter a **Download Name**.
STEP 5. Select a **File Type**.

✎*Note that the Maximum Limitation is 20MB for your uploaded file.*

Go back to Hijack Files tab, you can view detailed information about the file you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. To download it, you can click the cloud icon under File, and wait a few seconds for the file to be downloaded to your local folder. You can search file list based on ID and Name. Enter search conditions in search bar and click search icon.

Decryption Mirror Profiles

You also can mirror proxied traffic (decrypted) to third-party servers by referring a traffic mirror profile. The destination servers are described with VLAN Tag or MAC addresses, traffic will be load balanced over multiple servers of one profile.

You can manage the profile by the following procedure:

STEP 1. Select **Profiles** > **Proxy** > **Decryption Mirror Profiles** tab, and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.
STEP 3. Select VLAN or MAC as your **Connectivity** from drop-down.
STEP 4. Enter **VLAN ID/MAC**. Make sure to input valid mirror destination MAC address.

Go back to Traffic Mirror Profiles tab, you can view detailed information about the profile you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. You can search profile list based on ID and Name. Enter search conditions in search bar and click search icon.

Decryption Profile

A Decryption Profile includes three parts: Certificate Checks, Dynamic bypass and Protocol Version.

Certificate Checks

Server certificate verification options allow you to customize certificate check approaches.

Common Name: TSG checks if the client hello's SNI extension matches CN and SAN of the certificate.

Issuer: TSG checks the certificate chain if the issuer is a trusted certificate authority list. See Certificate Managements > Trusted Certificate Authorities > Built-in for a complete list.

Self-signed: TSG checks if a certificate is self-signed.

Expiry Date: TSG checks if a certificate is expired with the system clock.

Fail Action: If certificate is considered invalid, the proxy will take the fail action:

- **Fail-Close**: Terminate the SSL session by close the TCP connection.
- **Pass-through**: For expired, untrusted issuer or self-signed certificate, TSG send a certificate that signed by the default untrusted keyring to client-side. Thus, the client-side browser raises an untrusted issuer warning. For mismatched common names, TSG send a certificate that signed by policy defined keyring, client-side browser raises a common name invalid warning.

Dynamic Bypass

Dynamic bypass options allow you to customize intercept exceptions on policy basis. If an SSL session matches an intercept policy, and has one of following enabled properties, further communication will be exempt from intercept. That is to say, with dynamic bypass enabled, client-side can visit normally.

EV Certificate: An Extended Validation (EV) Certificate is a certificate used for HTTPS websites and software that proves the legal entity controlling the website or software package. Obtaining an EV certificate requires verification of the requesting entity's identity by a certificate authority.

Certificate Transparency: Certificate Transparency (CT) is an internet security standard and open source framework for monitoring and auditing digital certificates.

Mutual Authentication: Mutual authentication is a process or technology in which both entities in a communications link authenticate each other. The server sends a client certificate request, and the client must response with a valid certificate. Proxy could not intercept SSL sessions with mutual authenticated, these sessions will be blocked when this option is disabled.

On Protocol Errors: Protocol errors are unsupported ciphers, communication exceptions and etc., enable this option will increase network availabilities.

Certificate Pinning: The application known the server certificate by hard-coding, and can then ignore the device's trust store and rely on its own. The proxy detects pinning by client alert and SSL handshake errors. The proxy can also determine whether the current connection is Pinning through the SSL fingerprint profile. The SSL Fingerprint profile will be checked in advanced than client alert and SSL handshake errors when proxy detects pinning. Proxy could not intercept SSL sessions with certificate pinning, these sessions will be blocked when this option is disabled. For more details, see **Dynamic Bypass when Certificate Pinning**.

Certificate Not Installed: Trusted Root Certificate is Not Installed on Client. For more details, see **Dynamic Bypass when Certificate is Not Installed**.

0.0.0.0.1 Dynamic Bypass when Certificate Pinning

Certificate pinning is the process of a client check the server certificate with its pre-configured certificate list, if the server certificate does not match then the client will prevent the session from taking place. This enforcement ensures that the user devices are communicating

only to the dedicated trustful servers. Applications, such as Facebook, Twitter and Apple App store, utilize certificate pinning approach.

In order for an SSL proxy to decrypt and re-encrypt traffic so that a proxy policy can be enforced it needs to intercept the server certificate sent by the server to the client. Once it has intercepted the server certificate it will replace the server certificates with keyring signed ones. If a site works in a browser but not in an app on the same device, you are almost certainly looking at an instance of certificate pinning.

In reality, MITM applications of certificate pinning will block their communications. Alternatively, you can configure SSL Proxy to automatically bypass the next connection when the first N attempts to establish a connection fails.

The following behavior are indications of application use certificate pinning:

• The proxy received an SSL ALERT Message from the client during the SSL handshake. The Alert is usually an "Unknown CA (48)" alert indicating Certificate Pinning.

• The proxy received no alerts, instead, it received a TCP reset after the handshake is done.

If the SSL connection establishment fails as above for 4 or more times in 5 minutes, the proxy will consider it as certificate pinning, following attributes will be recorded for bypassing further connections:

• Client IP address

• Server Name Indicator (SNI) of SSL handshake message, if any

• SSL fingerprints, e.g. cipher suites of SSL handshake message

Different applications often have different handshake fingerprints, and therefore the proxy will only bypass those use certificate pinning.

0.0.0.0.2 Dynamic Bypass when Certificate is Not Installed As a best practice, the trusted root certificate certificates should be installed on clients to ensure that the browsers/apps perform the certificate checks to validate the identity of the proxy before establishing a connection. When client does not install the trusted root certificate, intercept its SSL connection will be failed.

The challenge is the behavior of certificate not installed is very similar to certificate pinning. The proxy determines whether the current connection is Not Pinning by querying the SSL fingerprint profile. When an SSL connection fails like certificate

pinning, and its fingerprint status is Not Pinning, the application is not considered as certificate pinning. Following figure shows the process.

You can configure SSL Proxy to automatically bypass those applications, or alternatively, still intercept to make the client install the trusted root certificate.

Let's dig into the technical details by a use case. There are two clients, client A and B, reside in our network. They shared a same IPv4 address (NAT), where client A has trusted root certificate installed and uses Facebook app (pinning); client B has no root certificate installed and uses Chrome to visit Facebook website (not pinning). With dynamic bypass configuration:

- Certificate Pinning: Enabled
- Certificate Not Installed: Disabled

At the beginning, both Client A and B's SSL connections are failed for their own reasons. And then, the proxy identifies client B's SSL connection as MITMable by finding Chrome's SSL fingerprints status is Not Pinning in the SSL fingerprint profile. Finally, Client A is bypassed, and client B is not.

Protocol Version

Protocol Versions allows you to configure SSL/TLS versions. By default, Proxy mirrors the client versions. Note that some website disable SSLv3 supports for security concerns, set both minimum and maximum version to SSLv3 will interrupt communications.

HTTP/2 is a major revision of the HTTP network protocol that provide increased speed. If Allow HTTP/2 is enabled, user will have better experience, but requires third-party systems to be able to process decrypted HTTP/2 traffic.

Create a Decryption Profile

Perform the following to create a decryption profile:

STEP 1. Select **Profiles** > **Decryption** > **SSL Decryption Profile**, and click **Create**.
STEP 2. Add a **Name**. The name is case-sensitive and can have up to 128 characters.
STEP 3. Enable or disable the following certificate checks: **Common Name**, **Issuer**, **Self-signed** and **Expiry Date**. If you enable Common Name, select Fail-close or Pass-through as your **Fail Action**.

STEP 4. Enable or disable the following Dynamic bypass: **EV Certificate**, **Certificate Transparency**, **Mutual Authentication**, **On Protocol Errors**, **Certificate Pinning**, **Certificate Not Installed**.

STEP 5. Enable or disable the following Protocol Versions: **Mirrors Client Versions**, **Allow HTTP/2**. If you disable Mirrors Client Versions, you need to select the Min Version and Max Version from SSLv3.0, TLSv1.0, TLSv1.1, TLSv1.2 and TLSv1.3.

Go back to Decryption Profile tab, you can view detailed information about the profile you just created. To edit and delete, find the item you want to edit or delete in the list. Click **Edit** or **Delete** at the top left. You can search profile list based on ID and Name. Enter search conditions in search bar and click search icon.

Monitoring

To forestall potential issues and to accelerate incidence response when needed, TSG provides intelligence about traffic and user patterns using customizable and informative reports. The dashboard, logs, and reports on TSG allow you to monitor activity on your network. You can monitor the logs and filter the information to generate reports with predefined or customized views. For example, you can use the predefined templates to generate reports on user activities or analyze the reports and logs to interpret unusual behavior on your network and generate a custom report on the traffic pattern. For a visually engaging presentation of network activity, the dashboard chart, with which you can interact to find the information you care about.

> Use the Dashboard
> View and Manage Logs
> View and Manage Report
> Take Packet Captures

Use the Dashboard

The TSG Dashboard include two sub menus, Main board and Live Chart. Main board show general TSG system overview, endpoints, policy hits statistics. By default, the Main board shows information of the last 24 hours. However, you can customize time range by clicking the time widget. By default, the statistics on the screen will not refresh automatically. You can turn it on and the Minimum Refresh Time is 15s. The following table describes the Main board widgets:

Main Board		Description
System Overview	Traffic	Monitor traffic bandwidth over time. You can view Bits, Packets and Sessions number. Bits is the accumulated bits sent/received. Packets is the accumulated packets sent/received. Sessions is the total sessions blocked/allowed.
	New	Displays the count number of New connections, which go through the system.
	Live	Displays the count number of current active connections, which go through the system.
	Device Count	Device Count shows the devices number in TSG, and the devices number that are in Up, Down and Alarm status. And it provides a link to NEZHA system.
	Security Policy	Enabled count, total count and Disabled count of Security Policy. Hover over Total, and you can also get enabled and disabled count about Proxy Policy.
Security Policy Hits	Policy Hits by Action	Hits are the times that traffic matched the criteria you defined in the Security Policy rule. A display of Security Policy Hits by action within the current time scope. You can view the Number of hit action by Sessions, Packets and bytes.
	Top Hits	Displays Top 10 or 100 Security Policy Hits. You can view either the tables or bars.
Endpoints	Active Client IP	Displays Active Client IP by Sessions, Packets and bytes. You can view either the tables, pies or bars.

	Active Server IP	Displays Active Server IP by Sessions, Packets and bytes. You can view either the tables, pies or bars.
	Active Subscriber ID	Displays Active Subscriber ID by sessions, Packets and bytes. You can view either the tables, pies or bars.
	Top APP	Displays Top APP by Sessions, Packets and Bytes. You can view either the table, pie or bar.
	Top URLs	Displays Top URLs of Security and Proxy Policy Hits by session count. You can view either the tables or bars.
	Top Domains	Displays Top Domains by sessions, Packets and bytes. You can view either the tables, pies or bars.
Proxy Policy Hits	Policy Hits by action	Hits are the times that traffic matched the criteria you defined in the Proxy Policy rule. A display of Proxy Policy Hits by action within the current time scope. You can view the Number of hit action by Sessions, Packets and bytes number.
	Pinning	Displays Pinning information for SSL/TSL traffic.
	Top Hits	Displays Top 10 or 100 Proxy Policy Hits. You can view either the tables or bars.

Live Chart show network summary, survey statistics, throughput and protocol information. Live Chart shows information of last 12 hours, and you can also customize it. Each built-in dashboard contains multiple widgets which can be expanded for detail view. The following table describes the Live Chart widgets:

Live Chart	Description
Network Summary	Displays Unique Client IP count, Sessions, Date Rate and Throughput.
Survey Statistics	Displays the percentage and number of Uncategorized Traffic, One-side Connections, Sequence Gap Loss and Packet Fragmentation.

Protocols	Total Throughput	Displays Total Throughput in real-time by protocol. When you click the Protocol Composition pie chart, it updates dynamically. The first time to enter the page the X-axis displays 20% of the current time scope. (Sometimes the peak data is outside the display window, so the line graph will appear empty.) You can adjust the display time window as needed.
	Protocol Composition	Classify traffic by protocol shown in Sunburst chart view. The Sunburst chart shows the distribution (or percentage) of each level of protocol and the total bandwidth. Sunburst chart supports zoom in and out. You can also drag and reset it.
	Network Protocol Tree	It is like the Protocol Composition widget, but displayed in tree structure.
	Protocol Statistics	Displays the Sent Packets, Received Packets and Sessions of each protocol.

✎*You can click a specific protocol to display the detailed data about it shown in the adjacent widget: Total Throughput, Network Protocol Tree and Protocol Statistics. As you can see the three widgets change dynamically when you click different protocol. By default, the reports of ETHERNET will be displayed. For the full list of Network Protocol Tree, please refer to Appendix B Network Protocol Tree.*

View and Manage Logs

A log is an automatically generated, time-stamped file that provides an audit trail for network traffic events that TSG monitors. Log records contain columns, which are properties, activities, or behaviors associated with the logged event. Each log type records information for a different event type. You can see the following 6 log types in the Log pages.

- Security Events
- Proxy Events

- Session Records
- Radius Logs
- VoIP Records
- GTP-C Records

Log Types

Security Events and Proxy Events

- Security Events and Proxy Events data provides the ability to validate rule additions and rule changes and to monitor the time frame when a rule was used. The log gives you the information to determine whether a rule is effective for access enforcement.

Session Records

- Session records regardless of policy configuration; it shows all traffic that is allowed on your network. Traffic logs display an entry for the start and end of each session. TSG Session records display Transaction records when clicking details. Session records also consist of GTP, MPLS information. You can view live sessions in session records, but reports do not include live sessions.

Radius Logs

- Remote Authentication Dial-In User Service (RADIUS) is a broadly supported networking protocol that provides centralized authentication and authorization. TSG will keep track of radius traffic information, including Packet Type, Account, Nas IP, Framed IP, ACC Status Type and so on. You can use the Account information in radius log to create Subscriber ID object.

VoIP Records

- Voice over IP (VoIP) requires faster speeds and time-sensitive, real-time delivery. It mainly uses RTP as its media protocol to deliver multimedia sessions and Session Initiation Protocol (SIP) for signaling. SIP can open dynamic pinholes in the firewall where NAT is enabled. TSG only supports t VoIP calls using SIP for signaling and RTP for delivering audio data. TSG will keep track of VoIP traffic regarding general info, action, source, destination, application, transmission and SIP fields.

GTP-C Records

• GTP-C records is composed of GTP-C version (v1 or v2), International Mobile Equipment Identity (IMEI),International Mobile Subscriber Identity (IMSI), APN and Phone Number.

Please refer to Appendix C Logs Fields Description for more details.

DoS Events

• Dos Events provides detailed statistics on the detected Dos attacks. Currently supports DNS flood, TCP SYN flood, UDP flood and ICMP flood. You can view the Source Countries, Destination Countries, Start Time, End Time, Attack Type, Severity, Sessions/s, Packets/s, Bits/s, etc.

View Logs

You can view the different log types on TSG in a tabular format.

STEP 1. Select **Logs**. Select a log type from the list.
STEP 2. (Optional) Customize the log column display.

1. Click the gear icon to the left of any column header.
2. Select columns to display from the list. The log updates automatically to match your selections.

STEP 3. View additional details about log entries.

1. Click the **Log ID** for a specific log entry. The Detailed Log View has more information about the source and destination of the session, as well as a list of sessions related to the log entry.
2. (Optional)If you are viewing Security Events and Proxy Events, hover over the **Policy ID**, you can view the Policy Name, Action, Application, Destination, Source and Filter. Hover over the **Client IP** and **Server IP**, you can view the corresponding IP, Port, ASN and Location.
3. (Optional) If you are viewing Radius Log, hover over the **Subscriber ID**, you can view Subscriber ID, related Security Events and Session Records. Hover over **Framed IP**, you can also view related Security Events and Session Records.

Filter Logs

The ability to filter logs is useful for focusing on events on your TSG that possess particular attributes. Filter logs by columns that are associated with individual log records. For example, filtering by the Policy ID makes it easier to pinpoint the specific rule you want to locate, even among many similarly rules. If your ruleset is very large and contains many rules, using the rule's Policy ID as a filter spotlights the particular rule you need to find without having to navigate through pages of results.

TSG log filter supports search by multiple fields in AND/OR relation. You can perform exact match search and fuzzy search using Operators. To filter logs, follow these steps:

STEP 1. Select **Logs**. Select a log type from the list. For example, **Proxy Events**.

STEP 2. Select the time period picker. By default, it shows logs of **Last 1 hour**.

STEP 3. Click **Add Filter** to add search term. The supported search fields are: Log ID, Policy ID, Subscriber ID, IMEI, IMSI, Phone Number, Client IP, Internal IP, Client Port, Server IP, Server Port, External IP, Action, Sled IP, Schema Type, Data Center, Application Label, FQDN Category, Session ID, TCP Client ISN, TCP Server ISN, Http.URL, Http.Domain, SSL.SNI and SSL. JA3 hash etc. Then, select **Operator**, such as =, !=, in, not in, like, not like, notEmpty, empty, HAS. And input the value. If you wish to add multiple search fields, click Add Filter again, and proceed. TSG support **AND/OR** relations between search fields. For example, enter Client IP 192.168.50.62 and Action Deny to display only entries that contain both fields in the log.

Export Logs

You can export the contents of a log type to a xlsx file. Firstly, Filter Logs according to time and other conditions. Then, Click the Log Export icon on the right. Wait a few seconds for the file to be generated and downloaded to your local folder.

✎*Maximum export log records are 100000.*

View and Manage Reports

The reporting capabilities on TSG allow you to keep a pulse on your network and focus your efforts on maintaining network security for keeping your users safe and productive.

The purpose of the report is to summarize a large amount of log data. Based on the configured report parameters, the data is extracted and displayed graphically, making the data easier and faster to understand. The models and trends revealed in the report already exist in the form of data in the database, but manually locating, cross-referencing and analyzing multiple log files is very difficult and takes a lot of time, especially when the user does not know what kind of trends and models to look for. User-customized reports are created based on the TSG report interface. The reports will summarize logs and generate user-customized graphical reports, providing you with fast and detailed network data. You can then use this information to better understand network behavior or improve network security.

✎*Note that the report itself does not provide any recommendations or give any indication of problems. Users must analyze and consider network problems based on report data and graphs.*

TSG includes a number of predefined datasets, charts and reports. It can basically meet the needs of most users. You can quickly view network activities based on preset reports for event or behavior analysis. For more details about predefined reports, please refer to Appendix D Predefined Reports.

You can use predefined reports as-is, or you can build custom reports that meet your needs for specific data and actionable tasks. Create and schedule custom reports that show exactly the information you want to see by filtering on conditions and columns to include. You can also include aggregate function for more specific drill down on report data.

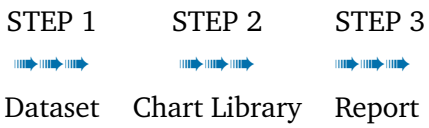
In order to create purposeful custom reports, you must consider the attributes or key pieces of information that you want to retrieve and analyze. Reports can be sent for email delivery or FTP service if you enable notification when creating a report.

Report is a set of data placed in orderly charts. A chart contains two elements:

- The data set is a SELECT query that extracts the specified data from the database.
- In what format the data is displayed (for example: pie chart, bar chart and table).

Each chart is associated with one dataset. When you generate a report, the dataset associated with each chart extracts data from the logs and populates the charts. Each dataset requires a specific log type.

To customize a report, you need to follow three steps.



This consideration guides you in making the following selections in a custom report. For example, create a report about the top 10 IP Addresses that visited Google last month by sessions.

STEP 1. Create a Dataset. Dataset specify what data to extract from logs.

1. Select **Reports** > **Datasets** menu, and click **Create**.
2. Enter a **Name**.
3. Select a **Log Type** from the following: Security Event, Proxy Event, Session Records and Radius. Because session records show all traffic that is allowed on your network, select session records.
4. Select the **Group by**. Fields are the result fields of a dataset collection, include group by fields and metric fields. Fields are the dimensions of your report. This will serve as the X-Axis Data Binding options for Chart Libraries. The report mainly displays client IP, so select client IP.
5. Specify the **Data Bindings**, add Field, Aggregate, and Label, which serve as legend of your Chart Libraries. Variable consists of log field. You can add at least one or multiple variables. Aggregate is metric function and the available options are sum, min, max, avg, count, and count distinct. Label is the legend that will show in your Chart Libraries. In this case, Field select Sessions; Aggregate select sum; and sessions is filled in for Label by default.
6. (optional)Specify **Filter** and **Having**. Query conditions, only calculated within the data range of a certain field or a few fields that match a specific expression. All lines that do not match the conditions will be excluded from the collections of data. Your selections will automatically generate a SQL. Filter conditions is before data aggregation, and Having is after data aggregation. Since the visiting target is Google, you can set Feild as Http.Domain, set Aggregate as Suffix and set Value to “google.com”.
7. Click **OK**.

Please view the following table for details about new dataset.

Field	Description
Name	Enter a name for the dataset.
Log Type	Data source, select a log type from the dropdown list. The following log types are available: • Security Event • Proxy Event • Session Records • Radius Logs • VoIP Records • GTP-C Records
Group by	Collect data from multiple records and group the results by one or more fields (columns). Click the "+" button to add variable. ✎*Field in Group by will serve as the X-Axis Data Binding options for Chart Libraries.* For the detailed meaning of each Log Field, please view **Appendix C Log Fields Description**.
Data Bindings	Data Bindings are the result fields of a dataset collection, include group by log fields. Click the "+" button to add field, aggregate, and description information. • Field: log fields For the detailed meaning of each Log Field, please view **Appendix C Log Fields Description**. • Aggregate: Metric function, Select a value from the dropdown list. The available options vary depending on the selected variable. – sum – min – max – avg – count – Count Distinct • Label: description information

Filter	Query conditions, only calculated within the data range of a certain field or a few fields that match a specific expression. All lines that do not match the conditions will be excluded from the collections of data. Click the "+ Add Condition" button to add field, operator, and value. • Field: log field • Operator: Match operator, Select a value from the dropdown list. The available options vary depending on the selected field. – = – != – >= – < – >= – <= – Like – Not Like – notEmpty • Value: Enter a value for field
Having	Query conditions after data aggregation. Click the "+ Add Condition" button to add field, aggregate, operator and value.
SQL	Generate SQL based on log type, grouping field, result field, and filter conditions, for user reference only.

The following table lists different Group by, Data Bindings and Filter fields for different log types.

Log Type	Group by	Data Bindings (Metrics)	Filter

Session Records	Server IP Client IP Internal IP External IP Sled IP Device ID Client Location Server Location Subscriber ID Client Port Server Port Schema Type L4 Protocol L7 Protocol Data Center Client ASN Server ASN Start Time End Time IMEI IMSI Phone Number Application Label Http.Host Http.Domain Http.URL SSL.SNI SSL.JA3 hash Receive Time	Server IP Client IP Internal IP External IP Subscriber ID Sled IP Device ID Packets Sent Packets Received Bytes Sent Bytes Received Sessions Duration(ms) Establish Latency(ms) Fragmentation Packets(c2s) Fragmentation Packets(s2c) Sequence Gap Loss(c2s) Sequence Gap Loss(s2c) Unorder Packets (c2s) Unorder Packets (s2c) IMEI IMSI Phone Number Application Label Http.Host Http.Domain Http.URL SSL.SNI SSL.JA3 hash Log ID	Address Type Server IP Client IP Internal IP External IP Client Port Server Port Client Location Server Location Subscriber ID Packets Sent Packets Received Bytes Sent Bytes Received Fragmentation Packets(c2s) Fragmentation Packets(s2c) Sequence Gap Loss(c2s) Sequence Gap Loss(s2c) Unorder Packets(c2s) Unorder Packets(s2c) L4 Protocol L7 Protocol Stream Direction Direction Data Center Sled IP Device ID Schema Type Client ASN Server ASN Start Time End Time Duration(ms) Establish Latency(ms) IMEI IMSI Phone Number Application Label Http.Host Http.Domain Http.URL SSL.SNI SSL.JA3 hash

Security Event	Server IP Client IP Internal IP External IP Policy ID Action Sled IP Device ID Client Location Server Location Subscriber ID Client Port Server Port Schema Type L4 Protocol L7 Protocol Data Center Client ASN Server ASN Start Time End Time IMEI IMSI Phone Number Application Label Http.Host Http.Domain Http.URL SSL.SNI SSL.JA3 hash SSL.Client Side Version SSL.Server Side Version Mail.Account Mail.From Mail.To Quic.SNI Receive Time	Server IP Client IP Internal IP External IP Subscriber ID Sled IP Device ID Sessions Packets Sent Packets Received Bytes Sent Bytes Received Duration(ms) Establish Latency(ms) IMEI IMSI Phone Number Application Label Http.Host Http.Domain Http.URL SSL.SNI SSL.JA3 hash SSL.Client Side Latency(ms) SSL.Server Side Latency(ms) Mail.Account Mail.From Mail.To Quic.SNI Log ID	Policy ID Action Address Type Server IP Client IP Internal IP External IP Client Port Server Port Client Location Server Location Subscriber ID Packets Sent Packets Received Bytes Sent Bytes Received L4 Protocol L7 Protocol Stream Direction Data Center Sled IP Device ID Direction Schema Type Client ASN Server ASN Start Time End Time Duration(ms) Establish Latency(ms) IMEI IMSI Phone Number Application Label Http.Host Http.Domain Http.URL Http.Content Type SSL.SNI SSL.JA3 hash SSL.Pinning SSL.Intercept State SSL.Client Side Version SSL.Server Side Version SSL.Certificate Verify SSL.Client Side Latency(ms) SSL.Server Side Latency(ms)

			Mail.Account Mail.From Mail.To Mail.Subject Quic.SNI
Radius Logs	Nas IP Framed IP Subscriber ID Receive Time	Framed IP Event Timestamp Packets Sent Packets Received Bytes Sent Bytes Received Log ID	Framed IP Subscriber ID Packet Type Acct Session ID Acct Multi Session ID ACC Status Type

Proxy Event	Server IP Client IP Internal IP External IP Policy ID Sub Action Sled IP Device ID Client Location Server Location Subscriber ID Client Port Server Port Schema Type Data Center Client ASN Server ASN IMEI IMSI Phone Number Http.Host Http.Domain Http.URL DoH.Host DoH.QNAME Receive Time	Server IP Client IP Internal IP External IP Subscriber ID Sled IP Device ID Sessions Bytes Sent Bytes Received Http.Host Http.Domain Http.URL DoH.Host DoH.QNAME Log ID Phone Number	Policy ID Sub Action Address Type Server IP Client IP Internal IP External IP Client Port Server Port Client Location Server Location Subscriber ID L4 Protocol Data Center Sled IP Device ID Client ASN Server ASN Direction Schema Type Http.Host Http.Domain Http.URL Http.Content Type DoH.Host DoH.QNAME Phone Number

			Address Type
			Server IP
			Client IP
			Internal IP
			External IP
		Server IP	Client Port
	Server IP	Client IP	Server Port
	Client IP	Internal IP	Client Location
	Internal IP	External IP	Server Location
	External IP	Subscriber ID	Subscriber ID
	Sled IP	Sled IP	Packets Sent
	Device ID	Device ID	Packets Received
	Client Location	Packets Sent	Bytes Sent
	Server Location	Packets Received	Bytes Received
	Subscriber ID	Bytes Sent	Fragmentation Packets(c2s)
	Client Port	Bytes Received	Fragmentation Packets(s2c)
	Server Port	Sessions	Sequence Gap Loss(c2s)
	Schema Type	Duration(ms)	Sequence Gap Loss(s2c)
	L4 Protocol	Establish	Unorder Packets(c2s)
	L7 Protocol	Latency(ms)	Unorder Packets(s2c)
	Data Center	Fragmentation	L4 Protocol
VoIP	Client ASN	Packets(c2s)	L7 Protocol
Records	Server ASN	Fragmentation	Stream Direction
	Start Time	Packets(s2c)	Direction
	End Time	Sequence Gap	Data Center
	SIP.Call-ID	Loss(c2s)	Sled IP
	SIP.Originator	Sequence Gap	Device ID
	SIP.Responder	Loss(s2c)	Schema Type
	SIP.User-Agent	Unorder Packets	Client ASN
	SIP.Server	(c2s)	Server ASN
	SIP.Duration	Unorder Packets	Start Time
	SIP.Bye	(s2c)	End Time
	RTP.Payload	SIP.Call-ID	Duration(ms)
	Type(c2s)	SIP.Originator	Establish Latency(ms)
	RTP.Payload	SIP.Responder	SIP.Call-ID
	Type(s2c)	SIP.User-Agent	SIP.Originator
	RTP.Direction	SIP.Server	SIP.Responder
	Receive Time	SIP.Duration	SIP.User-Agent
		Log ID	SIP.Server
			SIP.Duration
			SIP.Bye
			RTP.Payload Type(c2s)
			RTP.Payload Type(s2c)
			RTP.Direction

GTP-C Records	Server IP Client IP Internal IP External IP Sled IP Device ID Client Location Server Location Subscriber ID Client Port Server Port Schema Type L4 Protocol L7 Protocol Data Center Client ASN Server ASN Start Time End Time Version APN IMEI IMSI Phone Number Receive Time	Server IP Client IP Internal IP External IP Subscriber ID Sled IP Device ID Packets Sent Packets Received Bytes Sent Bytes Received Sessions Duration(ms) Establish Latency(ms) Fragmentation Packets(c2s) Fragmentation Packets(s2c) Sequence Gap Loss(c2s) Sequence Gap Loss(s2c) Unorder Packets (c2s) Unorder Packets (s2c) Version APN IMEI IMSI Phone Number Log ID	Address Type Server IP Client IP Internal IP External IP Client Port Server Port Client Location Server Location Subscriber ID Packets Sent Packets Received Bytes Sent Bytes Received Fragmentation Packets(c2s) Fragmentation Packets(s2c) Sequence Gap Loss(c2s) Sequence Gap Loss(s2c) Unorder Packets(c2s) Unorder Packets(s2c) L4 Protocol L7 Protocol Stream Direction Direction Data Center Sled IP Device ID Schema Type Client ASN Server ASN Start Time End Time Duration(ms) Establish Latency(ms) Version APN IMEI IMSI Phone Number End User Address V4 End User Address V6 Uplink TEID Downlink TEID Message Type

✎*Viewing the SQL query of an existing dataset.*

To view the SQL query for an existing dataset:

1. Go to Reports > Datasets.

2. Hover the mouse cursor over the dataset on the dataset list. The SQL query is displayed as a tooltip. You can also open the dataset to view the Query and detail information.

STEP 2. Create Chart Library. Chart set the display form of the dataset.

1. Enter a **Name** and **Description**.
2. Select the **Dataset** you just created.
3. Select **Chart Type** from the following: Line, Pie, Bar, Area and Table. This selection affects the rest of the available selections. To create Line and Area chart, you must select Group by Receive Time. Here select Bar.
4. Set the **Data Bindings** for X-Axis and Y-Axis. For X-Axis, set Data Binding to Client IP, set Label to Client IP and set Show Top to 10. For Y-Axis, set Data Binding to Sessions, set Format to Counter(K/M/G) and set Label to Sessions.
5. Click **OK**.

Please view the following table for details about new chart library.

Field	Description
Name	Enter a name for the dataset.
Description	Enter a description of the chart.
Dataset	Select a dataset from the dropdown list.
Chart Type	Select a graph type from the dropdown list, one of Table, Bar, Pie, Line and Area. This selection affects the rest of the available selections. ✎*For Line and Area type, Receive Time field must be chosen as one of Group By – Variable in selected dataset to generate meaningful report.*
Data Bindings	The data bindings vary depending on the chart type selected.
Table	
Table Type	Regular or Drilldown.
Add Column	Click to add Column. Up to 15 columns can be added for a Regular table. Drilldown tables have three columns.

Columns	The following column settings must be set: • Title: Enter a title for the column. • Width: Enter the column width as a percentage. • Data Binding: Select a value from the dropdown list. The options vary depending on the selected dataset. • Format: Select a value from the dropdown list. You can select Bar to create an intuitional chart. • Legend: Description information. • Click the "+" button to add more data bindings to the column. Every column must have at least one data binding. The maximum of columns varies depending on the table type.
Order by	Decide what to order the table by. The available options vary depending on the selected dataset.
Show Top	Enter a numerical value. Only the first 'X' items are displayed. Other items can be bundled to be displayed as Others when you check the **Bundle rest into 'Others'** which is above the **Order By**.
Drilldown Top	Enter a numerical value. Only the first 'X' items are displayed. This option is only available for Drilldown tables.
Bar	
X-Axis	• Data Binding: Select a value from the dropdown list. The available options vary depending on the selected dataset. • Label: Enter a label for the axis. • Show Top: Enter a numerical value. Only the first 'X' items are displayed. • Other items are bundled into the Others category.
Y-Axis	• Data Binding: Select a value from the dropdown list. The available options vary depending on the selected dataset. • Format: Select a format from the dropdown list: Bandwidth, Counter, Default and Percentage. • Label: Enter a label for the axis.
Bundle rest into "Others"	Other items are bundled into the Others category if check this option. This reflects in 'X' Label of a Bar chart.
Pie	

Category	• Data Binding: Select a value from the dropdown list. The available options vary depending on the selected dataset. • Label: Enter a label for the axis. • Show Top: Enter a numerical value. Only the first 'X' items are displayed. • Other items can be bundled into the Others category.
Series	• Data Binding: Select a value from the dropdown list. The available options vary depending on the selected dataset. • Format: Select a format from the dropdown list: Bandwidth, Counter, Default and Percentage. • Label: Enter a label for the axis.
Bundle rest into "Others"	Other items are bundled into the Others category if check this option. This reflects in Category of a Pie chart.
Line or **Area**	
X-Axis	• Data Binding: Select a value from the dropdown list. The available options vary depending on the selected dataset. The selected dataset should include time field. ✎*Group by must include Receive Time in selected dataset.* • Format: Select the format Time from the dropdown list.
Lines	• Data Binding: Select a value from the dropdown list. The available options vary depending on the selected dataset. • Format: Select from the dropdown list, one of: Bandwidth, Counter and Default. • Type: Select from the dropdown list, one of: Line Up and Line Down. ✎*If data in Y-Axis has different units, please select Line Up and Line Down respectively.* • Legend: Enter the legend text for the line.
Add line	Click "+" button to add more lines.

✎*Viewing datasets associated with charts*

To view datasets associated with charts:

1. *Go to Reports > Chart Libraries.*
2. *In the Chart list pane, find the name of the dataset associated with the chart in the Dataset.*
3. *Go to Reports > Datasets.*
4. *In the Search box, type the name of the dataset.*
5. *Select the dataset that is found, and click View in the toolbar to view it.*

Or you can perform the following:

1. *Edit chart.*
2. *Click the Dataset Input.*
3. *Click the Edit pencil of the dataset in the dataset list, and you can view the dataset.*

✎*Hover the mouse cursor over the chart name on the chart libraries list, you can view the key information about this chart.*

STEP 3. Create a Report. You can generate reports directly, or you can use schedule to generate reports regularly. Predefined reports are actually report templates with basic default setting configurations.

1. Enter a **Name**.
2. Select a **Time Period**. In this case, select 'last' '1' 'month'.
3. (Optional)If you **Enable Schedule**, select a schedule from drop-down or create a schedule. For more details, see **Policies** > **Schedules**. Here don't Enable Schedule. ✎*The time period defines the time scope of the data in the report. And schedule defines when the report will begin to generate. For example, set Time Period to last 1 hour, and Enable Schedule set to 14:00-18:00 every day. The report will begin to generate at 14:00 every day and the report data comes from 13:00-14:00 of that day. Take another example, set Time Period to today and Enable Schedule to 14:00-18:00 every day. TSG will create a statistical task for the report at 14:00 but it will complete at 24:00 and the report data only comes from that day.*
4. (Optional)If you **Enable Notification**, you can select FTP or Email as Output Profile.
   - Email: Enter recipient's e-mail.
   - FTP: Enter server, port, username, password and directory.
5. Select the **Chart Library** you just created. Enter number and select time unit for **Time Granularity**. The available options vary depending on the

selected chart, only applies to charts with time parameters. This will affect the data density of X-Axis. Here, let it grey by default since it is a Bar chart.

6. (optional)Add **Filter** if you have related requirements. You can apply log message filters to reports and charts. If add multiple charts, the filter field is limited to the common fields of multiple charts. Here don't add any Filter.
7. Click **OK**.
8. Wait a while for the generation of the report. Click button (▶) at the left of the report row to get the details of the result. After the status reach 100%, click **View** and you'll see: firstly, the overviews of traffic statistics, then the traffic trend in the time period and finally the results of your custom selections.

Please view the following table for details about new report.

Field	Description
Name	Enter a name for the dataset.
Time Period	The time period the report covers. Select a time period or select customize to manually specify the start and end date and time. • today: from 0:00-24:00 today • yesterday: from 0:00-24:00 yesterday • this: current time period. For example, this 1 hour means from yyyy-mm-dd hh:00:00 to yyyy-mm-dd hh:59:59. • last: last time period. For example, last 1 hour means from yyyy-mm-dd hh-1:mi:ss to yyyy-mm-dd hh:mi:ss. • previous: previous time period. For example, previous 1 hour means yyyy-mm-dd hh-1:00:00 to yyyy-mm-dd hh:00:00. • customize: If you select customize, you can't Enable Schedule.
Enable Schedule	Select to enable schedules.
Schedule	Select a schedule from the dropdown list to run the report.
Enable Notification	Setting report notification profile.

Output Profile	Select the output profile from the dropdown list: • Email: Enter recipient's e-mail. • FTP: Enter server, port, username, password and directory.
Chart Library	Click the button "+" and select a chart from the slide page Chart Library.
Time Granularity	Enter number and select time unit. The available options vary depending on the selected chart, only applies to charts with time parameters.
Add Chart	Click the "+ Add Chart" button to add more charts.
Filter	You can apply log message filters to reports and charts. If add multiple charts, the filter field is limited to the common fields of multiple charts. Click the "+ Add Condition" button to add log Field, Operator, and Value. • Field: Variables in log field for reports. • Operator: Match operator. Select a value from the drop-down list. The available options vary depending on the selected field. – = – != – >= – < – >= – <= – Like – Not Like – notEmpty • Value: Enter a value for log Field.

✎*To base a report on a predefined template, select the predefined Dataset or Chart Libraries. If you Enable Schedule for a report, the execution button (►) under Operation will not be available and (-) will show in column Operation. You can view the create time of a result for the report, when you click button (►) (switch to button (▼) to indicate folding back) at the left of a row to unfold the report details. Create time shows the time that the report generated this cycle according to the schedule, not the*

time you configure the report. The reports are displayed in descending order by create time. For example, you create a report with Time Period "today" and Enable Schedule "Daily, Start Time 14:00 & End Time 18:00" at 9:00 am. In this case, the first result of report which is shown in (▼) will be created at 14:00 today with Create Time: YYYY-MM-DD 14:00. Meanwhile the percentage which indicates the ready status of the result will reach 100% after 24:00:00 since the report configuration was set Time Period as "today". And you can get a new result at the 14:00 and view the report after midnight every day from now on. The report list displays Last Modified Time and Last Execution Time and you can click the column to make the list display in descending or ascending order.

For more details about reports best practice, please refer to Appendix F Make Your Own Reports

Advantages of TSG report:

- Flexible customization: If the predefined reports do not meet the needs of users, you can add new customized report models at any time according to different scenarios. Reduce the waiting period for statistical requirements, demand communication, development and online.
- Schedule statistics: support report schedules and it can generate reports regularly. The schedule is flexible, and reports can be generated regularly by day, week, month, and year.
- Report archive: Support online sending or offline exporting to report archive.

Take Packet Captures

TSG captures packets for all traffic or for specific traffic based on filters that you define. For example, you can configure TSG to only capture packets to and from a specific source and destination IP address or port. You may need to take packet captures when creating a custom application, because you have to gather information about the application.

STEP 1. Before you start a packet capture, identify the attributes of the traffic that you want to capture.

For example, to determine the source IP address, and the destination IP address for traffic between two systems. If you wish to troubleshoot a Security Event Log, go to **Logs** > **Security Event Log** and locate the traffic log for the two

systems. Click the **Log ID** to view the Client IP, Client Port, Server IP, Server Port.

STEP 2. Set packet capture conditions, so TSG only captures traffic you are interested in.

1. Select **Setting** > **Trouble Shooting** menu, and select Packet Capture tab.
2. Enter a descriptive **Name**.
3. Select **Address Type**, IPv4 or IPv6.
4. Enter **Client IP**, **Client Port**, **Server IP**, **Server Port**.
5. Select TCP, UDP or Any as **Protocol**.
6. Select **Effective Devices** leave the value set to any.
7. Enter **Captured Number** and **Capture Duration**.
8. Click **OK**.

STEP 3. Packet Capture will automatically start after you created it. Generate traffic that matches the filters that you defined. It will stop after your capture duration expires, or you can turn it off manually after TSG captures the data that you want to analyze.

STEP 4. Click the triangle icon to the left in the list to expand the item, then click **Download** to download packets or **Delete** to delete packets.

STEP 5. View the packet capture files using a network packet analyzer.

DoS Detection

A denial-of-service attack (DoS attack) is a cyber-attack in which the perpetrator seeks to make a machine or network resource unavailable to its intended users by temporarily or indefinitely disrupting services of a host connected to the Internet. Denial of service is typically accomplished by flooding the targeted machine or resource with superfluous requests in an attempt to overload systems and prevent some or all legitimate requests from being fulfilled.

In a distributed denial-of-service attack (DDoS attack), the incoming traffic flooding the victim originates from many different sources. This effectively makes it impossible to stop the attack simply by blocking a single source.

The implications for victims range from a nuisance to millions of dollars in lost revenue.

TSG supports DoS or DDoS detection that includes TCP flood (SYN, ACK, invalid flag combination), UDP flood, ICMP flood, DNS Amplification.

> DoS Overview
> DoS Detection

DoS Overview

Computer network security is a challenge as old as the Internet itself. In such a DoS attack, the hacker attempts to consume all the resources of a networked system so that no other users can be served. In distributed denial of service (DDoS) attacks, attackers write a program that will covertly send itself to dozens, hundreds, or even thousands of other computers. These computers are known as 'bots' or 'zombies', because they act on behalf of the hackers to launch an attack against target systems. A network of these computers is called a botnet. An administrator of such a botnet is called a botmaster. the botmaster will cause all of these bots to attempt repeated connections to a target site. If the attack is successful, it will deplete all system or network resources, thereby denying service to legitimate users or customers.

To circumvent detection, attackers are either increasingly mimicking the behavior of a large number of clients or actually using a large number of IoT clients. The resulting attacks are hard to defend against with standard techniques, as the malicious requests differ from the legitimate ones in intent but not in content. Because each attacking system looks innocent, advanced techniques are required to separate the 'bad' traffic from the 'good' traffic.

DoS Detection

TSG supports DoS or DDoS detection that includes TCP flood (SYN, ACK, invalid flag combination), UDP flood, ICMP flood, DNS Amplification.

TSG will display DDoS detection result in two ways: DoS Event Logs and DoS Threat Map. DoS event logs support log retrieval by Source IPs, Destination IP, Attack Type, Start Time and End Time. DoS threat map allows attack trend display within given time span. The attack source and destination are obtained by mapping source IPs, destination IP with geography in IP libraries. DoS threat map displays attacks of last 1 hour by Attack Type, Severity, Top Source Countries, Top Destination and Top Victims.

DoS or DDoS logs are aggregated based on thresholds and baseline. For example, within 30s, there are more than 10000 TCP SYN in destination IP, will be ruled as SYN Flood. To detect Dos attacks, you can customize threshold conditions by creating DoS Detection Profile for Target IPs. The threshold units are Packets/second, Bits/second and Sessions/ second.

Advanced Setting

This section contains information about configuring TSG advanced features, including:

> Proxy TCP Options
> System Usage
> System Appearance

Proxy TCP Options

TSG provides TCP default option which keeps the Enable TCP Passthrough and Bypass Duplicated Packet off. However, you can Create your own Proxy TCP Options for special situations. Under certain boundary conditions of network transmission, some network parameters need to be renegotiated during network transmission, such as MTU. The network equipment will renegotiate network parameters through some mechanisms, and the negotiation process may not under precise monitor of TSG. And it may affect related policies. In this case, the affected network parameters need to be preset through the PROXY TCP OPTION.

STEP 1. Select **Profiles** > **Proxy** > **Proxy TCP Options**, and click **Create**.

STEP 2. Specify Proxy TCP Options name and effective scope.

1. Enter a descriptive **Name**.
2. Specify single or multiple **Source** IP Addresses/ Subscriber IDs or leave the value set to any.
3. Specify one or multiple **Destination** IP Addresses or leave the value set to any.
4. (Optional) Specify a **Filter**. For example, add **SNI** and select a **Category** or an **FQDN**. If you select a category, only traffic is destined for that specified category will be effective.

STEP 3. Enable Proxy TCP Options.

1. (Optional)**Enable TCP Passthrough**. When enabled, traffic will bypass TSG proxy system but will not be decrypted. You can enable it, when troubleshooting.
2. (Optional)Enable **bypass Duplicated Packet**.
3. Specify Client-side Parameters and Server-side Parameters respectively. They have the same options.
   a) (Optional)Enable **Override MSS**. The MSS range: 536-1460.
   b) (Optional)Enable **No Delay**.
   c) Specify **TTL**. The TTL range: 1-255.
   d) (Optional)Enable **Keepalive**. Specify **Probe Number**, **Idle Time** and **Interval**. Probe Number is the maximum number of keepalive probes TCP should send before dropping the connection. Idle Time is the time (in seconds) the connection needs to remain idle before TCP starts sending keepalive probes, if the socket option SO_KEEPALIVE has been set on this socket. Interval is the time (in seconds) between

individual keepalive probes. These three options should not be used in code intended to be portable.

e) Define **User Timeout**. The User Timeout range: 20060000.

4. Click **OK**.

You can **Edit** or **Delete** your customized Proxy TCP Options.

System Usage

System usage displays policy and object usage in bar chart and also shows storage usage of files, traffic logs, reports and metrics. Files are unstructured logs carried by session records and will store at least one month by default. Traffic Logs include all four types of logs and will also store at least one month by default. Reports and Metrics are the predefined or customized data aggregated based on multiple dimensions and will store at least one year by default.

TSG automatically deletes logs and reports that exceed the expiration period. When TSG reaches the storage quota, it automatically deletes older data. Deleting operates every day at 24:00.

You can estimate server storage space and configure storage parameters. For your convenience, provide the following formula to calculate your required server count.

Server Count = Logs/sec*Storage Days *86400*Log Size *0.4*RF/ (Disk storage per server*0.7)

• Logs/sec: log rate, the log records count generated per second.

• Log Size: the average size of each log, and 1KB will be used in storage calculating as average size.

• Compression Rate: For TSG logs storage, it is 0.4.

• RF: Replication Factor, it is the count of copies on your server. E.g., setting RF=2 means the storage usage will be double of the real size of logs, files, reports and metrics.

• Interval in seconds: 86400, Seconds of a day.

• Disk Safety Factor: it is normally 0.7. It means 70% of the storage will be used for runtime data of TSG, including logs, files, reports and metrics. The others will occupy the remaining 30%, such as OS, Raid5 and temporary data.

Following are some calculation for a typical scenario:

Suppose there is a data center which is configured to handle the traffic of mainstream website and applications by TSG. The log rate of session records is 2000/s; the log rate of event logs is 1500/s; the planed storage days is 30 for event logs with files, so does the storage of session records. The RF of event logs storage is set to 2 since event logs, which is raised by hit policies, require more attention. While the RF of session records is set to 1.

• Storage of session records: 2000*30*86400*1KB*0.4*1 = 1.93TB

• Storage of event logs: 1500*30*86400*1KB*0.4*2 = 2.9TB

In this scenario, the storage space for Files is generally smaller than 150% of storage of session records. The storage space for metrics, which is the data of calculation to set up Mainboard and Live Chart, is less than 100MB every day while the storage space required for reports depends on their complexity and quantity.

To configure Storage Usage and expiration period:

STEP 1. Select **System** > **System Usage**. It displays Storage Usage percentage for Reports and Metrics, Files and Traffic Logs. You can view History Log Storage by time.

✍*Warning: Please take caution when performing STEP 2 and STEP 3, because this operation is not recoverable.*

STEP 2. (Optional)Click **Setting**. Check ☑I understand the risks. Enter the Max Days for each type and Click **OK**. TSG will delete logs and reports that exceed the expiration period. It may take a long time, during which the maximum sustainable log rate will degrade.

STEP 3. (Optional)To clear data once, click **Clear** to Clear Traffic Logs, Clear Reports and Metrics, Clear Files or Clear All Data.

System Appearance

System appearance enables user to customize firewall's logo, title and default language. Go to **System** > **System Appearance** to upload your own logo, favicon, specify title, and select default language.

Appendix A Built-in Category

Built-in Category

Dimension	Label	Description
Function	adv	All about advertising: This includes sites offering banners and banner creation as well as sites delivering banners to be shown in webpages. Advertising companies are listed, too.
	anonvpn	This category covers sites providing VPN services to the public. The focus is on VPN sites used to hide the origin of the traffic like tor nodes.
	audio-video	Some audio and video sites.
	bitcoin	Sites for bitcoin mining.
	cleaning	Sites to disinfect, update and protect computers.
	downloads	This covers mostly filesharing, p2p and torrent sites. Other download sites (for software, wallpapers...) are included as well.
	filehosting	Websites which host files (pictures, video, ...)
	forums	Discussion sites. Covered explicit forum sites and some blogs. Sites where people can discuss and share information in a non-interactive/real-time way. Realtime discussions are covered with -> chat.
	jobsearch	Site to looking for job.
	redirector	Sites that actively help to bypass URL filters by accepting URLs via webform and play a proxying and redirecting role.
	remotecontrol	Sites offering the service to remotely access computers, especially (but not limited to going) through firewalls. This includes using a third-party computer. Traditional VPN is not covered.
	social networks	Sites bringing people together (social networking) be it for friendship or for business.
	searchengines	Collection of search engines and directory sites.

	tracker	Site keeping an eye on where you surf and what you do in a passive. Covers web bugs, counters and other tracking mechanism in web pages that do not interfere with the local computer yet collecting information about the surfing person for later analysis. Sites actively spying out the surfer by installing software or calling home sites are not covered with tracker but with -> spyware.
	translation	Sites for translation
	url shortener	Domains that can be used to shorten long URLs. The original (long) URL will be accessed after the short URL has been requested from the shortener. This distinguishes this category from redirector where the original URL is never accessed directly.
	webmail	Webmail sites (hot mail like...)
	webphone	Sites that enable user to phone via Internet/WWW. Any site where users can voice-chat with each other (normal chat sites, where users type their messages are part of chat, not web phone).
	webradio	Sites that offer listening live radio streams.
	CDN	Content Delivery Networks
	blog	Some blog sites.
	chat	Sites for real-time chatting and instant messaging. Everything that is not real-time is included in -> forum.
National Security	aggressive	Sites of obvious aggressive content. This covers hate speech and all kinds of racism.
	Alcohol & Tobacco	Sites which legally promote or sell alcohol/tobacco products and accessories.
	science	Sites of institutions as well as of amateurs about all topics of astronomy, chemistry and science related
	astrology	Sites to provide daily horoscopes, online tarot readings, psychic readings
	bank	Online bank; Websites that feature banking and/or financial services.
	celebrity	Famous people, actors, and magazine which talk about them
	child	Any website allowed to child
	cooking	Sites concerning food and food preparation.

drugs	Sites offering drugs or explain how to make drugs (legal and non-legal). Covers tabaco as well as viagra and similar substances.
humor	Humorous pages, comic strips, funny stories, everything which makes people laugh.
lingerie	Sites selling and presenting sexy lingerie or lingerie in a sexy manner.
gambling	Gambling and games sites, casino, etc.; Sites offering the possibility to win money. Poker, Casino, Bingo and other chance games as well as betting sites. Differs from ->games in the aspect of winning or losing money or being lured to do so.
games	Sites about online games (all kinds of browser-based games). The games are for fun only (no gamble).
mixed adult	Websites which contain adult sections unstructured
movies	Sites offering cinema programs, information about movies and actors. Sites for downloading video clips/movies (as long it is legal) are included as well.
music	Sites that offer the download of music, information about music groups or music in general.
news	Sites presenting news. Homepages from newspapers, magazines and journals as well as some blogs.
politics	Sites of political parties, political organizations and associations; sites with political discussions.
porn	Sites about all kinds of sexual content ranging from bare bosoms to hardcore porn and sm.
press	Any press (informational) site
sexualeducation	Websites which talk about sexual education, and can be mis-detected as porn; Sites explaining the biological functions of the body concerning sexuality as well as sexual health; this, too, covers sites for teenagers with questions about first love, first sex, and subjects related to these topics. This category does not cover porn.
shopping	Sites offering online shopping and price comparisons.

sports	All about sports: sports teams, sport discussions as well as information about sports people and the various sports themselves.
travel	Sites with information about foreign countries, travel companies, travel fares, accommodations and everything else that has to do with travel including travel blogs.
traffic	Sites concerning the business of commercial transportation by land, sea or air.
violence	Sites about killing and harming people. Covers anything about brutality and bestiality.
Real Estate	Information on buying, selling or renting real estate or properties. Serve for buyers, sellers, landlords, renters, agents, and other home professionals. Work as a marketing channel for the professional service providers that contribute to the transaction. Realtors, mortgage professionals, banks, and inspectors can advertise on the website to generate leads.
religion	Sites with religious content: all kind of churches, sects, religious interpretations and so on.
weapons	Sites offering all kinds of weapons or accessories for weapons: Firearms, knifes, swords, bows... Armory shops are included as well as sites holding general information about arms (manufacturing, usage).
Government and Legal Organizations	Government: Sites sponsored by branches, bureaus, or agencies of any level of government, except for the armed forces, including courts, police institutions, city-level government institutions. Legal Organizations: Sites that discuss or explain laws of various government entities.
financial	Sites relative financial information.
dating	Dating, matching site for single person; Sites to contact people for love and live together. He seeks her, she seeks him and so on.
health	Sites about treatments for feeling internally and externally healthy
art	Sites that cater to fine arts, cultural behaviors and backgrounds including conventions, artwork and paintings, music, languages, customs, etc.

	computers	Sites for computers
	reference	Sites for reference
	Business	Sites for business
	Home	Sites about everything required to create a cozy home
	Recreation	Sites for recreation
Regional Related	Justice Ministry's federal list	A list of works that were banned in Russia, primarily based on the Russian Internet Restriction Bill.
Network Security	warez	Collection of sites offering programs to break license keys, license keys themselves, cracked software and other copyrighted material.
	Phishing	Counterfeit web pages that duplicate legitimate business web pages for the purpose of eliciting financial, personal or other private information from the users.
	Malicious Websites	Sites that host software that is covertly downloaded to a user's machine to collect information and monitor user activity, and sites that are infected with destructive or malicious software, specifically designed to damage, disrupt, attack or manipulate computer systems without the user's consent, such as virus or trojan horse.
	malware	Any website which delivers malware
	hacking	Sites that depict illicit activities surrounding the unauthorized modification or access to programs, computers, equipment and sites.
	Domain Parking	Sites that simply are place holders of domains without meaningful content.
Minority Language	Arabic	Arabic sites
	Uyghur	Uyghur sites

Appendix B Predefined Applications

App Name	App ID	Category	Subcategory	Description
Unknown _TCP	100	unknown		
Unknown _UDP	101	unknown		
Unknown _Other	102	unknown		
DNS	103	networking	infrastructure	(Domain Name System) The Internet's system for converting alphabetic names into numeric IP addresses.
FTP	104	general-internet	file-sharing	(File Transfer Protocol) A protocol used to transfer files over an IP network.
FTPS	105	general-internet	file-sharing	(FTP Secure) Transferring files over the secure sockets layer (SSL) protocol.
HTTP	106	general-internet	internet-utility	(HyperText Transfer Protocol) The communications protocol used to connect to Web servers on the Internet or on a local network (intranet).
HTTPS	107	general-internet	internet-utility	(HyperText Transport Protocol Secure) The protocol used to access a secure Web server.
ICMP	108	networking	ip-protocol	(Internet Control Message Protocol) A TCP/IP protocol used to send error and control messages.

IKE	109	networking	encrypted-tunnel	(Internet Key Exchange) A method for establishing a security association (SA) in the IPsec protocol. IKE authenticates users, negotiates the encryption method and exchanges the secret key.
MAIL	110	collaboration	email	
IMAPS	111	collaboration	email	IMAP (Internet Message Access Protocol) Secure
IPSEC	112	networking	encrypted-tunnel	(IP SECurity) A security protocol from the IETF that provides authentication and encryption over the Internet.
XMPP	113	networking	instant-messaging	Extensible Messaging and Presence Protocol
L2TP	114	networking	ip-protocol	
NTP	115	networking	infrastructure	
POP3S	117	collaboration	email	Pop3 (Post Office Protocol) Secure
PPTP	118	networking	infrastructure	
QUIC	119	networking	infrastructure	
SIP	120	collaboration	voip-video	(Session Initiation Protocol) A very simple text-based application-layer control protocol. It creates, modifies, and terminates sessions with one or more participants. Such sessions include Internet telephony and multimedia conferences.
SMB	121	business-systems	storage-backup	(Server Message Block) The file sharing protocol in DOS and Windows.
SMTPS	123	collaboration	email	(Simple Mail Transfer Protocol) Secure
SPDY	124	networking	encrypted-tunnel	(SPeeDY) A more efficient Web protocol from Google that is the basis of HTTP/2.

SSH	125	networking	encrypted-tunnel	(Secure SHell) A security protocol for logging into a remote server. SSH provides an encrypted session for transferring files and executing server programs. Also serving as a secure client/server connection for applications.
SSL	126	networking	encrypted-tunnel	(Secure Sockets Layer) An earlier security protocol that was the Internet standard prior to TLS. Developed by Netscape, SSL was widely used to validate the identity of a website and create an encrypted connection.
SOCKS	127	networking	proxy	(SOCKetS) A circuit-level proxy server for IP networks from the IETF. SOCKS is a general-purpose proxy that is typically used for non-HTTP traffic such as FTP, Telnet, etc. SOCKS4 handles TCP connections while SOCKS5 added support for UDP, ICMP, user authentication and hostname resolution.
TELNET	128	networking	remote-access	The Internet standard protocol for remote login. Runs on top of TCP/IP. Defined in STD 8, RFC 854 and extended with options by many other RFCs.
DHCP	129	networking	infrastructure	(Dynamic Host Configuration Protocol) The automatic assigning of IP addresses to client machines logging into an IP network.
RADIUS	130	business-systems	auth-service	(Remote Authentication Dial-In User Service) An authentication, authorization and accounting (AAA) protocol developed by Livingston Enterprises (later acquired by Lucent). RADIUS uses a challenge/response method for authentication and has been widely used prior to Diameter.

OPENVPN	131	networking	encrypted-tunnel	An open source virtual private network (VPN) protocol that is widely used by VPN services to provide anonymity and privacy to users. OpenVPN uses the OpenSSL library for encryption and authentication.
STUN	132	networking	infrastructure	(Session Traversal Utilities for NAT) An IETF protocol for real-time voice, video and messaging in an IP network. STUN provides the mechanism to communicate with users behind a network address translation (NAT) firewall, which keeps their IP addresses private within the local network (LAN).
TEREDO	133	networking	routing	
DTLS	134	networking	encrypted-tunnel	
DOH	135	networking	encrypted-tunnel	
ISAKMP	136	networking	encrypted-tunnel	(Internet Security Association and Key Management Protocol) A framework for establishing trusted interactions among entities using TCP/IP.
MDNS	137	networking	infrastructure	(Multicast DNS) Using IP multicast with DNS to provide the capabilities of a DNS server for service discovery in a small network that does not have a DNS server.
NETBIOS	138	networking	infrastructure	(Network Basic Input/Output System) It provides services related to the session layer of the OSI model allowing applications on separate computers to communicate over a local area network.

NETFLOW	139	business-systems	management	Routers and switches that support NetFlow can collect IP traffic statistics on all interfaces where NetFlow is enabled, and later export those statistics as NetFlow records toward at least one NetFlow collector - typically a server that does the actual traffic analysis.
RDP	140	networking	encrypted-tunnel	(Remote Desktop Protocol) An input/output presentation protocol from Microsoft that governs the keyboard, mouse and screen formatting between clients and servers in a thin client environment. Based on the T.share protocol, RDP is used by Microsoft's Terminal Services and Remote Desktop Services to provide thin client, centralized architecture for the datacenter.
RTCP	141	media	photo-video	RTCP is a companion protocol to RTP that is used to maintain QoS. RTP nodes analyze network conditions and periodically send each other RTCP packets that report on network congestion.
RTP	142	media	photo-video	(Realtime Transport Protocol) An IP protocol that supports real-time transmission of voice and video. It is widely used for IP telephony and audio and video streaming.
SLP	143	networking	infrastructure	(Service Location Protocol) An IETF standard used to announce and discover services such as printers and file shares on an IP network. Apple used SLP prior to Mac OS 10.2, but migrated to its Bonjour technology.

SNMP	144	networking	infrastructure	(Simple Network Management Protocol) A widely used network monitoring and control protocol. Data are passed from SNMP agents, which are hardware and/or software processes reporting activity in each network device (hub, router, bridge, etc.) to the workstation console used to oversee the network. The agents return information contained in a MIB (Management Information Base), which is a data structure that defines what is obtainable from the device and what can be controlled (turned off, on, etc.).
SSDP	145	networking	infrastructure	(Simple Service Discovery Protocol) A standard for advertising services on a TCP/IP network and discovering them.
TFTP	146	general-internet	file-sharing	(Trivial File Transfer Protocol) A lightweight version of the FTP protocol that has no directory browsing or password capability.
BJNP	147	business-systems	management	
LDAP	148	business-systems	auth-service	(Lightweight Directory Access Protocol) A protocol used to access a directory listing in a TCP/IP network. LDAP is used to query network directories, email servers and other information repositories.
RTMP	149	media	photo-video	(Real-Time Messaging Protocol) A proprietary streaming protocol from Adobe that is supported in its Flash Media Server/Adobe Media Server platforms. RTMP defines "virtual channels" that operate independently (audio, video, control messages, etc.).

RTSP	150	media	photo-video	(RealTime Streaming Protocol) An application layer protocol used to transmit streaming audio, video and 3D animation over the Internet. It enables the user's client software to provide remote control of the server with functions such as pause, rewind and fast forward.
SSL with ESNI	8008	networking	encrypted	SSL with Encrypted SNI

App Name	App ID	Category	Subcategory
alipay	1100	business-systems	management
baiduhd	1200	collaboration	instant-messaging
taobao	1300	general-internet	internet-utility
eleme	1400	general-internet	marketing
meituan	1500	general-internet	marketing
mobileqq	1600	collaboration	instant-messaging
MobileQQ	1700	general-internet	web-posting
pinduoduo	1800	general-internet	internet-utility
qq	1900	collaboration	instant-messaging
QQ.VoiceCalls	1901	collaboration	voip-video
QQ.VideoCalls	1902	collaboration	voip-video
QQ.Message	1903	collaboration	instant-messaging
QQ.MessageFile	1904	general-internet collaboration	file-sharing voip-video
QQ.Login	1905	business-systems	auth-service
qqfile	2500	general-internet	file-sharing
sougouseries	2600	collaboration	instant-messaging
webqq	2700	collaboration	instant-messaging
txweiyun	2800	general-internet	file-sharing
youdaonote	2900	business-systems	office-programs
baiduclouddisk	3000	general-internet	file-sharing
applepush	3100	general-internet	internet-utility
badoo	3200	collaboration	social-networking
Badoo.VideoCalls	3201	collaboration	voip-video
Badoo.Message	3202	collaboration	instant-messaging

Badoo.Live	3203	collaboration	photo-video
Badoo.SurfingWebs	3204	collaboration	social-networking
drugvokrug	3700	collaboration	instant-messaging
Drugvokrug.Message	3701	collaboration	instant-messaging
firechat	3900	collaboration	instant-messaging
googlepush	4000	general-internet	internet-utility
Mail.ruAgent	4100	collaboration	email
Mail.ruAgent.VideoCalls	4101	collaboration	email
Mail.ruAgent.Message	4102	collaboration	instant-messaging
MicrosoftUpdate	4400	business-systems	general-business
Odnoklassniki	4500	collaboration	social-networking
Odnoklassniki.VideoCalls	4501	collaboration	social-networking
Odnoklassniki.Message	4502	collaboration	instant-messaging
SignalPrivateMessenger	4800	collaboration	instant-messaging
SignalPrivateMessenger. MessageFile	4801	collaboration	instant-messaging
Skype	5000	collaboration	voip-video
Skype.VideoCalls	5001	collaboration	voip-video file-sharing
Skype.Message	5002	collaboration	instant-messaging
Skype.MessageFile	5003	collaboration	voip-video
Snapchat	5400	collaboration	instant-messaging
Telegram	5500	collaboration	instant-messaging
Telegram.VideoCalls	5501	collaboration	voip-video
Twitter	5700	collaboration	auth-service social-networking
Twitter.Message	5701	collaboration	social-networking
viber	5800	collaboration	voip-video
Viber.VoiceCalls	5801	collaboration	voip-video
Viber.Message	5802	collaboration	auth-service voip-video
Viber.MessageFile	5803	collaboration	file-sharing
Vkontakte	6300	collaboration	instant-messaging auth-service social-networking
Vkontakte.VideoCalls	6301	collaboration	voip-video
Vkontakte.Live	6302	collaboration	photo-video

Vkontakte.VOD	6303	collaboration	photo-video
WeChat	6700	collaboration	auth-service instant-messaging
WeChat.VideoCalls	6701	collaboration	voip-video
WeChat.Message	6702	collaboration	instant-messaging
WeChat.MessageFile	6703	general-internet	file-sharing
whatsapp	7100	collaboration	auth-service instant-messaging
WhatsApp.VideoCalls	7101	collaboration	voip-video instant-messaging
WhatsApp.Message	7102	collaboration	instant-messaging
WhatsApp.MessageFile	7103	general-internet	file-sharing
CMBMobileBank	7400	business-systems	general-business
ICBCMobileBank	7500	business-systems	general-business
bilibili	7600	media	photo-video
douyutv	7700	media	photo-video
facebook	7800	media collaboration	photo-video auth-service social-networking
Facebook.VideoCalls	7801	collaboration	voip-video
Facebook.Message	7802	collaboration	instant-messaging
Facebook.Live	7803	media	photo-video
Facebook.MessageFile	7804	media	photo-video
iqiyi	8300	media	photo-video
wangyiyunmusic	8400	media	audio-streaming
periscope	8500	media	photo-video
qqmusic	8600	media	audio-streaming
youtube	8700	media	photo-video
YouTube.Live	8701	media	photo-video
YouTube.VOD	8702	media	photo-video
YouTube.UploadFiles	8703	media	photo-video
YouTube.Search	8704	collaboration	web-posting
ArcadeVPN	9200	networking	encrypted-tunnel
Betternet	9300	networking	encrypted-tunnel
BrowsecVPN	9400	networking	encrypted-tunnel
DubkiVPN	9500	networking	encrypted-tunnel
EasyVPN	9600	networking	encrypted-tunnel

FishVPN	9700	networking	encrypted-tunnel
FoxVPN	9800	networking	encrypted-tunnel
VPNFrance	9900	networking	encrypted-tunnel
freegate	10000	networking	proxy
FreeVPNProxy	10100	networking	proxy
FreeVPNUnlimitedProxy	10200	networking	proxy
VPNGermany	10300	networking	proxy
GooseVPN	10400	networking	proxy
Heywire	10500	networking	proxy
HidemeVPN	10600	networking	proxy
hidemyass	10700	networking	proxy
HolaVPN	10800	networking	encrypted-tunnel
HotSpotshieldVPN	10900	networking	encrypted-tunnel
HotSpotVPN	11000	networking	encrypted-tunnel
LunaVPN	11100	networking	encrypted-tunnel
LuxVPN	11200	networking	encrypted-tunnel
VPNMalaysia	11300	networking	encrypted-tunnel
orbot	11400	networking	encrypted-tunnel
PandaVPN	11500	networking	encrypted-tunnel
psiphon	11600	networking	proxy
PureVPN	11700	networking	encrypted-tunnel
SecureVPN	11800	networking	encrypted-tunnel
SetupVPNChromeExtension	11900	networking	encrypted-tunnel
SuperVPN	12000	networking	encrypted-tunnel
SurfSharkVPN	12100	networking	proxy
Talk2	12200	networking	proxy
ThunderVPN	12300	networking	proxy
TOROrbot	12400	networking	encrypted-tunnel
TunnelBear	12500	networking	encrypted-tunnel
TurboVPN	12600	networking	encrypted-tunnel
USAVPN	12700	networking	encrypted-tunnel
UVPN	12800	networking	encrypted-tunnel
VPNCanada	12900	networking	encrypted-tunnel
VPNIndonesia	13000	networking	encrypted-tunnel
VPNIndia	13100	networking	encrypted-tunnel
VPNJapan	13200	networking	encrypted-tunnel
VPNKorea	13300	networking	encrypted-tunnel

VPNProxyMaster	13400	networking	encrypted-tunnel
AvastVPN	13500	networking	encrypted-tunnel
VPNRussia	13600	networking	encrypted-tunnel
VPNTap2free	13700	networking	encrypted-tunnel
VPNUkraine	13800	networking	encrypted-tunnel
VPNUSA	13900	networking	encrypted-tunnel
VyprVPN	14000	networking	encrypted-tunnel
WolfVPN	14100	networking	encrypted-tunnel
Ultrasurf	14200	networking	encrypted-tunnel
ZenmateVPN	14300	networking	proxy
SecurityKiss	14400	networking	encrypted-tunnel
VPNOneClick	14500	networking	encrypted-tunnel
NordVPN	14600	networking	encrypted-tunnel
HoxxVPN	14700	networking	encrypted-tunnel
WindscribeVPN	14800	networking	encrypted-tunnel
TouchVPN	14900	networking	encrypted-tunnel
VPN365	15000	networking	encrypted-tunnel
ProtonVPN	15100	networking	encrypted-tunnel
OVPNspider	15200	networking	encrypted-tunnel
BusinessVPN	15300	networking	encrypted-tunnel
VPNTT	15400	networking	encrypted-tunnel
FoxFlyRouter	15500	networking	encrypted-tunnel
LaowangVPN	15600	networking	encrypted-tunnel
SurfVPN	15700	networking	encrypted-tunnel
BaiJingVPN	15800	networking	encrypted-tunnel
FlexVPN	15900	networking	encrypted-tunnel
LionVPN	16000	networking	encrypted-tunnel
VPNIsrael	16100	networking	encrypted-tunnel
PandaVPNPro	16200	networking	encrypted-tunnel
LieBaoJiaSuQi	16300	networking	encrypted-tunnel
AceVPN	16400	networking	encrypted-tunnel
ShotVPN	16500	networking	encrypted-tunnel
McAfeeVPN	16600	networking	encrypted-tunnel
SmartVPN	16700	networking	encrypted-tunnel
RusVPN	16800	networking	encrypted-tunnel
VPNify	16900	networking	encrypted-tunnel
UnlockSecureFreeVPN	17000	networking	encrypted-tunnel

BestVPNProxy	17100	networking	encrypted-tunnel
MelonVPN	17200	networking	encrypted-tunnel
FreeUnlimitedVPNProxy	17300	networking	encrypted-tunnel
FreeVPNUnblockProxy	17400	networking	encrypted-tunnel
VPNProxyMasterPro	17500	networking	encrypted-tunnel
VPNProxyMasterLite	17600	networking	encrypted-tunnel
CyberGuardVPN	17700	networking	encrypted-tunnel
TopVPN	17800	networking	encrypted-tunnel
FreeZenVPN	17900	networking	encrypted-tunnel
VPNEasy	18000	networking	encrypted-tunnel
SetupVPN	18100	networking	encrypted-tunnel
VPNHub	18200	networking	encrypted-tunnel
FreeVPNWhiteKey	18300	networking	encrypted-tunnel
FreeVPNColorKey	18400	networking	encrypted-tunnel
EutVPN	18500	networking	encrypted-tunnel
UFOVPNBasic	18600	networking	encrypted-tunnel
FinchVPN	18700	networking	encrypted-tunnel
FlashVPN	18800	networking	encrypted-tunnel
LunaVPNFreeProxy	18900	networking	encrypted-tunnel
StarVPN	19000	networking	encrypted-tunnel
PlexVPN	19100	networking	encrypted-tunnel
DailyVPN	19200	networking	encrypted-tunnel
CyberGhostVPN	19300	networking	encrypted-tunnel
VPNLite	19400	networking	encrypted-tunnel
SnapVPN	19500	networking	encrypted-tunnel
VPNRobot	19600	networking	encrypted-tunnel
BESTVPN	19700	networking	encrypted-tunnel
JapanVPNMasterFree	19800	networking	encrypted-tunnel
IPVanishVPN	19900	networking	encrypted-tunnel
UnlimitedVPN	20000	networking	encrypted-tunnel
OneVPN	20100	networking	encrypted-tunnel
AppsverseAvVPN	20200	networking	encrypted-tunnel
CloudVPN	20300	networking	encrypted-tunnel
VPNInf	20400	networking	encrypted-tunnel
MelonVPNNew	20500	networking	encrypted-tunnel
GeckoVPN	20600	networking	encrypted-tunnel
Seed4MeVPN	20700	networking	encrypted-tunnel

MatrixVPN	20800	networking	encrypted-tunnel
ElectroVPN	20900	networking	encrypted-tunnel
BVPN	21000	networking	encrypted-tunnel
WASELProVPN	21100	networking	encrypted-tunnel
IWASEL	21200	networking	encrypted-tunnel
CatVPN	21300	networking	encrypted-tunnel
PhoneGuardianVPN	21400	networking	encrypted-tunnel
VpnBeaver	21500	networking	encrypted-tunnel
SpeedVPN	21600	networking	encrypted-tunnel
TroidVPN	21700	networking	encrypted-tunnel
SmallRabbitVPN	21800	networking	encrypted-tunnel
ArmadaVPN	21900	networking	encrypted-tunnel
GOVPN	22000	networking	encrypted-tunnel
IvacyVPN	22100	networking	encrypted-tunnel
VPNGO	22200	networking	encrypted-tunnel
Try2CatchVPN	22300	networking	encrypted-tunnel
BoxPN	22400	networking	encrypted-tunnel
VPNTunnel	22500	networking	encrypted-tunnel
VPNAI	22600	networking	encrypted-tunnel
FreeVPN	22700	networking	encrypted-tunnel
YinHeVPN	22800	networking	encrypted-tunnel
FreeSSVPN	22900	networking	encrypted-tunnel
FreeMikaVPN	23000	networking	encrypted-tunnel
AntVPN	23100	networking	encrypted-tunnel
WaspVPN	23200	networking	encrypted-tunnel
RejinderiFreeVPN	23300	networking	encrypted-tunnel
GoatVPN	23400	networking	encrypted-tunnel
SoloVPN	23500	networking	encrypted-tunnel
HouWangVPN	23600	networking	encrypted-tunnel
Nash	23700	networking	encrypted-tunnel
HOTVPN	23800	networking	encrypted-tunnel
WORLDVPN	23900	networking	encrypted-tunnel
HexatechVPN	24000	networking	encrypted-tunnel
R-VPN	24100	networking	encrypted-tunnel
SatuVPN	24200	networking	encrypted-tunnel
PolestarVPN	24300	networking	encrypted-tunnel
VPNMonster	24400	networking	encrypted-tunnel

BuddyVPN	24500	networking	encrypted-tunnel
VPNEARTH	24600	networking	encrypted-tunnel
FastFreeVPN	24700	networking	encrypted-tunnel
liquiduRocketVPN	24800	networking	encrypted-tunnel
TunnelLight	24900	networking	encrypted-tunnel
NetVPN	25000	networking	encrypted-tunnel
SmartLockVPN	25100	networking	encrypted-tunnel
BoxVPN	25200	networking	encrypted-tunnel
TheBestVPNHotspotMaster	25300	networking	encrypted-tunnel
VPNChina	25400	networking	encrypted-tunnel
VPNBlue	25500	networking	encrypted-tunnel
BlueVPN	25600	networking	encrypted-tunnel
VPNPlus	25700	networking	encrypted-tunnel
VPNBlue	25800	networking	encrypted-tunnel
VPNCity	25900	networking	encrypted-tunnel
FreshVPN	26000	networking	encrypted-tunnel
PhoenixVPN	26100	networking	encrypted-tunnel
LightVPN	26200	networking	encrypted-tunnel
BESTVPN	26300	networking	encrypted-tunnel
LvYeVPN	26400	networking	encrypted-tunnel
VPNBOTTLE	26500	networking	encrypted-tunnel
VPNExpress	26600	networking	encrypted-tunnel
PrivatixVPN	26700	networking	encrypted-tunnel
WhatsVPN	26800	networking	encrypted-tunnel
OrangeVPN	26900	networking	encrypted-tunnel
SuperVPN	27000	networking	encrypted-tunnel
LiteVPN	27100	networking	encrypted-tunnel
JustVPN	27200	networking	encrypted-tunnel
HoxxVPN	27300	networking	encrypted-tunnel
OmanVPN	27400	networking	encrypted-tunnel
TaiwanVPN	27500	networking	encrypted-tunnel
WangVPN	27600	networking	encrypted-tunnel
EgyptVPN	27700	networking	encrypted-tunnel
BrazilVPN	27800	networking	encrypted-tunnel
SpainVPN	27900	networking	encrypted-tunnel
NewZealandVPN	28000	networking	encrypted-tunnel
PhilippinesVPN	28100	networking	encrypted-tunnel

IndonesiaVPN	28200	networking	encrypted-tunnel
VPNBestVPN	28300	networking	encrypted-tunnel
FROGVPN	28400	networking	encrypted-tunnel
BlackVPN	28500	networking	encrypted-tunnel
ChinaFastVpn	28600	networking	encrypted-tunnel
RussiaFastVPN	28700	networking	encrypted-tunnel
LaoSongVPN	28800	networking	encrypted-tunnel
IranVPN	28900	networking	encrypted-tunnel
ZVPN	29000	networking	encrypted-tunnel
OMGVPN	29100	networking	encrypted-tunnel
AmigoVPN	29200	networking	encrypted-tunnel
FreeVPN	29300	networking	encrypted-tunnel
PhilippinesVPN	29400	networking	encrypted-tunnel
EgyptVPN	29500	networking	encrypted-tunnel
PurpleVPN	29600	networking	encrypted-tunnel
MambaVPN	29700	networking	encrypted-tunnel
AirVPN	29800	networking	encrypted-tunnel
LunaVPN	29900	networking	encrypted-tunnel
UFOVPN-NetVPN	30000	networking	encrypted-tunnel
ArvinVPN	30100	networking	encrypted-tunnel
OwlVPNFree	30200	networking	encrypted-tunnel
HalleyVPN	30300	networking	encrypted-tunnel
HotFreeVPN	30400	networking	encrypted-tunnel
VPNPro	30500	networking	encrypted-tunnel
FaceTime.VideoCalls	30600	collaboration	voip-video
zello	30700	collaboration	social-networking
Zello.MessageVoice	30701	collaboration	voip-video
appstore	30900	general-internet	internet-utility
google	31000	general-internet	internet-utility
instagram	31100	media	photo-video
Instagram.VideoCalls	31101	media	photo-video
Instagram.Message	31102	media	photo-video
Instagram.Live	31103	media	photo-video
JingDong	31500	business-systems	general-business
microsoftteam	31600	collaboration	internet-conferencing

Operamini	31700	general-internet networking	internet-utility encrypted-tunnel
GooglePlayMarket	31800	general-internet	internet-utility
GooglePlayMarket. DownloadFiles	31801	general-internet	internet-utility
GooglePlayMarket.Search	31802	general-internet	internet-utility
QQWebVideo	32100	collaboration	voip-video
QQVideoChat	32200	collaboration	voip-video
Weibo	32300	collaboration	social-networking
BaiDuTieBa	32400	collaboration	social-networking
WPS	32500	business-systems	general-business
Yandex	32600	general-internet	file-sharing
Office365	32700	business-systems	office-programs
slack	32800	collaboration	instant-messaging voip-video
slack.Login	32901	collaboration	social-business
slack.Logout	32902	collaboration	social-business
YandexTurbo	33100	general-internet	file-sharing
Teamviewer	33200	networking	photo-video
Teamviewer.TransferFile	33201	networking	file-sharing
Teamviewer.RemoteControl	33202	networking	remote-access
XiangRiKui	33500	networking	remote-access file-sharing
Xiangrikui.Login	33501	networking	auth-service
HuaTianOA	33700	business-systems	general-business
PHPOA	33800	collaboration	web-posting
78OA	33900	collaboration	web-posting
TongDaOA	34000	collaboration	web-posting
WeaverOA	34100	general-internet	gaming
RanZhiOA	34200	collaboration	web-posting
JinDieOA	34300	business-systems	management
Tita	34400	business-systems	management
IIEARP	34500	business-systems	general-business
WindowsRemote	34600	networking	remote-access
HuaWeiCloudWeLink	34700	business-systems	voip-video management

HuaWeiCloudWeLink.VideoCalls	34701	business-systems	office-programs
FeiShu	34900	business-systems	management
dingding	35000	business-systems	management
DingDing.VideoCalls	35001	business-systems	management
CSTCloud	35200	business-systems	internet-conferencing
CSTCloud.Login	35201	business-systems	internet-conferencing
ShiMoDocs	35400	business-systems	office-programs
WeiXinWork	35500	collaboration	instant-messaging
WeiXinWork.VideoCalls	35501	collaboration	voip-video
TencentDocs	35700	business-systems	office-programs
YinXiangNotes	35800	business-systems	office-programs
YouDaoNotes	35900	business-systems	office-programs
TencentMeeting	36000	collaboration	internet-conferencing
TencentMeeting.VideoCalls	36001	collaboration	internet-conferencing
Trello	36200	business-systems	management
Quip	36300	business-systems	general-business
XYLink	36400	collaboration	internet-conferencing
XYLink.VideoCalls	36401	collaboration	internet-conferencing
XYLink.Login	36402	collaboration	internet-conferencing
Teambition	36700	business-systems	office-programs
Teambition.Login	36701	business-systems	office-programs
vpngate	36900	meida	gaming
webex	37000	collaboration	internet-conferencing
webex.login	37001	collaboration	internet-conferencing
tunnel_behavior	37200	networking	encrypted-tunnel

Appendix C Log Fields Description

✎*The column with * is the default display column after logging in to the system for the first time. Once the user has made the configuration, the configuration will be saved in the browser's local data. The next time the user logs in to the system through the same machine and the same browser, it will display columns that the user has previously configured. The fields with bold font are the Log Fields used when creating reports.*

Log Type

Log Type	Schema Type
Security Events	All types
Proxy Events	Base, HTTP and DoH
Session Records	All types except Radius
Radius Logs	Base and Radius
VoIP Records	Base, SIP and RTP
GTP-C Records	Base and GTP-C

Dos Events

Field	Description
Start Time	The start time of the attack
End Time	The end time of the attack
Log ID	A log entry identifier incremented sequentially; each log has a unique number

Attack Type	The type includes: • TCP SYN Flood, • UDP Flood, • ICMP Flood, • DNS Flood, • DNS Amplification.
Severity	Critical Severe Major Warning Minor
Conditions	The conditions that trigger the event
Destination IP	The destination IP address
Destination Country	The destination country name
Source IPs	The source IP addresses
Source Countries	The source country names
Sessions/s	Session rate
Packets/s	Packet rate
Bits/s	Bit rate

Base

Field	Description
Log ID *	A log entry identifier incremented sequentially; each log has a unique number
Receive Time *	Time the log was received
Subscriber ID *	Identifier of RADIUS Accounting for Subscriber Access (if applicable)
Client IP *	Original session client IP address.
Internal IP	Internal region IP of the session (if applicable)
Client Port	Client port utilized by the session
L4 Protocol	Transport layer protocol associated with the session
Address Type	IP protocol version associated with the session, 4 or 6

Server IP *	Original session server IP address
Server Port *	Server port utilized by the session
External IP	External region IP of the session (if applicable)
Action	Action taken for the session; possible values are: • Allow - session was allowed by policy. • Deny - session was denied by policy. • Monitor - session was allowed by policy and a log will be generated when matched. • Intercept - Intercept HTTP/HTTPS traffic for proxy. If the traffic use SSL/TSL, it will be decrypted. • Redirect - The Proxy redirect matched HTTP session to a predefined URL. • Replace - The Proxy Searches in a given HTTP part to Find a given string, and Replace any matches with another given string. • Hijack - The Proxy hijack a downloading file. • Insert - The Proxy insert a “js” or “css” scripts to webpages.
Direction	Indicates session client-to-server direction, possible values are: Egress—Internal to External Ingress—External to Internal
Sled IP	IP of sled which the session was processed
Client Location	Geographic location the client IP
Client ASN	BGP Autonomous system number the client IP
Server Location	Geographic location the server IP
Server ASN	BGP Autonomous system number the server IP
Sessions	Number of sessions with same client IP, server IP, Application, seen within 5 seconds
Packets Sent	Number of client-to-server packets for the session
Packets Received	Number of server-to-client packets for the session
Bytes Sent	Number of bytes in the client-to-server direction of the session

Bytes Received	Number of bytes in the server-to-client direction of the session
Sub Action *	Sub Action taken for action; possible values are: drop—session was dropped by deny action block—session was blocked by deny action alert—session was alerted by deny action allow—session was allowed by intercept action deny—session was denied by intercept action monitor—session was monitored by intercept action redirect—session was redirected by intercept action replace—session was replaced by intercept action hijack—session was hijacked by intercept action insert—session was inserted by intercept action
Device ID	Unique identifier of devices on which the session was logged
Data Center	Name of data center on which the session was logged
Application Label	Application label associated with the session
Protocol Label	Protocol associated with the session
L7 Protocol	Layer 7 Protocol associated with the session
Start Time	Time of session start
End Time	Time of session end
Establish Latency	Establish time of the session
Duration(ms)	Elapsed time of the session
Stream Direction	Captured packet direction of the session, possible values are: c2s, s2c, double
Session ID	An internal numerical identifier applied of the session
Fragmentation Packets(c2s)	Number of IP fragment packets in client-to-server direction of the session
Fragmentation Packets(s2c)	Number of IP fragment packets in server-to-client direction of the session
Sequence Gap Loss(c2s)	Number of TCP gap loss packets in client-to-server direction of the session

Sequence Gap Loss(s2c)	Number of TCP gap loss packets in server-to-client direction of the session
Unorder Packets(c2s)	Number of TCP out of order packets in client-to-server direction of the session
Unorder Packets(s2c)	Number of TCP out of order packets in server-to-client direction of the session
TCP Client ISN	TCP uses a three-way handshake to establish a reliable connection. The connection is full duplex, and both sides synchronize (SYN) and acknowledge (ACK) each other. The client chooses an initial sequence number, set in the first SYN packet. Initial sequence numbers (ISN) refers to the unique 32-bit sequence number assigned to each new connection on a TCP-based data communication. An ISN is unique to each connection and separated by each device. Now use a random number in ISN selection process to defeat malicious attacks.
TCP Server ISN	The server also chooses its own initial sequence number, set in the SYN/ACK packet. Each side acknowledges each other's sequence number by incrementing it.

Log Fields per Protocol

HTTP

Field	Description
Http.URL	Uniform Resource Locator (URL) of current HTTP session
Http.Host	Host of current HTTP session
Http.Domain	Domain of current HTTP session
Http.Cookie	Cookie of current HTTP session
Http.Referer	Reference of current HTTP session
Http.User Agent	User-Agent of current HTTP session
Http.Content Length	Content Length of current HTTP session

Http.Content Type	Content Type of current HTTP session
Http.Set Cookie	Set Cookie of current HTTP session
Http.Version	HTTP protocol version of current HTTP session, possible values are: http1, http2
Http.Response Latency(ms)	HTTP response latency of current HTTP session
Http.Action File Size	HTTP sub action profile size of current HTTP session, possible sub actions are: alert hijack insert
Http.Session Duration(ms)	Duration time of current HTTP session
HTTP.Request Body	The request body to current HTTP session
HTTP.Response Body	The response body to current HTTP session

MAIL

Field	Description
Mail.Protocol Type	Specific mail protocol type of current Mail session, possible values are: smtp, imap, pop
Mail.Account	Account of current Mail session
Mail.From CMD	From information of current Mail session
Mail.To CMD	To information of current Mail session
Mail.From	Sender account of current mail session
Mail.To	Receiver account of current mail session
Mail.CC	Mail carbon copy account of current mail session
Mail.BCC	Mail blind carbon copy account of current mail session
Mail.Subject	Subject of current mail session
Mail.Attachment	Attachment name of current mail session
Mail.EML File	EML file download link of current mail session

DNS

Field	Description
Dns.Message ID	Message identifier of current DNS session
Dns.QR	Query/Response Flag of current DNS session

Dns.OPCODE	Operation Code of current DNS session
Dns.AA	Authoritative Answer Flag of current DNS session
Dns.TC	Truncation Flag of current DNS session
Dns.RD	Recursion Desired of current DNS session
Dns.RA	Recursion Available of current DNS session
Dns.RCODE	Response Code of current DNS session
Dns.QDCOUNT	Question Count of current DNS session
Dns.ANCOUNT	Answer Record Count of current DNS session
Dns.NSCOUNT	Authority Record Count of current DNS session
Dns.ARCOUNT	Additional Record Count of current DNS session
Dns.QNAME	Query domain name of current DNS session
Dns.QTYPE	Query type of current DNS session
Dns.QCLASS	Query class of current DNS session
Dns.cname	Canonical Name Record of current DNS session
Dns.SUB	Extension description of current DNS session, possible values are: dns, dnssec
Dns.RR	Resource records of current DNS session

SSL

Field	Description
SSL.SNI	Server Name Indication of current SSL session
SSL.SAN	Subject Alternative Name of current SSL session
SSL.CN	Common Name of current SSL session
SSL.Pinning	Pinning status of current SSL session, possible values are: 0-not pinning 1-pinning 2-maybe pinning
SSL.Intercept State	Intercept State of current SSL session, possible values are: 0-passthrough: not decrypted 1-intercept: decrypted 2-shutdown: connection due to exception
SSL.Server Side Latency(ms)	Server side establish latency of current SSL session

SSL.Client Side Latency(ms)	Client side establish latency of current SSL session
SSL.Server Side Version	Server-side SSL version latency of current session, possible values are: SSLv3, TLSv1, TLSv1.1, TLSv1.2, TLSv1.3, unknown
SSL.Client Side Version	Client-side SSL version latency of current session, possible values are: SSLv3, TLSv1, TLSv1.1, TLSv1.2, TLSv1.3, unknown
SSL.Certificate Verify	Certificate verify status of current SSL session, possible values are: 0-passed 1-failed
SSL.Error	Error message of current SSL session
SSL.JA3 hash	JA3 Fingerprint is hashed with MD5
SSL.Connection Latency(ms)	SSH handshake latency time
SSL.Issuer	SSL certificate issuer
SSL.Subject	SSL certificate owner

QUIC

Field	Description
Quic.Version	Version of current QUIC session
Quic.SNI	Server Name Indication of current QUIC session
Quic.User Agent	User Agent of current QUIC session

DoH

Field	Description
DoH.URL	Uniform Resource Locator (URL) of current DoH session
DoH.Host	Host of current DoH session
DoH.Cookie	Cookie of current DoH session
DoH.Referer	Referer of current DoH session
DoH.User Agent	User Agent of current DoH session
DoH.Version	HTTP Version of current DoH session, possible values are: http1, http2

DoH.Message ID	Message ID of current DoH session
DoH.QR	Query/Response Flag of current DoH session
DoH.OPCODE	Operation Code of current DoH session
DoH.AA	Authoritative Answer Flag of current DoH session
DoH.TC	Truncation Flag of current DoH session
DoH.RD	Recursion Desired of current DoH session
DoH.RA	Recursion Available of current DoH session
DoH.RCODE	Response Code of current DoH session
DoH.QDCOUNT	Question Count of current DoH session
DoH.ANCOUNT	Answer Record Count of current DoH session
DoH.NSCOUNT	Authority Record Count of current DoH session
DoH.ARCOUNT	Additional Record Count of current DoH session
DoH.QNAME	Query domain name of current DoH session
DoH.QTYPE	Query type of current DoH session
DoH.QCLASS	Query class of current DoH session
DoH.CNAME	Canonical Name Record of current DoH session
DoH.SUB	Extension description of current DoH session, possible values are: dns, dnssec
DoH.RR	Resource records of current DoH session

FTP

Field	Description
Ftp.Account	Account of current ftp session
Ftp.URL	Uniform Resource Identifier of current ftp session
Ftp.Content	File content of current ftp session
FTP.Link Type	FTP link type

SIP

Field	Description
SIP.Call-ID	Call identifier of current SIP session
SIP.Originator	Originator description of current session
SIP.Responder	Responder description of current session

SIP.User-Agent	A user agent is a logical network endpoint that sends or receives SIP messages and manages SIP sessions.
SIP.Server	A network server with UAC and UAS components
SIP.Originator IP	IP address of the originator
SIP.Originator Port	Port of the originator
SIP.Originator Media Type	The media type of the originator
SIP.Originator Content	The content information of the originator, using Session Description Protocol(SDP)
SIP.Responder IP	IP address of the responder
SIP.Responder Port	Port of the responder
SIP.Responder Media Type	The media type of the responder
SIP.Responder Content	The content information of the responder using Session Description Protocol(SDP)
SIP.Duration	Duration time of current SIP session
SIP.Bye	Signal termination of a dialog and end a call. This message may be sent by either endpoint of a dialog.

RTP

Field	Description
RTP.Payload(c2s)	Payload type encodings sequence number of c2s
RTP.Payload(s2c)	Payload type encodings sequence number of s2c
RTP.PCAP	Packet capture path; there are maybe two paths for asymmetric traffic.
RTP.Direction	Originator direction. 0: unknown 1: c2s 2: s2c

GTP-C

Field	Description
Version	GTP-C version number
APN	Access Point Name
IMEI	International Mobile Equipment Identity

IMSI	International Mobile Subscriber Identity
Phone Number	Phone Number
Uplink TEID	Uplink TEID
Downlink TEID	Downlink TEID
Message Type	Create, modify, delete
End User Address V4	End user IPv4 address
End User Address V6	End user IPv6 address

RADIUS

Field	Description
Packet Type	Packet type of current radius session, possible values are: 1- Access-Request 2- Access-Accept 3- Access-Reject 4- Accounting-Request 5- Accounting-Response 11-Access-Challenge
Account	RADIUS account of current radius session
Nas IP *	NAS device IP of current radius session
Framed IP *	IP address assigned by RADIUS server to user of current radius session
Session Timeout	Timeout of current radius session
Idle Timeout	Maximum idle time of current radius session
ACC Status Type *	Account status type of current radius session, possible values are: 1-Start 2-Stop 3-Interim-Update

Acct Terminate Cause	Account terminating reason of current radius session, possible values are: 1-User Request 2-Lost Carrier 3-Lost Service 4-Idle Timeout 5-Session Timeout 6-Admin Reset 7-Admin Reboot 8-Port Error 9-NAS Error 10-NAS Request 11-NAS Reboot 12-Port Unneeded 13-Port Preempted 14-Port Suspended 15-Service Unavailable 16-Callback 17-User Error 18-Host Request
Event Timestamp	Dial-in user request event time stamp of current radius session
Service Type	Dial-in user request authentication type, possible values are: 2(Framed): A Framed Protocol should be started for the User, such as PPP or SLIP. 5(Outbound): The user should be granted access to outgoing devices. 10(Call Check): Used by the NAS in an Access-Request packet to indicate that a call is being received and that the RADIUS server should send back an Access-Accept to answer the call.
Nas Port	NAS-Port of current radius session

Framed Protocol	Framed Protocol of current radius session, possible values are: 1-PPP 2-SLIP 3-AppleTalk Remote Access Protocol (ARAP) 4-Gandalf proprietary SingleLink/MultiLink protocol 5-Xylogics proprietary IPX/SLIP 6-X.75 Synchronous
Callback Number	A dialing number to be used for callback of current radius session
Callback ID	Callback Identifier of current radius session
Termination Action	Action the NAS should take when the specified service is completed, possible values are: 0-Default 1-RADIUS-Request
Called Station Id	NAS device Identifier of current radius session
Calling Station Id	Client Identifier of current radius session
Acct Delay Time	Indicates how many seconds the user has been trying to send record of current radius session
Acct Session ID	A unique accounting ID of current radius session
Acct Multi Session ID	A unique Accounting ID to make it easy to link together multiple related sessions
Acct Input Octets	Input bytes of current radius account
Acct Output Octets	Output bytes of current radius account
Acct Input Packets	Input Packets of current radius account
Acct Output Packets	Output Packets of current radius account
Acct Session Time	Indicates how many seconds the user has received service for
Acct Link Count	Count of links which are known to have been in a given multilink session at the time the accounting record
Acct Interim Interva	l Indicates the number of seconds between each interim update in seconds

APP

Field	Description
APP.Extra Info	Detail information about the application.

Appendix E TSG Packet Flow

Overview

This document describes the packet handling sequence in TSG. TSG Firewall performs stateful checks, TSG proxy performs decryption.

The Ingress stage, session stage and egress stage are the three stages of TSG traffic processing. The ingress and egress stages handle network functions and make packet forwarding decisions on a per-packet basis. The remaining stages are session-based security functions such as application identification and content inspection. TSG does not change the operation of the MPLS, 802.1Q, IPv4, IPv6 protocols regardless of the configured policies.

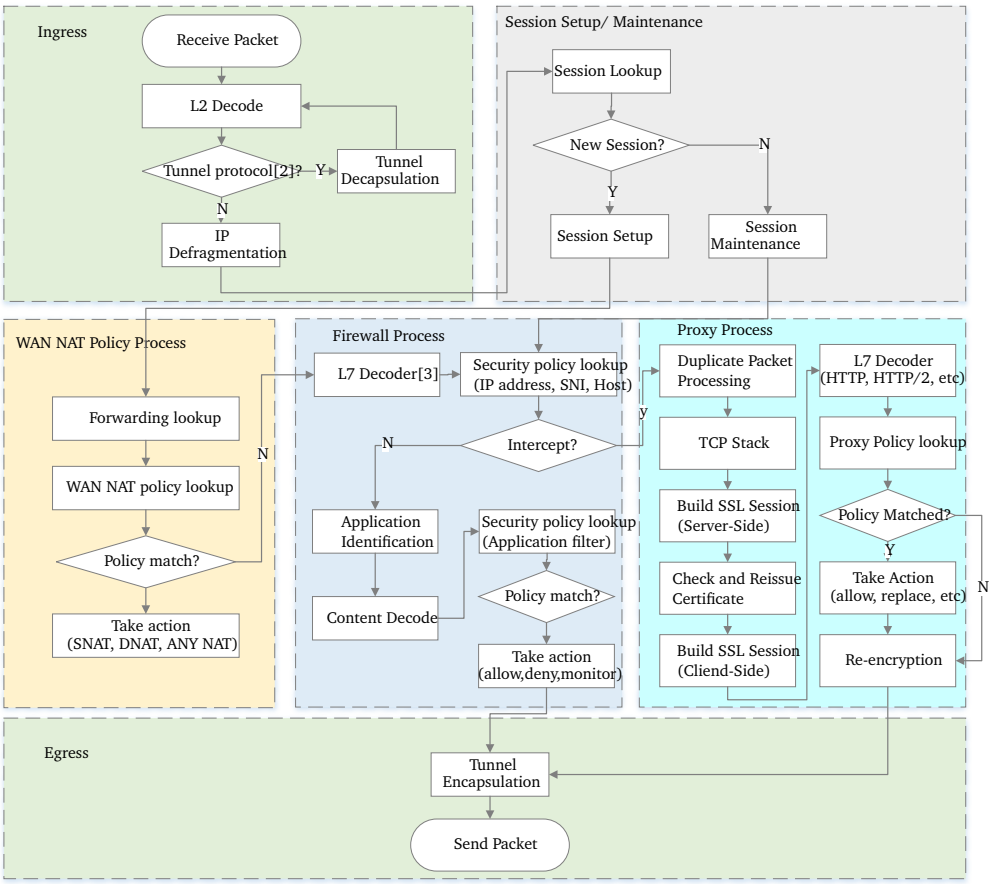
Packet Handling Sequence

Ingress Stage

The ingress stage receives packets from the network interface, parses those packets, and then determines whether a given packet is subject to further inspection. If the packet is subject to further inspection, the system continues with a session lookup and the packet enters the session stage. Otherwise, the system forwards the packet to the egress stage.

Layer 2 Decode

Packet parsing starts with the Ethernet (Layer-2) header of the packet received from the wire. VLAN, MPLS, MAC_IN_MAC headers are parsed here. The ingress port, 802.1q tag, and destination MAC address are used as keys to lookup the ingress logical interface.



[1] L2-L4 Decoded Protocols: Ethernet, VLAN, MPLS, MAC_IN_MAC, IPv4, IPv6, TCP, UDP
[2] Tunnel protocols: PPPOE, IPIP, GRE, PPTP, L2TP, Teredo, GTP
[3] L7 Decoded Protocols: HTTP, SSL/TLS, DNS, MAIL(SMTP, POP3, IMAP), FTP and QUIC

Tunnel Decapsulation

If the packet has PPPOE, IPIP, GRE, PPTP, L2TP, Teredo, GTP encapsulations, they are decoded at this stage.

IP Defragmentation

After the IP header is parsed (Layer-3), the TSG parses IP fragments, reassembles using the defragmentation process, and then feeds the packet back to the parser starting with the IP header. At this stage, a fragment may be discarded due to tear-drop attack (overlapping fragments), fragmentation errors, or if the TSG hits system limits on buffered fragments (hits the max packet threshold).

Session Setup

If the packet is subject to firewall inspection, it performs a flow lookup on the packet. In TSG 's implementation, the firewall identifies the flow using a 5-tuple key:

• Source and destination IP addresses: IP addresses from the IP packet.

• Source and destination ports: Port numbers from TCP/UDP protocol headers. For non-TCP/UDP, different protocol fields are used (e.g., for ICMP the ICMP identifier and sequence numbers are used).

• Protocol: The IP protocol number from the IP header is used to derive the flow key.

The firewall stores active flows in the flow lookup table. When a packet is determined to be eligible for firewall inspection, the firewall extracts the 5-tuple flow key from the packet and then performs a flow lookup to match the packet with an existing flow. Each flow has a client and server component. For TCP, the client is the sender of the TCP SYN packet of the session from firewall's perspective. For UDP, the client is the sender with smaller Port number. The server is the receiver of this first packet.

If the packet belongs to on new session, firewall performs a session setup.

The firewall uses the IP address of the packet to query mapping tables.

• Subscriber-IP mapping table: The corresponding user information is fetched.

• ASN-IP mapping table: The autonomous system number is fetched.

• Geo-IP mapping table: The geographical location is fetched.

There is a chance that above information is not available at this point. In that case, policies with these conditions cannot be enforced.

Session Maintenance

A packet that matches an existing session will enter the session maintenance stages. This stage starts with Layer 2 to Layer 4 firewall processing:

• If the session is in close wait state, then the firewall discards the packet by forwarding to egress stage.

• If the session is active, refresh session timeout.

• If the packet is a TCP FIN/RST, the session TCP half closed timer is started if this is the first FIN packet received (half closed session) or the TCP Time Wait timer is started if this is the second FIN packet or RST packet. The session is closed as soon as either of these timers expire.

If an application uses TCP as the transport, the firewall processes it by the TCP reassembly module before it sends the data stream into the security-processing module. The TCP reassembly module will also perform window check, buffer out-of-order data while skipping TCP retransmission. The firewall drops the packets if there is a reassembly error or if it receives too many out-of-order fragments, resulting in the reassembly buffers filling up.

A packet matching an existing session is subject to further processing if packet has TCP/UDP data (payload). If the firewall does not detect the session application, it performs application identification. If the identification is non-conclusive, the content inspection module runs known protocol decoder checks and heuristics to help identify the application. The application identification result could change throughout the life of the session. Once a traffic attribute is parsed, it's subject to security policy enforcement.

Firewall Process

Layer 7 Decode

The firewall decodes Layer 7 protocols such as HTTP, SSL/TLS and DNS, to get traffic attributes.

Application Identification

The firewall identifies application with built-in and customized signature. The firewall uses protocol decoding in the content inspection stage to determine if an application changes from one application to another. After the firewall identifies the session application, security policy will be enforced as configured.

Content Decode

The firewall decodes the flow and parses attributes and content, then scan them for keywords, e.g., email attachments that are text-based. If it results in keywords detection, then the corresponding security policy action is taken. Application identification is still on at this stage, as the more traffic attributes are parsed, application identification result may be changed.

Security Policy Lookup

The firewall uses application ANY, IP, port, Geographic and Subscriber ID to perform the lookup and check for a rule match. In case of a rule match, if the policy action is set to 'deny', the firewall drops the packet.

If security policy action is set to intercept and the application is SSL or HTTP, the firewall sends session packets to Proxy to perform decryption.

After more packet transferred, identified application as well as FQDN, URL and applicable protocol fields in the session are used as key to find rule match. If the session matches a security rule, and the rule has logging enabled, the firewall generates a security event log at the session end.

Proxy Process

Duplicate Packet Processing

Packet usually travels a route one time only, otherwise it's a waste of bandwidth. But network of real world is complicated: tunnels, policy ACLs and human mistakes all induce packet travels a link more than one time.

This will cause problems on Proxy function. When Proxy terminates server-side SSL connections, as a client, it sends packet to server with original 4-tuple, and that packet goes through TSG again. Since this packet has the same 4-tuple and sequence number as the client-side, TSG will drop it as a client-side retransmission packet, which causes server-side timeout.

The proxy fixes this problem by following two mechanisms:

• Duplicate packet identification: For TSG sent packets, TSG adds its 4-tuple, IPID, acknowledge number, sequence number and timestamp (if applicable) to the Bloom

filter. For received packet, TSG will query the Bloom filter with the aforementioned protocol fields. If a packet is considered duplicate, it will be bypassed, if not, it will be delivered to TCP stack.

• Duplicate flow detection: Since not all traffic flows have duplicate packets, do duplicate identification on each packet is a waste of CPU and memory, moreover, it induces performance overhead. Hence, only the TCP flows with duplicate SYN or SYN/ACK packet will conduct duplicate packet identification.

TCP Stack

Opening a TCP connection involves a three-way handshake packets: the client contacts the server, the server acknowledges the client, and the client acknowledges the server. The proxy's TCP stack attempts to connect server-side immediately after receiving the client's initial connection request, but waits to return the server acknowledgement until determining whether or not the server-side connection succeeds. The TCP stack act as transparent proxy and keep the same TCP connection source and destination IP and ports. This provides greater transparency, as the client receives either an RST or no response, which mirrors what is sent from a server when connections fail.

Build SSL Session

The proxy first builds server-side SSL session, if the server certificate verification is failed, the proxy will sign certificate with untrusted root certificate. That's because the proxy wants the untrusted certificate to warn users that they are trying to access potentially unsafe sites.

Proxy Policy Lookup

After decode Layer 7 protocols, traffic attributes like HTTP URL, headers are subject to proxy policy lookup, then according action is taken.

Re-encryption

The content is compressed and encrypted, and send to egress stage

Egress Stage

The system identifies encapsulation information for the packet, including MAC addresses, VLAN ID and tunnel headers. And find the egress interface. Finally, the packet is transmitted out of the physical egress interface.

Appendix F Best Practices

Security Policy

To improve your overall security posture, use the guidelines in this section to plan, deploy, and maintain your internet gateway best practices security policy. Apply security best practices to gain visibility into traffic, prevent threats, and protect your network, users, and data.

The first use case is to deny SSL traffic based on SNI. For example, users are forbidden to visit Netflix pages.

1. Without any policies, visit Netflix website and you can view it normally.
2. Create a Deny Security Policy:
   a) After logging into the system, create a Security Policy.
   b) Policy names can be defined according to user habits.
   c) Action select Deny.
   d) Specify client IP as matching criteria for the source field.
   e) Application select SSL.
   f) Specify the Netflix page you wish to deny as matching criteria for SNI field.
   g) Select Drop as Sub Action.
   h) Enable the policy and submit.
3. Visit Netflix again and pages cannot be accessed.

The second use case is monitor mails based on specified keywords. This Security Policy will generate a Security Event Log when matched.

1. Create a Monitor Security Policy:
   a) After logging into the system, create a Security Monitor Policy.
   b) Policy names can be defined according to user habits.
   c) Action select Monitor.
   d) Specify client IP as matching criteria for the source field.
   e) Application select MAIL.

f) Specify the keywords you wish to monitor as matching criteria for Content field.

g) Enable the policy and submit.

2. When the policy is matched, you can view Logs > Security Events and see the mail content.

Proxy Policy

The growth in SSL/TLS encrypted traffic traversing the internet is on an explosive upturn. We are going to take five use case to demonstrate proxy policy best practices.

The first use case is to specifically deny HTTPS traffic based on the URL. For example, users are forbidden to visit specific pages of YouTube, but other pages of YouTube website are not affected.

1. Without any policies, visit YouTube website and you can view it normally.
2. Create a Deny Policy:
   a) After logging into the system, create a policy.
   b) Policy names can be defined according to user habits.
   c) Action select Deny.
   d) Specify client IP as matching criteria for the source field. Specify the YouTube page you wish to deny as matching criteria for URL field.
   e) You can select the Response Content of the blocking page as an optional parameter.
   f) Enable the policy and submit.
3. Visit YouTube again. The specified pages cannot be accessed and other pages can be accessed normally.

The second use case is URL redirect, such as redirect the traffic accessing Google to Bing.

1. Without any policies, you can visit Google normally and search normally.
2. Create a redirect policy. The above has demonstrated how to create a policy, to save time, simply enable the policy that which was created before. Then click Edit to view the content of the policy.
   a) Action is Redirect.
   b) Filter is Google URL.

c) Redirect URL is Bing URL.

d) The policy is Enabled and click OK.

3. Visit Google again and the page is redirected to Bing.

The third use case is content Replace. For example, search china in Google. Create a content Replace policy to make the search results only come from www.news.cn.

1. Without any policies, search china in Google and you'll see the search results come from different websites, wikis, etc.
2. Create a Replace policy:
   a) Action select Replace.
   b) Add Google URL as Filter.
   c) Specify Action Parameters:
      i. Replace "q=china" in HTTP Request URI with "q=china+site
      ii. Replace "china site:www.news.cn" in HTTP Response Body with "china".
   d) Enable the policy and submit.
3. Visit Google again, search china and you'll see the search results only come from www.news.cn.

The fourth use case is hijack file. For example, download the WhatsApp installation package and replace it with other files, such as the WhatsApp installation package with trojan virus. Now replace it with test.exe is used Instead.

1. Without any policies, it is normal to download the WhatsApp installation package.
2. Create a Hijack policy:
   a) Action select Hijack.
   b) Select WhatsApp download URL as Filter.
   c) Select test.exe as Hijack File for Action Parameters.
   d) Enable the policy and submit.
3. Download WhatsApp again and find that the actual download file is test.exe. If you do not want to be found easily, you can change the hijack file into the same name as the original.

The fifth use case is script injection. For example, visit Facebook, insert the script into Facebook page, and the script will be executed on the client side. Here we use a JS script of pop-up window will show up to demonstrate.

1. Without any policies, visit Netflix website and it displays normally without any pop-up window.

2. Create an Insert policy:
    a) Action select Insert.
    b) Select Netflix URL as Filter.
    c) Select the JS script of pop-up window as Insert Script for Action Parameters.
    d) Enable the policy and submit.
3. Visit the Netflix website again, and the pop-up window appears.

✎*The Watch feature only support policy and object. You can add them to your watch list. Select the checkbox for objects and policies in the list and Click* ***Watch*** *at the bottom to add to Watch List. And then you can click the star icon in the bottom right and select Object/Policy tab to view the Watch List. You can search objects and policies by ID and Name in the list. For policy, you can check policy in the list and enable or disable it. You can also export and clear the list.*

Custom Application

A policy object consists of one item or a set of collective items that groups discrete identities such as IP addresses, URLs, applications, or accounts. The relationship between each item is “or”. However, for keyword, within each item, there can be “and” expression and the item can have up to 4 substrings. You can reference the object in policy as conditions. The relationship between each condition within policy is “and”. There can be eight conditions at most for every policy.

Application is a little special as a kind of object in TSG. Applications increase the value of our TSG firewalls by making it easier and faster to determine the exact identity of applications traversing the network, enabling teams to set and enforce the right policies. Applications can reduce complexity and minimize human error.

The following introduced predefined application attributes:

TCP/UDP/IP Attributes

Attribute Name	Layer	Stage	Value Type
IP Address ip.src	Packet	0	ip
ip.dst	Packet	0	ip
tcp.payload.c2s_first_data	Session	1	string
tcp.payload.s2c_first_data	Session	2	string
tcp.analysis.create_with_syn	Session	1	Bool

ip.payload	Packet	0	string
tcp.payload	Packet	0	string
tcp.srcport	Packet	0	Numeric
tcp.dstport	Packet	0	Numeric
tcp.syn.fingerprint	Session	1	string
tcp.sack.fingerprint	Session	1	string
udp.payload.c2s_first_data	Session	0	string
udp.payload.s2c_first_data	Session	1	string
udp.srcport	Packet	0	Numeric
udp.dstport	Packet	0	Numeric

SSL Attributes

Attribute Name	Layer	Stage	Value Type
ssl.handshake.extensions_server_name	Session	1	string
ssl.handshake.cert.fingerprint	Session	2	string
ssl.handshake.cert.serial_number	Session	2	string
ssl.handshake.certificate.issuer_common_name	Session	2	string
ssl.handshake.certificate.issuer_organization_name	Session	2	string
ssl.handshake.certificate.issuer_country_name	Session	2	string
ssl.handshake.certificate.subject_common_name	Session	2	string
ssl.handshake.certificate.subject_organization_name	Session	2	string
ssl.handshake.certificate.subject_country_name	Session	2	string
ssl.handshake.certificate.not_valid_before	Session	2	string
ssl.handshake.certificate.not_valid_after	Session	2	string
ssl.handshake.certificate.algorithm_id	Session	2	string
ssl.analysis.use_session_resumption	Session	3	Numeric
ssl.analysis.use_selfsigned_certificate	Session	3	Numeric
ssl.analysis.incomplete_certificate_chain	Session	3	Numeric

✎*Client Hello is in Stage 1, and Server Hello is in Stage 2. SSL Attributes are all in Session Layer.*

HTTP Attributes

Attribute Name	Layer	Stage	Value Type
http.host	Session	1	string
http.uri	Session	1	string

http.user_agent	Session	1	string
http.content_type	Session	1	string
http.content_encoding	Session	1	string
http.referer	Session	1	string
http.cookie	Session	1	string
http.set_cookie	Session	3	string

✎*HTTP Request Header, Request Body, Response Header, Response Body are in Stage 1, Stage 2, Stage 3, Stage 4 in sequence. HTTP attributes are all in Session Layer.*

Other Attributes

Attribute Name	Layer	Stage	Value Type
quic.sni	Session	1	string
dns.qry.name	Session	1	string
general.session.analysis.app_id	Session	2	numeric
general.c2s_session_size	Session	2	string
general.s2c_session_size	Session	2	string

Make Your Own Reports

The following guides you to set a report with one type of chart. Of course, you can create a report with all types of charts by TSG function **Reports**.

The 1st: Bar

Create a Bar Report. For example, make a report about the top 10 server IP that our security policy matched last week and the sum of bytes sent and bytes received from that server IP.

STEP 1. Create a Dataset. Dataset specify what data to extract from logs.

1. Select **Reports** > **Datasets** menu, and click **Create**.
2. Enter a **Name**, security event bytes by server IP dataset.
3. Select a **Log Type**: Security Event.

4. Select the **Group by**. This will serve as the X-Axis Data Binding options for Chart Libraries. The report mainly displays Server IP, so select Server IP.
5. Specify the **Data Bindings**, add field, aggregate, and label, which serve as Y-Axis Data Binding options of your Chart Libraries. Field consists of the Group by field you just added. You can add one or multiple fields. Field select bytes Sent and bytes Received. Aggregate select sum. Label set to bytes-Sum.
6. Specify **Filter** and **Having**. For Filter, select Action as Field; select "=" as Operator; select Monitor and Intercept as Value. That is to say, only security policies with monitor and intercept actions will be counted. For Having, select Log ID as Field; select count as Aggregate; select ">=" as Operator; select 10 as value. This means only security policies that match more than 10 times (including 10) will be counted. Your selections will automatically generate a SQL.
7. Click **OK**.

STEP 2. Create Chart Library. Chart set the display form of the data.

1. Enter a **Name** and **Description**. Complete name, security event bytes by server IP Chart Library. Leave description set to empty.
2. Select the **Dataset** you just created.
3. Select **Chart Type**: Bar.
4. Set the **Data Bindings** for X-Axis and Y-Axis. Set Show Top to 10. Set Label, which will be displayed in your report.
5. Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, security event bytes by server IP report.
2. Select a **Time Period**. Here select last 1 week, which is from yyyy-mm-dd-7 hh:mi:ss to yyyy-mm-dd hh:mi:ss.
3. Select the **Chart Library** you just created.
4. Click **OK**.

The 2nd: Line

Create a Line Report. For example, make a report about the security event sessions count previous day.

STEP 1. Create a Dataset. Dataset specify what data to extract from logs.

1. Select **Reports** > **Datasets** menu, and click **Create**.
2. Enter a **Name**, security event sessions by time dataset.
3. Select a **Log Type**: Security Event.
4. Select the **Group by**. For Line Chart, you must select Receive Time. This will serve as the X-Axis Data Binding options for Chart Libraries.
5. Specify the **Data Bindings**, add field, aggregate, and label, which serve as Lines Data Binding options of your Chart Libraries. Field consists of the Group by field you just added. You can add one or multiple fields. Field select Log ID. Aggregate select count. Label set to count.
6. Click **OK**.

STEP 2. Create Chart Library. Chart set the display form of the data.

1. Enter a **Name** and **Description**. Complete name, security event sessions by time Chart Library. Leave description set to empty.
2. Select the **Dataset** you just created.
3. Select **Chart Type**: Line.
4. Set the **Data Bindings** for X-Axis and Lines. For Lines, select count as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input count as Legend.
5. Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, security event sessions by time report.
2. Select a **Time Period**. Here select previous 1 day, which means from yyyy-mm-dd-1 00:00:00 to yyyy-mm-dd-1 23:00:00 if the Time Granularity is 1 hour.
3. Select the **Chart Library** you just created. Enter number and select time unit for **Time Granularity**. Here set it to 1 hour.
4. Click **OK**.

The 3rd: Drilldown

Create a Drilldown Report. For example, make a report about the session count for top 100 Http.Domain, Drilldown on Subscriber ID and show Drilldown top 5 that our security policy matched last week.

STEP 1. Create a Dataset. Dataset specify what data to extract from logs.

1. Select **Reports** > **Datasets** menu, and click Create.
2. Enter a **Name**, security event session top domain and subscriber IDs drilldown dataset.
3. Select a **Log Type**: Security Event.
4. Select the **Group by**. The report mainly displays Http.Domain and Subscriber ID, so add Http.Domain and Subscriber ID.
5. Specify the **Data Bindings**, add field, aggregate, and label, which serve as Columns options of your Chart Libraries. Field consists of the Group by field you just defined and you can add other fields. Here add Log ID; select count as Aggregate and set Label to sessions.
6. Specify **Filter** and **Having**. For Filter, select Address Type as Field; select "=" as Operator; select IPv4 as Value. That is to say, only security policies with IPv4 address type will be counted. For Having, select Log ID as Field; select count as Aggregate; select ">=" as Operator; select 10 as value. This means only security policies that match more than 10 times (including 10) will be counted. Your selections will automatically generate a SQL.
7. Click **OK**.

STEP 2. Create Chart Library. Chart set the display form of the data.

1. Enter a **Name** and **Description**. Complete name, security event session top domain and subscriber IDs drilldown chart. Leave description set to empty.
2. Select the **Dataset** you just created.
3. Select **Chart Type**: Table.
4. For **Data Bindings**, select Drilldown as Table Type. Drilldown tables have three columns and you must select one as drilldown column. Complete Title, Width, Data Binding, Format and Legend for each column. And enable Drilldown Columns for Subscriber ID. Select order by sessions descending. Show Top 100 Http.Domains. Drilldown Top 5 on Subscriber ID.
5. Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, security event session top domain and subscriber IDs drilldown report.

2. Select a **Time Period**. Here select last 1 week, which is from yyyy-mm-dd-7 hh:mi:ss to yyyy-mm-dd hh:mi:ss.
3. **Enable Notification**, select Email as Output Profile.

   Email: Enter recipient's e-mail.
4. Select the **Chart Library** you just created.
5. Click **OK**.

The 4th: Double-lines

Create a top N Line Report. For example, make a report about the session count group by Http.Domain (top 5) every day that our security policy matched in the last month.

STEP 1. Create a Dataset. Dataset specify what data to extract from logs.

1. Select **Reports** > **Datasets** menu, and click **Create**.
2. Enter a **Name**, security event counter by domain dataset.
3. Select a **Log Type**: Security Event.
4. Select the **Group by**. For Line Chart, you must select Receive Time. This will serve as the X-Axis Data Binding options for Chart Libraries. The report mainly displays Http.Domain, so add Http.Domain. This will be added to your Group by option in your Chart Libraries.
5. Specify the **Data Bindings**, add field, aggregate, and label. Field consists of the Group by field you just added. You can add other fields. Field select Log ID. Aggregate select count. Label set to counter.
6. Click **OK**.

STEP 2. Create Chart Library. Chart set the display form of the data.

1. Enter a **Name** and **Description**. Complete name, security event counter by domain chart. Leave description set to empty.
2. Select the **Dataset** you just created.
3. Select **Chart Type**: Line.
4. Set the **Data Bindings** for X-Axis and Lines. For Lines, select counter as Data Binding; select Default as Format; select Line Up as Type; input counter as Legend. Check Group by and select Http.Domain. Check Order by and select counter. Check Descending. Max line is 5.
5. Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, security event counter by domain report.
2. Select a **Time Period**. Here select last 1 month, which is from yyyy-(mm-1)-dd hh:mi:ss to yyyy-mm-dd hh:mi:ss.
3. Select the **Chart Library** you just created. Enter number and select time unit for **Time Granularity**. Here set it to 1 day.
4. Add **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This means, empty domain name will not be counted.
5. Click **OK**.

The 5th: Area

Create an Area Report. For example, make a report about the sum of bytes Sent and bytes Received and the sum of Packets Sent and Packets Received every day that our security policy matched in last month.

STEP 1. Create a Dataset. Dataset specify what data to extract from logs.

1. Select **Reports** > **Datasets** menu, and click **Create**.
2. Enter a **Name**, security event bytes and packets dataset.
3. Select a **Log Type**: Security Event.
4. Select the **Group by**. For Area Chart, you must select Receive Time. This will serve as the X-Axis (TimeLine) Data Binding options for Chart Libraries. You can add other Fields, which will be added to your Group by options in your Chart Libraries.
5. Specify the **Data Bindings**, add field, aggregate, and label. Field consists of the Group by field you just added. You can add other fields. Field select bytes Sent and bytes Received; aggregate select sum; label set to bytes. Field select Packets Sent and Packets Received; aggregate select sum; label set to Packets.
6. Click **OK**.

STEP 2. Create Chart Library. Chart set the display form of the data.

1. Enter a **Name** and **Description**. Complete name, security event bytes and packets chart. Leave description set to empty.
2. Select the **Dataset** you just created.
3. Select **Chart Type**: Area.

4. Set the **Data Bindings** for X-Axis and Lines. For Lines, select bytes as Data Binding; select Bandwidth (Kbps/Mbps/Gbps) as Format; select Line Up as Type; input bytes as Legend. Select Packets as Data Binding; select Bandwidth (Kpps/Mpps/Gpps) as Format; select Line Down as Type; input Packets as Legend.
5. Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, security event bytes and packets report.
2. Select a **Time Period**. Here select last 1 month, which is from yyyy-(mm-1)-dd hh:mi:ss to yyyy-mm-dd hh:mi:ss.
3. Select the **Chart Library** you just created. Enter number and select time unit for **Time Granularity**. Here set it to 1 day.
4. Click **OK**.

The 6th: Network Behavior Analysis for Specific User

Create a report to analyze network behavior of specific user, such as specific IP address or Subscriber ID. For example, create a report for Internal IP 192.168.50.2. It will include 9 charts and tables, that display traffic trend, top access domain and top access URLs based on sessions.

STEP 1. Create 3 Datasets. Select **Reports** > **Datasets** menu, and click **Create**. Select the same Log Type for the 3 datasets: Session Records.

1. Create a Dataset with **Name** Traffic-Metrics-Trend.
   a) Select Receive Time as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings once and reference the same dataset when creating multiple charts. Now add 7 items.
      i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to bytes.
      ii. Field select Bytes Sent; aggregate select sum; Label set to Bytes Sent.
      iii. Field select Bytes Received; aggregate select sum; Label set to Bytes Received.
      iv. Field select sessions; aggregate select sum; Label set to New sessions.

v. Field select Packets Sent and Packets Received; aggregate select sum; Label set to Packets.

vi. Field select Packets Sent; aggregate select sum; Label set to Packets Sent.

vii. Field select Packets Received; aggregate select sum; Label set to Packets Received.

c) Click **OK**.

2. Create a Dataset with **Name** Traffic-Top-Access-Domain.

a) Select Http.Domain as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. Field select Sessions; aggregate select sum; label set to Sessions.

c) Specify **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This configuration will exclude empty domains.

d) Click **OK**.

3. Create a Dataset with **Name** Traffic-Top-Access-URL.

a) Select Http.URL as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. Field select Sessions; aggregate select sum; label set to Sessions.

c) Specify **Filter**. Select Http.URL as Field; select notEmpty as Operator. This configuration will exclude empty URLs.

d) Click **OK**.

STEP 2. Create 9 Chart Libraries. Chart set the display form of the data. Select **Reports** > **Chart Library** menu, and click Create.

1. Create a Chart Library with **Name** Traffic Bandwidth Trend.

a) Select **Dataset** Traffic-Metrics-Trend.

b) Select **Chart Type**: Area.

c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. You can add multiple Data Binding for Lines.

i. Select Bytes as Data Binding; select Bandwidth (Kbps/Mbps/Gbps) as Format; select Line Up as Type; input total as Legend.

ii. Select Bytes Sent as Data Binding; select Bandwidth (Kbps/Mbps/Gbps) as Format; select Line Up as Type; input out as Legend.

iii. Select Bytes Received as Data Binding; select Bandwidth (Kbps/Mbps/Gbps) as Format; select Line Up as Type; input in as Legend.

d) Click **OK**.

2. Create a Chart Library with **Name** Traffic Packet Trend.
   a) Select **Dataset** Traffic-Metrics-Trend.
   b) Select **Chart Type**: Area.
   c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. You can add multiple Data Binding for Lines.
      i. Select Packets as Data Binding; select Bandwidth(Kpps/Mpps/Gpps) as Format; select Line Up as Type; input total as Legend.
      ii. Select Packets Sent as Data Binding; select Bandwidth (Kpps/Mpps/Gpps) as Format; select Line Up as Type; input out as Legend.
      iii. Select Packets Received as Data Binding; select Bandwidth (Kpps/Mpps/Gpps) as Format; select Line Up as Type; input in as Legend.
   d) Click **OK**.
3. Create a Chart Library with **Name** Traffic New Sessions Trend.
   a) Select **Dataset** Traffic-Metrics-Trend.
   b) Select **Chart Type**: Line.
   c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. Select New Sessions as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input New Sessions as Legend.
   d) Click **OK**.
4. Create a Chart Library with **Name** Traffic Top 30 Access Domains Detail.
   a) Select **Dataset** Traffic-Top-Access-Domain.
   b) Select **Chart Type**: Table.
   c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Domain as Title; select Domain as Data Binding; input Domain as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Select order by Sessions descending. Show Top 30.
   d) Click **OK**.
5. Create a Chart Library with **Name** Traffic Top 30 Access Domains Distribution.

a) Select **Dataset** Traffic-Top-Access-Domain.

b) Select **Chart Type**: Bar.

c) Set the **Data Bindings** for X-Axis and Y-Axis. For X-Axis, select Domain as Data Binding; set Domain as Label; set Show Top to 30. For Y-Axis, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions.

d) Click **OK**.

6. Create a Chart Library with **Name** Traffic Top 10 Access Domains Percent.

   a) Select **Dataset** Traffic-Top-Access-Domain.

   b) Select **Chart Type**: Pie.

   c) Set the **Data Bindings** for Category and Series. For Category, select Domain as Data Binding; set Domain as Label; set Show Top to 10. For Series, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions. Check Bundle rest into 'Others'.

   d) Click **OK**.

7. Create a Chart Library with **Name** Traffic Top 30 Access URLS Detail.

   a) Select **Dataset** Traffic-Top-Access-URL.

   b) Select **Chart Type**: Table.

   c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in URL as Title; select URL as Data Binding; input URL as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Select order by Sessions descending. Show Top 30.

   d) Click **OK**.

8. Create a Chart Library with **Name** Traffic Top 30 Access URLS Distribution.

   a) Select **Dataset** Traffic-Top-Access-URL.

   b) Select **Chart Type**: Bar.

   c) Set the **Data Bindings** for X-Axis and Y-Axis. For X-Axis, select URL as Data Binding; set URL as Label; set Show Top to 30. For Y-Axis, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions.

   d) Click **OK**.

9. Create a Chart Library with **Name** Traffic Top 10 Access URLS Percent.

   a) Select **Dataset** Traffic-Top-Access-URL.

b) Select **Chart Type**: Pie.

c) Set the **Data Bindings** for Category and Series. For Category, select URL as Data Binding; set URL as Label; set Show Top to 10. For Series, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions. Check Bundle rest into 'Others'.

d) Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, Traffic Report with Internal IP 192.168.50.2.
2. Select a **Time Period**. Select last 1 day.
3. Select the **Chart Library** you just created. Click Add Chart to add multiple charts.
   a) Select Traffic Bandwidth Trend. Set 5 minutes for **Time Granularity**.
   b) Select Traffic Packets Trend. Set 5 minutes for **Time Granularity**.
   c) Select Traffic New Sessions Trend. Set 5 minutes for **Time Granularity**.
   d) Select Traffic Top 30 Access Domains Detail.
   e) Select Traffic Top 30 Access Domains Distribution.
   f) Select Traffic Top 10 Access Domains Percent.
   g) Select Traffic Top 30 Access URLS Detail.
   h) Select Traffic Top 30 Access URLS Distribution.
   i) Select Traffic Top 10 Access URLS Percent.
4. Specify **Filter**. Select Internal IP as Field; select "=" as Operator; set Value to 192.168.50.2.
5. Click **OK**.

The 7th: Website Access Analysis for Specific Domain

Create a report to analyze website access of specific domain, such as google.com. For example, create a custom report to analyze security events which hit google.com. It will include 5 charts and tables, that display traffic trend based on bandwidth and client IP number, security event action hit sessions trend and and top Server IP based on sessions.

STEP 1. Create 3 Datasets. Select **Reports** > **Datasets** menu, and click **Create**. Select the same **Log Type** for the 3 datasets: Security Event.

1. Create a Dataset with **Name** Security-Intercept-Event-Metrics-Trend.

a) Select Receive Time as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 2 items.

i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.

ii. Field select Client IP; aggregate select count distinct; Label set to Unique Client IP.

c) Specify **Filter**. Select Action as Field; select "=" as Operator; set Value to Intercept. This configuration will only include Security Events with intercept action.

d) Click **OK**.

2. Create a Dataset with **Name** Security-Event-Action-Hit-Sessions-Trend.

a) Select Receive Time and Action as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. Field select Log ID; Label set to Sessions.

c) Click **OK**.

3. Create a Dataset with **Name** Security-Event-Top-Server-IP.

a) Select Server IP as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. Field select Log ID; Label set to Sessions.

c) Click **OK**.

STEP 2. Create 5 Chart Libraries. Chart set the display form of the data. Select **Reports** > **Chart Library** menu, and click Create.

1. Create a Chart Library with **Name** Security Intercept Event Metrics Trend.

a) Select **Dataset** Security-Intercept-Event-Metrics-Trend.

b) Select **Chart Type**: Line.

c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. For Lines, select Unique Client IP as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input Unique Client IP as Legend. Select Bytes as Data Binding; select Bandwidth(Kbps/Mbps/Gbps) as Format; select Line Down as Type; input bps as Legend.

d) Click **OK**.

2. Create a Chart Library with **Name** Security Event Action Hit Sessions Trend.

a) Select **Dataset** Security-Event-Action-Hit-Sessions-Trend.

b) Select **Chart Type**: Line.

c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. For Lines, select sessions as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input sessions as Legend. Check Group By and select Action. Check Order By, select sessions and check Descending. Show Top 10.

d) Click **OK**.

3. Create a Chart Library with **Name** Security Event Top 30 Server IP Detail.

a) Select **Dataset** Security-Event-Top-Server-IP.

b) Select **Chart Type**: Table.

c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Server IP as Title; select Server IP as Data Binding; input Server IP as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Select order by Sessions descending. Show Top 30.

d) Click **OK**.

4. Create a Chart Library with **Name** Security Event Top 30 Server IP Distribution.

a) Select **Dataset** Security-Event-Top-Server-IP.

b) Select **Chart Type**: Bar.

c) Set the **Data Bindings** for X-Axis and Y-Axis. For X-Axis, select Server IP as Data Binding; set Server IP as Label; set Show Top to 30. For Y-Axis, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions.

d) Click **OK**.

5. Create a Chart Library with **Name** Security Event Top 10 Server IP Percent.

a) Select **Dataset** Security-Event-Top-Server-IP.

b) Select **Chart Type**: Pie.

c) Set the **Data Bindings** for Category and Series. For Category, select Server IP as Data Binding; set Server IP as Label; set Show Top to 10. For Series, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions. Check Bundle rest into 'Others'.

d) Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, Security Event Report with Domain google.com.
2. Select a **Time Period**. Select last 1 day.
3. Select the **Chart Library** you just created. Click Add Chart to add multiple charts.
   a) Select Security Intercept Event Metrics Trend. Set 5 minutes for **Time Granularity**.
   b) Select Security Event Action Hit Sessions Trend. Set 5 minutes for **Time Granularity**.
   c) Select Security Event Top 30 Server IP Detail. Set 5 minutes for **Time Granularity**.
   d) Select Security Event Top 30 Server IP Distribution.
   e) Select Security Event Top 10 Server IP Percent.
4. Specify **Filter**. Select Http.Domain as Field; select "=" as Operator; set Value to google.com.
5. Click **OK**.

The 8th: Endpoints Access Analysis for Specific City

Customize a report to analyze endpoints access of specific Data Center. For example, create a report to analyze session records about endpoints information in specific Data Center. It will include 2 charts and tables, that display number of Unique External IP, Internal IP and Subscriber ID trend based on bandwidth and show Unique client IP number and Unique Subscriber ID number of Top 100 domains.

STEP 1. Create 2 Datasets. Select **Reports** > **Datasets** menu, and click **Create**. Select the same **Log Type** for the 2 datasets: Session Records.

1. Create a Dataset with **Name** Traffic-Endpoints-Metrics-Trend.
   a) Select Receive Time as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 5 items.
      i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.
      ii. Field select Sessions; aggregate select sum; Label set to Sessions.
      iii. Field select Internal IP; aggregate select count distinct; Label set to Unique Internal IP.

iv. Field select External IP; aggregate select count distinct; Label set to Unique External IP.

v. Field select Subscriber ID; aggregate select count distinct; Label set to Unique Subscriber ID.

c) Click **OK**.

2. Create a Dataset with **Name** Traffic-Top-Domain-with-Unique-Client-IP-and-Subscriber-ID.

a) Select Http.Domain as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 2 items.

i. Field select Client IP; aggregate select count distinct; Label set to Unique Client IP.

ii. Field select Subscriber ID; aggregate select count distinct; Label set to Unique Subscriber ID.

c) Specify **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This configuration will exclude Session Records with empty domain.

d) Click **OK**.

STEP 2. Create 2 Chart Libraries. Chart set the display form of the data. Select **Reports** > **Chart Library** menu, and click Create.

1. Create a Chart Library with **Name** Traffic Endpoints Metrics Trend.

a) Select **Dataset** Traffic-Endpoints-Metrics-Trend.

b) Select **Chart Type**: Line.

c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. For Lines, select Unique External IP as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input Unique External IP as Legend; select Unique Internal IP as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input Unique Internal IP as Legend; select Unique Subscriber ID as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input Unique Subscriber ID as Legend. Select Bytes as Data Binding; select Bandwidth(Kbps/Mbps/Gbps) as Format; select Line Down as Type; input bps as Legend.

d) Click **OK**.

2. Create a Chart Library with **Name** Traffic Top 100 Domains with Unique Client IP and Subscriber ID.

a) Select **Dataset** Traffic-Top-Domain-with-Unique-Client-IP-and-Subscriber-ID.

b) Select **Chart Type**: Table.

c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Domain as Title; select Http.Domain as Data Binding; input Http.Domain as Legend. Fill in Unique Client IP as Title; select Unique Client IP as Data Binding; select Counter(K/M/G) as Format; input Unique Client IP as Legend. Fill in Unique Subscriber ID as Title; select Unique Subscriber ID as Data Binding; select Counter(K/M/G) as Format; input Unique Subscriber ID as Legend. Select order by Unique Client IP descending. Show Top 100.

d) Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, Traffic Report with Data Center DC1 and DC2.
2. Select a **Time Period**. Select last 1 day.
3. Select the **Chart Library** you just created. Click Add Chart to add multiple charts.
   a) Select Traffic Endpoints Metrics Trend. Set 5 minutes for **Time Granularity**.
   b) Select Traffic Top 100 Domains with Unique Client IP and Subscriber ID.
4. Specify **Filter**. Select Data Center as Field; select "=" as Operator; set Value to DC1 and DC2.
5. Click **OK**.

The 9th: Endpoints Details Analysis for Intercept Action

Customize a report to analyze endpoints details of intercept action. For example, create a report to analyze Security Events about multiple dimensional endpoints information. It will include 8 charts and tables, that display endpoints statistics details, including Top Client IP, Server IP, Internal IP, External IP (by Sessions with Bandwidth), Top Domain Distribution (by Sessions with Bandwidth), Top Domain Drilldown Internal IP (by Sessions), Top Domain Drilldown Server IP (by Bandwidth), Top Subscriber ID Drilldown Domain (by Sessions). With the help of this example, you can have a better understanding of the meaning of Drilldown table and bar charts and how to create them.

STEP 1. Create 8 Datasets. Select **Reports** > **Datasets** menu, and click **Create**. Select the same **Log Type** for the 2 datasets: Security Event.

1. Create a Dataset with **Name** Security-Event-Top-Internal-IP-by-Sessions-with-Bandwidth.
   a) Select Internal IP as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 4 items.
      i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.
      ii. Field select Bytes Sent; aggregate select sum; Label set to Bytes Sent.
      iii. Field select Bytes Received; aggregate select sum; Label set to Bytes Received.
      iv. Field select Log ID; Label set to Sessions.
   c) Click **OK**.
2. Create a Dataset with **Name** Security-Event-Top-External-IP-by-Sessions-with-Bandwidth.
   a) Select External IP as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 4 items.
      i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.
      ii. Field select Bytes Sent; aggregate select sum; Label set to Bytes Sent.
      iii. Field select Bytes Received; aggregate select sum; Label set to Bytes Received.
      iv. Field select Log ID; Label set to Sessions.
   c) Click **OK**.
3. Create a Dataset with **Name** Security-Event-Top-Client-IP-by-Sessions-with-Bandwidth.
   a) Select Client IP as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 4 items.
      i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.
      ii. Field select Bytes Sent; aggregate select sum; Label set to Bytes Sent.

iii. Field select Bytes Received; aggregate select sum; Label set to Bytes Received.

iv. Field select Log ID; Label set to Sessions.

c) Click **OK**.

4. Create a Dataset with **Name** Security-Event-Top-Server-IP-by-Sessions-with-Bandwidth.

a) Select Server IP as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 4 items.

i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.

ii. Field select Bytes Sent; aggregate select sum; Label set to Bytes Sent.

iii. Field select Bytes Received; aggregate select sum; Label set to Bytes Received.

iv. Field select Log ID; Label set to Sessions.

c) Click **OK**.

5. Create a Dataset with **Name** Security-Event-Top-Domain-by-Sessions.

a) Select Http.Domain as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Field select Log ID; Label set to Sessions.

c) Specify **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This configuration will exclude Security Events with empty Domain.

d) Click **OK**.

6. Create a Dataset with **Name** Security-Event-Top-Domain-by-Internal-IP-and-Sessions.

a) Select Http.Domain and Internal IP as **Group by**.

b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Field select Log ID; Label set to Sessions.

c) Specify **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This configuration will exclude Security Events with empty Domain.

d) Click **OK**.

7. Create a Dataset with **Name** Security-Event-Top-Domain-by-Server-IP-and-Bandwidth.
   a) Select Http.Domain and Server IP as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.
   c) Specify **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This configuration will exclude Security Events with empty Domain.
   d) Click **OK**.
8. Create a Dataset with **Name** Security-Event-Top-Subscriber-ID-by-Website-Domains-and-Sessions.
   a) Select Http.Domain and Subscriber ID as **Group by**.
   b) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Field select Log ID; Label set to Sessions.
   c) Specify **Filter**. You can add multiple items. Select Http.Domain as Field; select notEmpty as Operator. Click add and select Subscriber ID as Field; select notEmpty as Operator. This configuration will exclude Security Events with empty Domain and Subscriber ID.
   d) Click **OK**.

STEP 2. Create 9 Chart Libraries. Chart set the display form of the data. Select **Reports** > **Chart Library** menu, and click **Create**.

1. Create a Chart Library with **Name** Top 20 Internal IP by Sessions with Bandwidth.
   a) Select **Dataset** Security-Event-Top-Internal-IP-by-Sessions-with-Bandwidth.
   b) Select **Chart Type**: Table
   c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Internal IP as Title; select Internal IP as Data Binding; input Internal IP as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Fill in Bytes as Title; select Bytes as Data Binding; select Bandwidth(KB/MB/GB) as Format; input Bytes as Legend. Fill in Bandwidth as Title; select Bytes Sent as Data Binding; select Bar as Format; input Bytes Sent as Legend; select Bytes Received as Data Binding; select Bar as Format; input Bytes Received as Legend. Select order by Sessions descending. Show Top 20.

d) Click **OK**.

2. Create a Chart Library with **Name** Top 20 External IP by Sessions with Bandwidth.
   a) Select **Dataset** Security-Event-Top-External-IP-by-Sessions-with-Bandwidth.
   b) Select **Chart Type**: Table
   c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in External IP as Title; select External IP as Data Binding; input External IP as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Fill in Bytes as Title; select Bytes as Data Binding; select Bandwidth(KB/MB/GB) as Format; input Bytes as Legend. Fill in Bandwidth as Title; select Bytes Sent as Data Binding; select Bar as Format; input Bytes Sent as Legend; select Bytes Received as Data Binding; select Bar as Format; input Bytes Received as Legend. Select order by Sessions descending. Show Top 20.
   d) Click **OK**.
3. Create a Chart Library with **Name** Top 20 Client IP by Sessions with Bandwidth.
   a) Select **Dataset** Security-Event-Top-Client-IP-by-Sessions-with-Bandwidth.
   b) Select **Chart Type**: Table
   c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Client IP as Title; select Client IP as Data Binding; input Client IP as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Fill in Bytes as Title; select Bytes as Data Binding; select Bandwidth(KB/MB/GB) as Format; input Bytes as Legend. Fill in Bandwidth as Title; select Bytes Sent as Data Binding; select Bar as Format; input Bytes Sent as Legend; select Bytes Received as Data Binding; select Bar as Format; input Bytes Received as Legend. Select order by Sessions descending. Show Top 20.
   d) Click **OK**.
4. Create a Chart Library with **Name** Top 20 Server IP by Sessions with Bandwidth.
   a) Select **Dataset** Security-Event-Top-Server-IP-by-Sessions-with-Bandwidth.
   b) Select **Chart Type**: Table

c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Server IP as Title; select Server IP as Data Binding; input Server IP as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Fill in Bytes as Title; select Bytes as Data Binding; select Bandwidth(KB/MB/GB) as Format; input Bytes as Legend. Fill in Bandwidth as Title; select Bytes Sent as Data Binding; select Bar as Format; input Bytes Sent as Legend; select Bytes Received as Data Binding; select Bar as Format; input Bytes Received as Legend. Select order by Sessions descending. Show Top 20.

d) Click **OK**.

5. Create a Chart Library with **Name** Top 20 Website Domains Sessions Distribution.

   a) Select **Dataset** Security-Event-Top-Domain-by-Sessions.

   b) Select **Chart Type**: Bar

   c) Set the **Data Bindings** for X-Axis and Y-Axis. For X-Axis, select Domain as Data Binding; set Domain as Label; set Show Top to 20. For Y-Axis, select Sessions as Data Binding; select Counter(K/M/G) as Format; set Label to Sessions.

   d) Click **OK**.

6. Create a Chart Library with **Name** Top 20 Website Domains by Internal IP and Sessions.

   a) Select **Dataset** Security-Event-Top-Domain-by-Internal-IP-and-Sessions.

   b) Select **Chart Type**: Table

   c) For **Data Bindings**, select Drilldown as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Domain as Title; select Domain as Data Binding; input Domain as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Fill in Internal IP as Title; select Internal IP as Data Binding; input Internal IP as Legend. Check Bundle rest into 'Others'. Select order by Sessions descending. Show Top 20. For Drilldown, Fill in Show Top 20.

   d) Click **OK**.

7. Create a Chart Library with **Name** Top 20 Website Domains by Server IP and Bandwidth.

a) Select **Dataset** Security-Event-Top-Domain-by-Server-IP-and-Bandwidth.

b) Select **Chart Type**: Table

c) For **Data Bindings**, select Drilldown as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Domain as Title; select Domain as Data Binding; input Domain as Legend. Fill in Server IP as Title; select Server IP as Data Binding; input Server IP as Legend. Fill in Bytes as Title; select Bytes as Data Binding; select Bandwidth(KB/MB/GB) as Format; input Bytes as Legend. Check Bundle rest into 'Others'. Select order by Bytes descending. Show Top 20. For Drilldown, Fill in Show Top 20.

d) Click **OK**.

8. Create a Chart Library with **Name** Top 20 Subscriber ID by Website domains and Sessions.

a) Select **Dataset** Security-Event-Top-Subscriber-ID-by-Website-Domains-and-Sessions.

b) Select **Chart Type**: Table

c) For **Data Bindings**, select Drilldown as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Domain as Title; select Domain as Data Binding; input Domain as Legend. Fill in Subscriber ID as Title; select Subscriber ID as Data Binding; input Subscriber ID as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Check Bundle rest into 'Others'. Select order by Sessions descending. Show Top 20. For Drilldown, Fill in Show Top 20.

d) Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, Security Event Intercept Endpoints Report.
2. Select a **Time Period**. Select last 1 day.
3. Select the **Chart Library** you just created. Click Add Chart to add multiple charts.

a) Select Top 20 Internal IP by Sessions with Bandwidth.

b) Select Top 20 External IP by Sessions with Bandwidth.

c) Select Top 20 Client IP by Sessions with Bandwidth.

d) Select Top 20 Server IP by Sessions with Bandwidth.

e) Select Top 20 Website Domains Sessions Distribution.

f) Select Top 20 Website Domains by Internal IP and Sessions.

g) Select Top 20 Website Domains by Server IP and Bandwidth.

h) Select Top 20 Subscriber ID by Website domains and Sessions.

4. Specify **Filter**. Select Action as Field; select "=" as Operator; set Value to Intercept.
5. Click **OK**.

The 10th: Traffic QoS Analysis for Specific Data Center

Customize a report to diagnose traffic statistics for specific data center. For example, create a report to analyze traffic QoS for Data Center DC2. It will include 4 charts and tables, that display network traffic QoS, including Estimated One-sided Connections, Internal IP at Top ADC Bandwidth Tend, SSL Certificate Installation Unique Client IP Trend and Top Internal IP Drill down Sled IP (by Sessions).

STEP 1. Create Datasets. Select **Reports** > **Datasets** menu.

1. Select a predefined Dataset with **Name** Estimated-One-sided-Connections. You can use the search function, click and select Name and enter the given name.
2. Create a Dataset with **Name** Traffic-Bandwidth-Trend-ADC-Internal-IP-192.168.50.2.
   a) Select Session Records for **Log Type**.
   b) Select Device ID and Receive Time as **Group by**.
   c) Specify the **Data Bindings**, add field, aggregate, and label. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.
   d) Specify **Filter**. Select Internal IP as Field; select "=" as Operator; set Value to 192.168.50.2. This configuration will restrict data source only come from ADC with internal IP 192.168.50.2.
   e) Click **OK**.
3. Create a Dataset with **Name** Unique-Client-Security-Intercept-Event-with-Bytes.
   a) Select Security Event for **Log Type**.
   b) Select Receive Time as **Group by**.
   c) Specify the **Data Bindings**, add field, aggregate, and label. You can add multiple items for data bindings. Now add 2 items.

i. Field select Bytes Sent and Bytes Received; aggregate select sum; Label set to Bytes.

ii. Field select Client IP; aggregate select count distinct; Label set to Unique Client IP.

d) Specify **Filter**. Select SSL.Intercept State as Field; select "=" as Operator; set Value to Intercept. This configuration will only include Security Events with intercept action.

e) Specify **Having**. Field select Bytes Sent and Bytes Received; Aggregate select sum; Operator select ">="; Value set to 65535.

f) Click **OK**.

4. Create a Dataset with **Name** Traffic-Top-Internal-IP-by-Sled-IP-and-Sessions.

a) Select Session Records for **Log Type**.

b) Select Internal IP and Sled IP as **Group by**.

c) Specify the **Data Bindings**, add field, aggregate, and label. Field select Sessions; aggregate select sum; Label set to Sessions.

d) Click **OK**.

5. Create a Dataset with **Name** Traffic-Top-Domain-by-Establish-Latency-Time.

a) Select Session Records for **Log Type**.

b) Select Http.Domain as **Group by**.

c) Specify the **Data Bindings**, add field, aggregate, and label. Field select Establish Latency(ms); aggregate select avg; Label set to Avg Establish Latency(ms).

d) Specify **Filter**. Select Http.Domain as Field; select notEmpty as Operator. This configuration will exclude empty domains.

e) Click **OK**.

STEP 2. Create 4 Chart Libraries. Chart set the display form of the data. Select **Reports** > **Chart Library** menu, and click Create.

1. Create a Chart Library with **Name** Estimated One-sided Connections with Bandwidth.

a) Select **Dataset** Estimated-One-sided-Connections.

b) Select **Chart Type**: Area.

c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. You can add multiple Data Binding for Lines.

i. Select Bytes as Data Binding; select Bandwidth (Kbps/Mbps/Gbps) as Format; select Line Down as Type; input bps as Legend.

ii. Select one_side_percent as Data Binding; select Default as Format; select Line Up as Type; input percent as Legend.

d) Click **OK**.

2. Create a Chart Library with **Name** Traffic Bandwidth Trend Top 10 ADC Internal IP 192.168.50.2.
   a) Select **Dataset** Traffic-Bandwidth-Trend-ADC-Internal-IP-192.168.50.2.
   b) Select **Chart Type**: Line.
   c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. For Lines, select Bytes as Data Binding; select Bandwidth(Kbps/Mbps/Gbps) as Format; select Line Up as Type; input bps as Legend. Group By Device ID; Order By Bytes Descending; Show Top 10.
   d) Click **OK**.
3. Create a Chart Library with **Name** Security Intercept Event Certificated Unique Client Trend.
   a) Select **Dataset** Unique-Client-Security-Intercept-Event-with-Bytes.
   b) Select **Chart Type**: Line.
   c) Set the **Data Bindings** for X-Axis and Lines. Select Receive Time defined in Group By from dataset as X-Axis. Select Time as Format. For Lines, select Unique Client IP as Data Binding; select Counter(K/M/G) as Format; select Line Up as Type; input Unique Client IP as Legend.
   d) Click **OK**.
4. Create a Chart Library with **Name** Traffic Top 20 Internal IP by Sled IP and Sessions.
   a) Select **Dataset** Traffic-Top-Internal-IP-by-Sled-IP-and-Sessions.
   b) Select **Chart Type**: Table
   c) For **Data Bindings**, select Drilldown as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Internal IP as Title; select Internal IP as Data Binding; input Internal IP as Legend. Fill in Sled IP as Title; select Sled IP as Data Binding; input Sled IP as Legend. Fill in Sessions as Title; select Sessions as Data Binding; select Counter(K/M/G) as Format; input Sessions as Legend. Select order by Sessions descending. Show Top 20. For Drilldown, Fill in Show Top 20.

d) Click **OK**.

5. Create a Chart Library with **Name** Top 100 Most Time Consuming Domains.
   a) Select **Dataset** Traffic-Top-Domain-by-Establish-Latency-Time.
   b) Select **Chart Type**: Table
   c) For **Data Bindings**, select Regular as Table Type. Click to add Column. Complete Title, Width, Data Binding, Format and Legend for each column. Fill in Domain as Title; select Domain as Data Binding; input Domain as Legend. Fill in Avg Establish Latency(ms) as Title; select Avg Establish Latency(ms) as Data Binding; input Avg Establish Latency(ms) as Legend. Select order by Avg Establish Latency(ms) descending. Show Top 100.
   d) Click **OK**.

STEP 3. Create a Report.

1. Enter a **Name**, Security Event Intercept Endpoints Report.
2. Select a **Time Period**. Select last 1 day.
3. Select the **Chart Library** you just created. Click Add Chart to add multiple charts.
   a) Select Estimated One-sided Connections with Bandwidth. Set 5 minutes for **Time Granularity**.
   b) Select Traffic Bandwidth Trend Top 10 ADC Internal IP 192.168.50.2. Set 5 minutes for **Time Granularity**.
   c) Select Security Intercept Event Certificated Unique Client Trend. Set 1 hour for **Time Granularity**.
   d) Select Traffic Top 20 Internal IP by Sled IP and Sessions.
   e) Select Top 100 Most Time Consuming Domains.
4. Specify **Filter**. Select Data Center as Field; select "=" as Operator; set Value to DC2.
5. Click **OK**.

Contact Information
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Last Revised
September 10, 2021